HYPER-RIDGE PERSONALEVO: META-LEARNING THE ARCHIVE GEOMETRY FOR PROVABLE PERSONALIZED ROUTING

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ABSTRACT

Personalized agents maintain libraries of prompt variants, skill modules, or orchestration policies, which we call *candidates*, collected in an *archive*.

The simplest strategy is *uniform deployment*: pick the single candidate with the highest average reward across all users.

We propose PersonalEvo, a contextual-bandit method that learns to route each user to their best-matching candidate online, and optionally grows the archive by evolving new candidates after failures.

We give PersonalEvo a provable guarantee: its reward decomposes exactly into four interpretable terms, the *default reward* of the best single candidate served to everyone, a *routing gain* Δ_{route} from matching each user to their own best candidate, an *evolution gain* from newly discovered candidates, and an *exploration cost* paid while learning user preferences.

This decomposition yields a sharp condition for when personalization helps, routing provably outperforms uniform deployment whenever the routing and evolution gains together exceed the exploration cost, together with a plug-in regret bound on the exploration cost.

We evaluate PersonalEvo on nine benchmarks: three synthetic personalized-agent environments, two agent-framework substrates (GEPA prompt archives (Agrawal et al., 2025), Hermes skill configurations (Team, 2025b)), and four real-data preference datasets (BESPOKE (Team, 2025c), PAHF Shopping and Embodied (Labs, 2025), and PrefEval (Zhao et al., 2025)).

Across all nine, PersonalEvo achieves higher mean reward than both the static baseline and random candidate selection.

Our results show that: (1) the routing gain Δ_{route} is larger than the evolution gain on every benchmark; (2) on the four real-data benchmarks, Δ_{route} ranges from 0.036 to 0.251, indicating that the value of personalized routing varies substantially across application domains; (3) growing the archive without gating hurts performance, because the added observed-reward residual outweighs the evolution gain. The saved base experiments report $m_t - r_t$, not the mean-reward exploration cost $m_t - \mu(U, a_t)$ in our theorem.

Building on this base, we introduce *Hyper-Ridge PersonalEvo*, which replaces independent isotropic regularization with a shared archive geometry. For known geometry, we prove a pathwise exploration bound with an explicit confidence radius and an effective-dimension potential term. We also prove finite-sample covariance control for a calibration-split variant that estimates the geometry from fresh candidates and freezes it before routing. With fixed problem constants, its explicit calibration charge and width-inflation remainder are $o(\sqrt{T})$, while its full exploration bound is $O(\sqrt{T}\log T)$. Neither theorem covers the periodically re-estimated router. In diagnostics, the true geometry lowers cumulative exploration cost by 46–59%, whereas periodic re-estimation raises it by 7–19%.

1 INTRODUCTION

Modern LLM agents increasingly choose among collections of prompts, skills, tools, and workflows instead of relying on one fixed policy. We call such a collection an *archive* and each deployable entry a *candidate*. Personalization methods learn user profiles and preference-conditioned behavior (Salemi et al., 2023; Zhao et al., 2025; Team, 2025c); prompt and agent evolution methods search for stronger candidates (Fernando et al., 2023; Guo et al., 2024; Zhou et al., 2024; Agrawal et al., 2025); and routing methods choose among available models or policies at inference time (Ong et al., 2024; Hu et al., 2024; Li, 2025; Tsai & Tran, 2026). These lines of work address three related decisions: what a user prefers, how the archive should improve, and which candidate should handle the next request.

Consider a support agent whose archive contains formal and casual policies for refund and shipping requests. Serving every user the best policy on average discards this useful diversity. A personalized router can learn which policy works for each user, but it must first try candidates whose value is uncertain. Archive growth makes this cost recur: whenever the agent adds a mutation after a failure, a conventional bandit sees a new arm and starts learning it from scratch. Yet the candidates are related. What the router learns about formality from a refund policy should inform a formal shipping policy, even though the two policies should retain different reward models. Archive growth therefore creates a tension: it can add better candidates while also adding enough exploration to erase their benefit.

Existing methods resolve only parts of this tension. Profile and memory methods condition an agent on the user, but do not control exploration over a growing archive. Evolution methods improve candidate quality, usually offline or at the population level, but do not learn which candidate to serve to each user online. Fixed-pool routers make that assignment, but do not account for archive growth. Standard disjoint linear bandits also estimate every candidate independently, so evidence about one candidate cannot reduce uncertainty about another. Pooling all candidates into one predictor goes too far in the other direction and erases real candidate-specific differences. What is missing is the *geometry of variation across the archive*: which behavioral directions recur across candidates, which differences remain candidate-specific, and when evidence can be transferred between them.

These observations lead to the question studied in this paper:

> *Can a self-evolving agent transfer evidence across related archive candidates while retaining a measurable guarantee that personalized routing is worth its exploration cost?*

We address this question in two steps. First, *PersonalEvo* separates the value of personalization from the cost of learning it. Let A_1 be the initial archive and let R_T be the router's average reward over T interactions. We prove the exact decomposition

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{1}$$

where $V_{\text{static}}(A_1)$ is the reward from serving everyone the best single initial candidate, $\Delta_{\text{route}}(A_1)$ is the gain from matching users to different initial candidates, $\bar{G}_T$ is the gain from candidates discovered after deployment, and $\bar{L}_T$ is the reward lost while the router learns. In the support example, these terms are the best universal reply policy, the value of matching tone and topic to each user, the value of newly evolved replies, and the cost of trying the wrong reply. The identity holds for any measurable router and gives the testable deployment criterion

$$\Delta_{\text{route}}(A_1) + \bar{G}_T \;>\; \bar{L}_T, \tag{2}$$

so personalized routing is worthwhile exactly when its routing and evolution gains cover its exploration cost.

Second, *Hyper-Ridge PersonalEvo* targets the only negative term in equation 1. Each candidate a has its own parameter vector θ_a, which maps d user and candidate features to expected reward. A disjoint LinUCB router estimates each vector with the same spherical regularizer and therefore encodes no relation among candidates (Abbasi-Yadkori et al., 2011; Chu et al., 2011). We instead model the candidate vectors as draws with a shared covariance Ω. This covariance is the archive geometry. Its dominant directions capture recurring factors such as tone and topic, while every candidate keeps its own reward model. Hyper-Ridge uses this geometry to regularize candidate estimates, so a newly evolved candidate starts with uncertainty shaped by the archive rather than an isotropic blank slate.

The gain is largest when candidate parameters vary mainly along a few shared directions even though the feature representation is high-dimensional. Let $\bar{d}_{\mathrm{eff},T}(\Omega)$ be a deterministic upper bound on the effective dimension of every candidate design through round T, and let $c_T^{\Omega} = 1 + \log(1 + TL_{\Omega}^2/\lambda)$. For known Ω, our theorem gives the explicit exploration budget

$$B_T^{\Omega}(\delta) = 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega)}, \tag{3}$$

where $\alpha_{\Omega,T}(\delta)$ is the confidence radius stated in Proposition 8. The base LinUCB budget has the same form with its own radius and the isotropic potential $d\log(1 + TL_x^2/(\lambda d))$. Thus geometry helps only when the complete budget in equation 3, including its confidence radius, is smaller. For $T \geq 2$, with fixed problem parameters and $\delta = 1/T$, both valid noisy-reward bounds scale as $O(\sqrt{T}\log T)$; the benefit is in their problem-dependent geometric factors, not a missing logarithm. The calibration-split theorem retains an estimated-geometry confidence radius; its separate calibration charge and width-inflation remainder are $o(\sqrt{T})$. It does not cover the periodically updated estimator in Algorithm 1.

Our experiments separate the motivation for Hyper-Ridge from evidence for its mechanism. Across nine benchmarks, the base PersonalEvo router has routing gains from 0.036 to 0.280, evolution gains from 0 to 0.061, and observed-reward residuals from 0.010 to 0.071. Because the simulator clips noisy rewards, these residuals are not identified with the theorem's mean-reward exploration cost. On separate planted rank-three problems, giving Hyper-Ridge the true geometry lowers cumulative exploration cost by 46–59%. Periodically estimating that geometry instead raises the cost by 7–19%. On seven adapter benchmarks, one-hot candidate features make the full-covariance router identical to its diagonal ablation, so their reward changes cannot show cross-candidate transfer. The evidence therefore supports the known-geometry mechanism while exposing the unresolved estimation and representation problem.

Contributions. This paper makes four contributions:

1. **A measurable formulation of personalized self-evolution.** We formalize routing over a growing candidate archive and prove the exact four-term decomposition equation 1, which turns the decision to personalize into the testable criterion equation 2.
2. **Archive-aware routing.** We introduce Hyper-Ridge PersonalEvo, which learns a covariance geometry across candidate reward models and uses it in a generalized-ridge UCB router. It shares evidence about important feature directions while retaining candidate-specific predictions.
3. **Effective-dimension guarantees with a clear boundary.** We prove a known-Ω bound with its full confidence radius and a finite-sample calibration-split theorem for estimated Ω. We do not transfer either guarantee to the unresolved periodic estimator.
4. **Evidence that separates mechanism from implementation.** We measure the structural terms and report the logged observed-reward residual on nine benchmarks, then test Hyper-Ridge in a controlled low-rank setting. The results show when the right geometry helps and why the current periodic estimator and one-hot representation do not yet realize that gain.

Paper organization. Section 2 positions the work relative to personalization, agent evolution, routing, and linear bandits. Section 3 defines the routing problem and Hyper-Ridge estimator. Section 5.1 develops the decomposition and effective-dimension analysis. Section 6 reports the base-router results, Hyper-Ridge diagnostics, and remaining evaluation gap. Appendix J proves the oracle geometry results, and Appendix L gives the calibration-split covariance theorem.

2 Related Work

Routing queries across a model or prompt pool to trade off cost and quality is now standard: RouteLLM (Ong et al., 2024) learns a strong/weak router from preference data and RouterBench

(Hu et al., 2024) benchmarks such routers, while LLM Bandit (Li, 2025) and NeuralUCB routing (Tsai & Tran, 2026) cast routing as an online contextual bandit.

These route at the *model* level over a fixed pool; PersonalEvo routes *per user* over an archive that may grow online, and decomposes the resulting reward rather than only reporting it.

Benchmarks such as LaMP (Salemi et al., 2023), PersonalLLM (Zollo et al., 2024), and PrefEval (Zhao et al., 2025) supply user-feedback signals, while PersonaAgent (Zhang et al., 2026) and persistent-memory frameworks (Westhäußer et al., 2025) maintain per-user personas and evolving profiles.

The closest personalized-bandit antecedent, PAB-F (Team, 2025c), models per-user reward as a factorized function of context and a fixed memory/tool/style action set, and studies regret against an in-archive oracle.

Prompt and agent evolution methods, including Promptbreeder (Fernando et al., 2023), EvoPrompt (Guo et al., 2024), Symbolic Learning (Zhou et al., 2024), and GEPA's reflective Pareto search (Agrawal et al., 2025), improve prompts or archives through offline search rather than online user-conditioned routing.

Hermes self-evolution (Team, 2025b;a) adds constraint-gated growth, which motivates our archive-growth gate.

Classical contextual-bandit methods such as OFUL and LinUCB (Abbasi-Yadkori et al., 2011; Chu et al., 2011; Li et al., 2010) provide standard finite-time tools for controlling exploration in linear bandits, and related structural-vs-online decompositions appear in contextual pricing, offline RL, and multi-task transfer.

3 Preliminaries: PersonalEvo Routing and the Hyper-Ridge Estimator

3.1 Overview

A personalized agent keeps a small library, an *archive*, of candidate prompts, skills, or routing templates.

At each round, for each user it routes one user at a time: it looks at the user, picks one candidate from the current archive, observes a reward, updates internal state, and may optionally append a child candidate to the archive after the round.

We compare this personalized routing policy to the simplest possible alternative: pick the single candidate with the best population-mean reward in the initial archive, and use it for everyone forever.

The remainder of this section formalizes the notation, introduces the hyper-ridge estimator and its generalized-ridge model, presents the routing algorithm, and states the base theoretical results.

3.2 Notation and setup

A user $U \sim \Pi$ is drawn from a population: $\mathcal{U}$ denotes the user space (in our benchmarks a finite set of user profiles; in general an abstract measurable space), and Π is a probability distribution over $\mathcal{U}$, so Π is not a vector or matrix and has no dimension of its own; it simply specifies how likely each user is to arrive. Practically, Π is the deployment's user mix, and expectations $\mathbb{E}_U[\cdot]$ below average over that mix.

The system operates over $T \in \mathbb{N}$ rounds (discrete interaction steps, e.g. successive requests) starting from a deterministic common initial archive A_1. The archive $A_1 = \{a^{(1)}, \ldots, a^{(K)}\}$ is a finite set of K candidates; each element a is one deployable unit of agent behavior (a prompt variant, skill module, or orchestration policy), and $K = |A_1|$ is its size. "Common" means every user starts from the same library. Later archives $A_t(U)$ may grow beyond A_1 as evolution adds candidates, up to the cap $K_{\max}$.

The analysis follows *one* user's trajectory and then averages over the population. A single user U is drawn once and interacts with the system for T rounds. Let $\mathcal{O}_{t-1}$ be the router's observable pre-action history: the user, the initial archive, all earlier actions and rewards, and the archive available at round t. For concentration arguments, let $\mathcal{F}_{t-1} \supseteq \mathcal{O}_{t-1}$ be an analysis filtration that also contains latent candidate parameters and proposer randomness once they are generated. The router's action is $\mathcal{O}_{t-1}$-measurable; enlarging the filtration for the proof does not give the router access to those latent variables. Population-level statements then average over U and all online randomness. Quantities with argument U, such as $A_t(U)$, $a_t(U)$, $G_t(U)$, and $L_t(U)$, are per-user random variables. Quantities with an expectation or a bar, such as V_{static}, V_{route}, $\bar{G}_T$, $\bar{L}_T$, and R_T, are population averages.

At round t, the user-specific archive $A_t(U)$ is available to the router and is required to be $\mathcal{O}_{t-1}$-measurable.

The router selects $a_t(U) \in A_t(U)$ and observes the reward

$$r_t \;=\; \mu(U, a_t(U)) + \xi_t,$$

Here r_t is the scalar feedback generated when user U receives candidate $a_t(U)$. Conditional on the user, archive construction, and latent candidate parameters, its remaining randomness is the observation noise ξ_t, with $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$. Unconditional statements may also average over proposer randomness and the random-effects model introduced below. The function $\mu(u, a) \in [0, 1]$ is the noise-free mean reward of serving candidate a to user u.

The current-archive oracle value is $m_t(U) = \max_{a \in A_t(U)} \mu(U, a)$.

We define the four terms of equation 6.

The *default reward* is $V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}_U[\mu(U, a)]$: average each candidate's reward over all users *first*, then pick the single best candidate and serve it to everyone. This is the performance of the simplest possible deployment, one that does no personalization at all (a single fixed prompt or policy shipped to every user), so it is the operating point a team is already at before adopting any routing method; it is the baseline any personalization system must beat to justify its cost.

The *oracle routing value* is $V_{\text{route}}(A) = \mathbb{E}_U[\max_{a \in A} \mu(U, a)]$, with the two operations in the opposite order: for *each* user pick that user's best candidate first, then average over users. This is the ceiling of personalization, the reward if user preferences were known perfectly and no exploration were ever needed. Practically it answers a question a team should ask before building anything: "if personalization worked perfectly on this archive, how much would we earn?" No online algorithm on the fixed archive can exceed it.

Their difference is the *routing gain* $\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A) \geq 0$: the value of swapping the order of $\max$ and $\mathbb{E}_U$, i.e., precisely the value of matching candidates to users instead of shipping one candidate to all. It is the budget available to pay for personalization. If users are nearly homogeneous, or one candidate dominates for everyone, then per-user best and population best coincide, $\Delta_{\text{route}} \approx 0$, and personalization cannot pay no matter how good the algorithm; if different user segments prefer different candidates (the terse-diff versus explanation developers of Section 1), Δ_{route} is large and routing is worth building. This is the practical situation the pair of definitions exists for: deciding, from a held-out user sample and before deployment, whether personalized routing is worth its exploration cost.

These quantities say *why* personalization matters but not *when* it pays off; the dominance condition of Corollary 3 answers the latter by weighing Δ_{route} against the exploration cost.

The *evolution gain* at round t is $G_t(U) = m_t(U) - m_1(U)$, measuring how much the archive has improved since initialization. We define it because self-modification is not free (each mutation adds an arm the router must explore), so a deployment needs to know how much reward the newly evolved candidates are actually adding; $G_t(U)$ is exactly that amount, the improvement of the best-available candidate for user U between round 1 and round t. It is measured by bookkeeping, not by extra experiments: whenever a candidate is added, evaluate the incumbent best and the new candidate for the affected user and record the difference of the two bests. Concretely, in the support-archive example: if user U's best launch-day template has mean satisfaction $m_1(U) = 0.70$ and at round t a mutated template raised the best available to $m_t(U) = 0.78$, then $G_t(U) = 0.08$; if evolution never produced anything better for this user, $G_t(U) = 0$. In our simulators μ is known so $m_t(U)$ is computed exactly; in a live system it is estimated by scoring the current archive on the user's held-out interactions.

The *exploration cost* is $L_t(U) = m_t(U) - \mu(U, a_t(U))$, the gap between the current-archive best and what the router actually picked. It is the price of learning and the only negative term in the decomposition. Continuing the same example, if the best current template has mean satisfaction 0.78 but the router serves one with mean 0.60, that round costs 0.18. The simulator contains both means needed to compute this quantity, but the saved headline tables use $m_t - r_t$ instead. A live system must estimate both means on held-out or logged interactions and report the resulting uncertainty separately.

The horizon-averaged terms are $\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[G_t(U)]$ and $\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[L_t(U)]$.

3.3 Model

For the Hyper-Ridge theory, write the bounded mean reward as

$$\mu(u, a) = b(u) + x(u, a)^\top \theta_a,$$

where $x(u, a) \in \mathbb{R}^d$ is the user-candidate feature vector and the known baseline $b(u)$ is common to every candidate offered to user u. We assume $\mu(u, a) \in [0, 1]$. A convenient compatible model is $b(u) = 1/2$ and $x(u, a)^\top \theta_a \in [-1/2, 1/2]$.

Instead of estimating each θ_a with ordinary ridge regression, assume the candidate parameters are random effects:

$$\mathbb{E}[\theta_a] = 0, \qquad \mathbb{E}[\theta_a \theta_a^\top] = \Omega,$$

where $\Omega \in \mathbb{R}^{d\times d}$ is a hyper-covariance matrix shared across archive candidates. The common baseline makes this zero-mean random-effects model compatible with nonnegative rewards. It also cancels from every candidate comparison.

The goal is to estimate Ω from the archive and use it to regularize routing.

3.4 Hyper-ridge estimator

For candidate a, define

$$X_{a,t} = \left[x_s^\top\right]_{s<t:a_s=a}, \qquad y_{a,t} = (r_s - b(u_s))_{s<t:a_s=a}.$$

Given an estimate $\widehat{\Omega}_t$, define the generalized ridge estimator

$$\widehat{\theta}_{a,t} = \arg\min_{\theta\in\mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda \theta^\top \widehat{\Omega}_t^{-1}\theta \right\}.$$

Equivalently,

$$\widehat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}\right)^{-1} X_{a,t}^\top y_{a,t}.$$

Define the generalized design matrix

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}.$$

Then the Hyper-Ridge UCB score is

$$\text{UCB}_t^{\text{hyper}}(u, a) = x(u, a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u, a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u, a)}.$$

The selected candidate is

Algorithm 1 Hyper-Ridge PersonalEvo routing

Require: Initial archive A_1, ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period $\tau \in \mathbb{N}$, regularizer $\epsilon > 0$

Initialize $\widehat{\Omega}_1 \leftarrow I_d$; for each candidate a, initialize $X_{a,1}$ and $y_{a,1}$ as empty
for $t = 1, 2, \dots, T$ **do**
  Observe user u_t and eligible candidate set $A_t(u_t)$
  for each candidate $a \in A_t(u_t)$ **do**
    Compute $V_{a,t}^{\text{hyper}} \leftarrow X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}$
    Compute $\widehat{\theta}_{a,t} \leftarrow \left(V_{a,t}^{\text{hyper}}\right)^{-1} X_{a,t}^\top y_{a,t}$
    Compute $\mathrm{UCB}_t^{\text{hyper}}(u_t, a) \leftarrow x(u_t, a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u_t, a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u_t, a)}$
  end for
  Select $a_t \in \arg\max_{a \in A_t(u_t)} \mathrm{UCB}_t^{\text{hyper}}(u_t, a)$ and observe reward r_t; set $y_t \leftarrow r_t - b(u_t)$
  Update per-candidate data: append $x(u_t, a_t)^\top$ to $X_{a_t,t+1}$ and y_t to $y_{a_t,t+1}$
  if $t \equiv 0 \pmod{\tau}$ **then**
    Re-estimate the hyper-covariance from the archive's candidate estimates:

$$\widehat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \widehat{\theta}_{a,t} \widehat{\theta}_{a,t}^\top + \epsilon I_d$$

  else
    $\widehat{\Omega}_{t+1} \leftarrow \widehat{\Omega}_t$
  end if
end for

$$a_t(u) \in \arg \max_{a \in A_t(u)} \mathrm{UCB}_t^{\text{hyper}}(u, a).$$

3.5 ALGORITHM

Algorithm 1 summarizes the resulting routing procedure. It mirrors the base PersonalEvo router round for round; the only changes are the generalized ridge geometry $\lambda \widehat{\Omega}_t^{-1}$ inside $V_{a,t}^{\text{hyper}}$ and the periodic re-estimation of the hyper-covariance from the archive's own per-candidate estimates.

Three clarifications about the data the algorithm accumulates. First, $y_{a,t}$ stacks centered observed rewards $r_s - b(u_s)$, not unknown mean rewards. Second, row s of $X_{a,t}$ is the feature vector for the user served at that round, so a shared router may pool observations from many users. Third, Algorithm 1 allows a stream of users, whereas the four-term decomposition follows one user and then averages over the population. Whenever we combine a Hyper-Ridge bound with that decomposition, we use the per-user specialization $u_t \equiv U$. The confidence and potential lemmas also hold for a predictable user stream, but the four-term conclusion does not automatically transfer to that setting.

The per-round cost is identical to base PersonalEvo up to the re-estimation step, which touches only the $d \times d$ matrix $\widehat{\Omega}_t$ and amortizes over the period τ. This periodic update is an experimental method. Proposition 8 analyzes the oracle variant that fixes $\widehat{\Omega}_t = \Omega$ and uses the stated confidence radius. Theorem 57 analyzes a second variant that estimates Ω on independent calibration candidates and freezes the estimate before routing. Neither result applies to the periodic update above.

The remainder of the paper turns to empirical evidence. Section 6 first reports the nine-benchmark results for the base PersonalEvo router and then separates two Hyper-Ridge questions: whether a known archive geometry can reduce exploration, and whether the current periodic estimator can recover that geometry from online data.

Algorithm 2 describes PersonalEvo's per-user routing loop.

At each round, a disjoint LinUCB bandit selects the best candidate for the current user based on UCB scores.

Algorithm 2 Per-user personalized routing over a growing archive.

Require: user u, common initial archive A_1, horizon T, exploration parameter α, gap threshold τ, cooldown κ, archive cap $K_{\max}$

```
A_1(u) ← A_1; for each a ∈ A_1(u) initialize A_a ← I, b_a ← 0
t_last ← −∞
for t = 1, ..., T do
    for each a ∈ A_t(u) do
        x_t(a) ← ψ(u, a)                                          ▷ feature map, equation 4
        θ̂_a ← A_a^{-1} b_a
        UCB_t(a) ← θ̂_a^⊤ x_t(a) + α sqrt( x_t(a)^⊤ A_a^{-1} x_t(a) )
    end for
    a_t(u) ← arg max_{a ∈ A_t(u)} UCB_t(a)
    observe r_t = μ(u, a_t(u)) + ξ_t
    A_{a_t(u)} ← A_{a_t(u)} + x_t(a_t(u)) x_t(a_t(u))^⊤;   b_{a_t(u)} ← b_{a_t(u)} + r_t x_t(a_t(u))
    if m_t(u) − r_t > τ and t − t_last > κ and |A_t(u)| < K_max then   ▷ oracle-assisted gate
        ã_t(u) ← F(u, A_t(u), trace_t); A_{t+1}(u) ← A_t(u) ∪ {ã_t(u)}; initialize A_ã, b_ã; t_last ← t
    else
        A_{t+1}(u) ← A_t(u)
    end if
end for
```

In the simulator, an oracle-assisted gate compares the unknown current-archive optimum with the observed reward and may propose a new candidate.

The feature map used in our experiments is

$$\psi(u,a) = \big[\, \mathrm{onehot}(\mathrm{cat}(a))\, ;\, w_u \,\big] \in \mathbb{R}^{2K_{\mathrm{cat}}}, \tag{4}$$

where $\mathrm{cat}(a)$ is the candidate's category and $w_u \in \mathbb{R}^{K_{\mathrm{cat}}}$ is the user preference vector exposed by the simulator.

As a concrete instance (used in our experiments), the $K_{\mathrm{cat}} = 5$ categories are $\{\mathrm{sorting}, \mathrm{searching}, \mathrm{formatting}, \mathrm{general}, \mathrm{calc}\}$, so for a candidate a in the `searching` category, $\mathrm{onehot}(\mathrm{cat}(a)) = (0,1,0,0,0)$ and the user block w_u records that user's affinity for each of the five categories.

We refer the reader to Appendix H for the full construction (categories, quality components, and feature concatenation), and to Team (2025c) for a richer factorized feature map over task type, memory mode, tool mode, and answer style.

The reflection trigger is the high-gap-failure rule

$$m_t(u) - r_t > \tau, \qquad t - t_{\mathrm{last}} > \kappa, \tag{5}$$

This gate is not deployable as written because $m_t(u)$ uses unknown mean rewards. The saved simulator also uses an oracle-assisted proposer: it reads the user's peak category and returns a child candidate biased toward that category.

The measured evolution gain therefore describes this idealized simulator. It is not a formal upper bound on a different proposer, and it does not show what an LLM-driven proposer would achieve.

Reward decomposition. Using Algorithm 2, we show that the average reward the router achieves over horizon T satisfies (the subscript 1 in A_1 refers to the round $t = 1$ initial archive shared across all users, not to a particular user):

$$\text{routed value} \;=\; V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{6}$$

This equation separates the router's performance into the four terms defined above. Computing all four terms requires the current-archive optimum and the selected candidate's mean reward; observed rewards alone are not enough.

Example. Consider two candidates and two equally likely user types with means $\mu(u_1, a_1) = 0.9$, $\mu(u_2, a_1) = 0.4$, $\mu(u_1, a_2) = 0.4$, $\mu(u_2, a_2) = 0.9$.

Then $V_{\text{static}} = 0.65$ (the best single candidate averages 0.65), while $V_{\text{route}} = 0.9$ (matching each user to their best gives 0.9), so $\Delta_{\text{route}} = 0.25$.

With no archive growth ($\bar{G}_T = 0$), the decomposition gives: routed value $= 0.65 + 0.25 - \bar{L}_T$.

Routing is profitable as long as the exploration cost $\bar{L}_T < 0.25$.

When a held-out reward model supplies $\mu(u, a)$, V_{static} and Δ_{route} can be estimated from a held-out user sample: $V_{\text{static}}(A_1) = \max_a N^{-1} \sum_u \mu(u, a)$ and $\Delta_{\text{route}}(A_1) = N^{-1} \sum_u \max_a \mu(u, a) - V_{\text{static}}(A_1)$.

Computing $\bar{G}_T$ and $\bar{L}_T$ also requires the archive means $m_t(u)$ and the selected mean $\mu(u, a_t(u))$. The saved tables use $m_t(u) - r_t$ instead, which is a different observed-reward residual.

4 METHOD

The preceding sections establish the base PersonalEvo framework: a routing decomposition over a self-evolving archive, guarantees for the fixed-archive router, and a repair-time bound for archive growth. All of these results are inherited unchanged. In this section we introduce the revision's contribution: a replacement for the router's per-candidate estimation scheme, motivated by a meta-learning perspective on the archive. The new method leaves the decomposition and its first three terms intact and targets only the exploration-cost term, by importing hyper-covariance estimation ideas from generalized ridge regression. We first record the two source documents that anchor this revision.

4.1 SUMMARY OF WHAT CHANGES

Before turning to experiments, we summarize how this section modifies the base paper. First, the router changes: the generalized-ridge router defined by $V_{a,t}^{\text{hyper}}$ and $\text{UCB}_t^{\text{hyper}}$ replaces the base per-candidate LinUCB router, which regularizes each θ_a independently with an isotropic ridge penalty. Second, the exploration-cost theory changes: the known-Ω theorem replaces the isotropic potential by an effective-dimension potential while retaining the full confidence radius. Appendix L proves finite-sample control for a calibration-split estimator that freezes $\widehat{\Omega}$ before routing. The periodically updated estimator in Algorithm 1 has no guarantee in this paper. Geometry improves the certificate only when the full Hyper-Ridge budget is smaller than the full LinUCB budget; low effective dimension alone is not sufficient. Finally, nothing else changes: the four-term decomposition $R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$ is inherited from the base framework exactly as proven there, and this revision only sharpens the bound on the $\bar{L}_T$ term.

5 THEORETICAL RESULTS FOR HYPER-RIDGE PERSONALEVO

5.1 THEORY: IMPROVED EXPLORATION COST

Throughout this section the base assumptions of Section 3 (Assumptions 1–4 below) remain in force, and we write $x(u, a)$ for the feature map denoted $\psi(u, a)$ there; the only new hypothesis is the shared archive geometry of Assumption 7. We first restate the inherited base results this revision builds on, then derive the sharpened exploration-cost bound.

The decomposition holds under minimal conditions (bounded rewards and measurability).

Below we state the assumptions needed for the finite-time guarantee.

Assumption 1 (Bounded mean reward)**.** *The mean reward is bounded:*

$$\mu(u, a) \in [0, 1] \quad \textit{for all } (u, a). \tag{7}$$

Assumption 2 (Conditionally sub-Gaussian noise)**.** *There exists $\sigma > 0$ with*

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \;\le\; \exp(\lambda^2 \sigma^2/2), \qquad \lambda \in \mathbb{R}. \tag{8}$$

Assumption 3 (Predictable archive)**.** *The user-specific archive is predictable:*

$$\sigma(A_t(U)) \subseteq \mathcal{O}_{t-1}. \tag{9}$$

Assumption 4 (Bounded features)**.** *The feature map satisfies*

$$\|\psi(u,a)\| \;\le\; L < \infty \quad \textit{for all } (u,a). \tag{10}$$

Assumptions 2–4 are not required for Theorem 2 (the main result); they are needed only for the conditional plug-in corollary that imports an exploration-cost bound from the linear bandit literature.

The first result establishes when routing gain is strictly positive:

Lemma 1 (Static-vs-routed value)**.** *For every finite archive A,*

$$\Delta_{\text{route}}(A) \;\ge\; 0, \tag{11}$$

with equality if and only if there exists $a^\star \in A$ such that

$$\mu(U,a^\star) = \max_{a \in A} \mu(U,a) \quad \textit{almost surely.} \tag{12}$$

Thus $\Delta_{\text{route}}(A) > 0$ exactly when no single candidate is optimal for almost every user.

The proof is in Appendix A.

Main theorem.

Theorem 2 (Exact decomposition on the common initial archive)**.** *Let A_1 be deterministic, finite, and shared by all users. For every round t and every horizon $T \ge 1$,*

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)], \tag{13}$$

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{14}$$

Proof sketch (full proof in Appendix A). By definition,

$$\begin{aligned} \mu(U, a_t(U)) \;&=\; m_t(U) - L_t(U), \\ m_t(U) \;&=\; m_1(U) + G_t(U). \end{aligned}$$

Combining and taking expectations over all model and online randomness,

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; \mathbb{E}[m_1(U)] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)].$$

Lemma 1 on A_1 gives $\mathbb{E}[m_1(U)] = V_{\text{route}}(A_1) = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1)$, which is equation 13. Averaging over t yields equation 14. □

Theorem 2 says that the reward achieved by the personalized routing policy decomposes into a structural piece on the initial archive ($V_{\text{static}} + \Delta_{\text{route}}$), an online discovery piece ($\bar{G}_T$), and a negative online learning piece ($-\bar{L}_T$).

In a simulator that logs mean rewards, $\bar{L}_T$ is computed from $m_t(u) - \mu(u, a_t(u))$. The saved headline artifacts instead contain $m_t(u) - r_t$, so they do not directly measure $\bar{L}_T$.

Dominance condition.

Corollary 3 (When personalized routing beats the best static initial candidate)**.** *For any horizon T,*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{15}$$

On a fixed archive (no growth, $\bar{G}_T = 0$), this collapses to

$$\Delta_{\text{route}}(A_1) > \bar{L}_T. \tag{16}$$

In other words, personalized routing is beneficial when the routing gain and evolution gain together exceed the exploration cost.

The condition can be estimated when the evaluation protocol records mean rewards for every current candidate. A held-out user sample estimates $\Delta_{\mathrm{route}}(A_1)$; the archive and selected-candidate means estimate $\bar{G}_T$ and $\bar{L}_T$. The saved experiments report the observed-reward residual instead, so they do not verify this condition.

The static-vs-routed inequality behind Lemma 1 is the folklore $\mathbb{E}[\max_a X_a] \geq \max_a \mathbb{E}[X_a]$; our contribution is the equality condition and the *corrected* static comparator $V_{\mathrm{static}}(A_1) = \max_a \mathbb{E}_U[\mu(U,a)]$, which (unlike the heuristic $\arg\max_a q(a)$ baseline in the original simulator) makes the dominance criterion equation 15 falsifiable rather than tautological.

What is specific to PersonalEvo is the four-term accounting for a user drawn from a population and then followed through one online trajectory.

Supporting lemmas. The two lemmas below support the main theorem: Lemma 4 ensures that the per-arm bandit statistics remain valid as the archive grows, and Lemma 5 bounds the number of mutations triggered by the failure rule.

Lemma 4 (Predictable birth-order archive). *Assume at most $K_{\max}$ candidates appear by round T, and index them by predictable birth order. Candidate slot k is created at a stopping time τ_k. Its identity and feature rule are $\mathcal{O}_{\tau_k - 1}$-measurable, while its latent parameter is fixed at birth and $\mathcal{F}_{\tau_k - 1}$-measurable. Before τ_k, pad that slot's feature and selection indicator by zero. Each padded process is predictable in the analysis filtration, so self-normalized concentration applies conditionally from its birth time. A union bound over the deterministic slots $k \leq K_{\max}$ is valid. The router remains adapted to $\mathcal{O}_t$ and does not observe the latent parameter.*

Lemma 5 (Failure-prioritization). *Under the trigger equation 5, the expected number of reflection mutations across T rounds is bounded by*

$$\mathbb{E}[\#\mathrm{mutations}] \leq \sum_{t=1}^{T} \Pr\big(L_t(U) + |\xi_t| > \tau\big) \leq T \sup_{t \leq T} \Pr\big(L_t(U) + |\xi_t| > \tau\big), \tag{17}$$

which controls archive bloat without changing the regret analysis of the router itself.

The failure-prioritization gate is the algorithmic counterpart of the constraint-gated self-evolution line reviewed in Section 2, but here it is embedded inside a user-conditioned routing rule.

Positioning of the decomposition. The comparison to personalized bandits can now be stated in the paper's notation.

A fixed in-archive oracle comparison folds the structural routing gain into the oracle value, whereas equation 14 factors it out as $\Delta_{\mathrm{route}}(A_1)$.

A fixed action set also removes the archive-improvement term, whereas PersonalEvo keeps $\bar{G}_T$ explicit and uses Lemma 4 to ensure that online archive growth does not depend on future randomness.

Offline prompt-evolution methods can improve the archive, but without user-conditioned routing they do not separate the routing gain $\Delta_{\mathrm{route}}(A_1)$ from the evolution gain $\bar{G}_T$.

Finite-time bound on the exploration cost. Theorem 2 is an exact equality but does not by itself bound the size of the exploration cost $\bar{L}_T$.

To obtain a finite-time lower bound on the router's average reward, we need a controlled estimate of $\bar{L}_T$.

Rather than re-deriving such a bound from scratch for our particular router, we state it as an external assumption (Assumption 5 below) and substitute it into the decomposition.

Assumption 5 (External exploration-cost bound). *There exists a deterministic sequence B_T such that*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_T \quad \textit{for every } T \geq 1. \tag{18}$$

Corollary 6 (Plug-in bound)**.** *Under Assumptions 1–4 and 5,*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mu(U,a_t(U))] \;\ge\; V_{\text{static}}(A_1)+\Delta_{\text{route}}(A_1)+\bar{G}_T-\frac{B_T}{T}. \tag{19}$$

For the theoretical LinUCB instantiation in Proposition 16, set the UCB multiplier to $\alpha=\beta_T(\delta)$, where

$$\beta_T(\delta)=\sigma\sqrt{2\log(K_{\max}/\delta)+d\log\left(1+\frac{TL_x^2}{\lambda d}\right)}+\sqrt{\lambda}S. \tag{20}$$

Then one may take

$$B_T=2\beta_T(\delta)\sqrt{2TK_{\max}d\log\left(1+\frac{TL_x^2}{\lambda d}\right)}+T\delta. \tag{21}$$

For $T\ge 2$, with $\delta=1/T$ and fixed problem parameters, including $K_{\max}$ and d, $B_T/T=O(\log T/\sqrt{T})$. The saved experiments use a fixed heuristic α and are not covered by this certificate.

Discovery-time bound. Discovery is the most fragile term in equation 6 because it depends on the proposer $\mathcal{F}$.

We replace the original geometric contraction theorem with a stopping-time statement under a per-attempt repair probability.

Assumption 6 (Repair probability)**.** *There exists $q_\epsilon\in(0,1]$ such that for each cluster g with current gap $\Delta_g>\epsilon$, each reflection attempt produces an ϵ-improving child with probability at least q_ϵ, independent of past attempts conditional on the trigger firing.*

Theorem 7 (First-useful-specialist time)**.** *Under Assumption 6, after m reflection attempts on cluster g,*

$$\Pr\big(\text{some attempt improves cluster } g \text{ by } \epsilon\big)\;\ge\;1-(1-q_\epsilon)^m. \tag{22}$$

The saved proposer uses privileged information, but the experiments do not estimate q_ϵ for a fixed ϵ. This bound is therefore not empirically tested.

The non-oracle case is left as future work.

Section 6 reports the first three terms in equation 14 and the separate observed-reward residual $\bar{L}_T^{\text{obs}}$.

We now turn to the new result. The four-term decomposition remains unchanged:

$$R_T=V_{\text{static}}(A_1)+\Delta_{\text{route}}(A_1)+\bar{G}_T-\bar{L}_T.$$

The new method changes only the certificate for $\bar{L}_T$. The valid comparison is between two complete finite-time budgets. Each budget contains a confidence radius and an elliptical-potential term. Hyper-Ridge can reduce the second term, but a small effective dimension does not by itself imply a smaller total budget. Proposition 8 states the known-geometry budget, and Corollary 9 compares it with the exact LinUCB certificate.

5.2 Why this matters

The original dominance condition is

$$\Delta_{\text{route}}(A_1)+\bar{G}_T>\bar{L}_T.$$

Therefore, any certified reduction in $\bar{L}_T$ lowers the margin needed for personalization to pay. The qualification "certified" matters: the theorem does not claim that an arbitrary covariance estimate, or the periodic estimate in Algorithm 1, reduces exploration.

We inherit the accounting identity from the Background section unchanged: Hyper-Ridge PersonalEvo changes only the statistical control of the exploration term, not the four-term decomposition

Algorithm 3 Hyper-Ridge PersonalEvo routing

Require: initial archive A_1, ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period τ, covariance floor $\epsilon > 0$
1: Initialize $\widehat{\Omega}_1 \leftarrow I_d$
2: Initialize $X_{a,1}$ and $y_{a,1}$ as empty for each $a \in A_1$
3: **for** $t = 1, \ldots, T$ **do**
4: Observe user u_t and eligible archive $A_t(u_t)$
5: Compute $\mathrm{UCB}_t^{\mathrm{hyper}}(u_t, a)$ from equation 26–equation 28 for every $a \in A_t(u_t)$
6: Select $a_t(u_t) \in \arg\max_{a \in A_t(u_t)} \mathrm{UCB}_t^{\mathrm{hyper}}(u_t, a)$
7: Observe reward r_t, set $y_t = r_t - b(u_t)$, append $x(u_t, a_t(u_t))^\top$ to $X_{a_t,t+1}$, and append y_t to $y_{a_t,t+1}$
8: **if** $t \equiv 0 \pmod{\tau}$ **then**
9: Set

$$\widehat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \widehat{\theta}_{a,t} \widehat{\theta}_{a,t}^\top + \epsilon I_d$$

10: **else**
11: Set $\widehat{\Omega}_{t+1} \leftarrow \widehat{\Omega}_t$
12: **end if**
13: **end for**

itself. The average routed value is

$$R_T^{\mathrm{hyper}} = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] \,. \tag{23}$$

By the decomposition in Section 3, the same quantity satisfies

$$R_T^{\mathrm{hyper}} = V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{24}$$

Thus the only theoretical change in this section is the replacement of the ordinary dimension-dependent control of

$$\bar{L}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L_t(U)] \tag{25}$$

by an archive-geometry-dependent control.

The routing rule analyzed below is the implementation target for the hyper-ridge revision. For candidate a, the generalized design matrix is

$$V_{a,t}^{\mathrm{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{26}$$

The generalized ridge estimate is

$$\widehat{\theta}_{a,t} = \left(V_{a,t}^{\mathrm{hyper}}\right)^{-1} X_{a,t}^\top y_{a,t}. \tag{27}$$

The corresponding upper-confidence score is

$$\mathrm{UCB}_t^{\mathrm{hyper}}(u, a) = x(u,a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u,a)^\top \left(V_{a,t}^{\mathrm{hyper}}\right)^{-1} x(u,a)}. \tag{28}$$

Assumption 7 (Archive geometry and complexity cap)**.** *There is a known candidate-independent baseline $b(u)$ such that*

$$\mu(u,a) = b(u) + x(u,a)^\top \theta_a \in [0,1]. \tag{29}$$

The feature vector satisfies $x(u,a) \in \mathbb{R}^d$. There is a positive definite matrix Ω and finite constants S_Ω, L_Ω such that

$$\Omega \succ 0, \qquad \|\theta_a\|_{\Omega^{-1}} \le S_\Omega, \qquad x(u,a)^\top \Omega x(u,a) \le L_\Omega^2 \tag{30}$$

for every candidate that can appear through round T. Candidate births, identities, parameters, and features satisfy the predictable birth-order conditions of Lemma 4. For the fixed-user trajectory, let

$$\mathcal{A}_T(U) = \bigcup_{t=1}^{T} A_t(U) \tag{31}$$

be the set of candidates that appear through round T. For a candidate design matrix X, define

$$d_{\text{eff}}(\Omega; X) = \operatorname{tr}\left[X^\top X \left(X^\top X + \lambda \Omega^{-1}\right)^{-1}\right]. \tag{32}$$

There is a deterministic complexity cap $\bar{d}_{\text{eff},T}(\Omega)$ such that, almost surely,

$$\max_{1 \le s \le T+1} \max_{a \in \mathcal{A}_T(U)} d_{\text{eff}}(\Omega; X_{a,s}) \le \bar{d}_{\text{eff},T}(\Omega). \tag{33}$$

The known-geometry theorem needs only the ellipsoid, feature-radius, and complexity-cap conditions. Interpreting Ω as a covariance requires the additional random-effects assumptions used by the calibration-split theorem: fresh candidate effects are independent, have mean zero and covariance Ω, and the regression uses the centered response $r - b(u)$.

Proposition 8 (Known-geometry exploration-cost bound)**.** *Assume Assumptions 1–3 and Assumption 7. Suppose $|\mathcal{A}_T(U)| \le K_{\max}$. For the decomposition-compatible result, draw one user U before routing and set $u_t = U$ for all t. Define*

$$c_T^{\Omega} = 1 + \log\left(1 + \frac{T L_\Omega^2}{\lambda}\right) \tag{34}$$

and, for $\delta \in (0,1)$,

$$\alpha_{\Omega,T}(\delta) = \max\left\{1,\ \sigma\sqrt{c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega) + 2\log(K_{\max}/\delta)} + \sqrt{\lambda}S_\Omega\right\}. \tag{35}$$

Consider the oracle router that sets $\widehat{\Omega}_t = \Omega$ for every round and uses the UCB multiplier $\alpha_{\Omega,T}(\delta)$. Then, with probability at least $1-\delta$,

$$\sum_{t=1}^{T} L_t(U) \le B_T^{\Omega}(\delta) := 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)}. \tag{36}$$

Consequently,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T^{\Omega}(\delta) + \delta T. \tag{37}$$

Proof sketch. Lemma 45 gives simultaneous confidence intervals after a union bound over the predictable birth-order slots. Lemma 46 bounds each slot's log-determinant and width sum by $c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)$. Optimism gives $L_t(U) \le 2\alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t)^\top (V_{a_t,t}^{\Omega})^{-1} x_t(a_t)}$ on the confidence event. Summing the clipped widths and applying Cauchy–Schwarz yields equation 36. Since $0 \le L_t(U) \le 1$, the complement of the confidence event contributes at most δT in expectation. Appendix K gives the full derivation. □

Remark 1 (Calibration-split guarantee)**.** *For the calibration-split router, Theorem 57 proves that the relative covariance error satisfies $\rho_{\Omega,T} = O(T^{-1/10}\sqrt{\log T})$ with high probability. It uses an explicit frequentist confidence radius. Under the theorem's schedule and fixed problem constants,*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_{T,0}^{\text{cal}} + \text{Rem}_T^{\text{cal}}. \tag{38}$$

Moreover,

$$B_{T,0}^{\text{cal}} = O\left(\sqrt{dK_{\max}\bar{d}_{\text{eff},T}(\Omega)T}\ \log T\right), \tag{39}$$

$$\text{Rem}_T^{\text{cal}} = O\left(T^{2/5}(\log T)^{3/2}\right) = o(\sqrt{T}). \tag{40}$$

This is a theorem for a fixed post-calibration archive. It is not a theorem for the periodic estimator. The leading term in equation 38 is not the oracle budget: its confidence radius already depends on the covariance-error bound. The displayed remainder isolates only the additional width inflation, failure probability, and, when charged, calibration pulls.

As a concrete scale calculation, suppose

$$K_{\max} = 16. \tag{41}$$

Suppose the ambient feature dimension is

$$d = 100. \tag{42}$$

Suppose the deterministic effective-dimension cap is

$$\bar{d}_{\text{eff},T}(\Omega) = 9. \tag{43}$$

If the two confidence radii and logarithmic potential factors are equal, the width-sum part changes by the ratio

$$\frac{\sqrt{K_{\max}\bar{d}_{\text{eff},T}(\Omega)T}}{\sqrt{K_{\max}dT}} = \sqrt{\frac{9}{100}} = 0.3. \tag{44}$$

This is a geometric illustration, not the final budget ratio. The final comparison must also include the two confidence radii.

Corollary 9 (Dominance in more regimes)**.** *Define the structural-and-evolution margin by*

$$M_T = \Delta_{\text{route}}(A_1) + \bar{G}_T. \tag{45}$$

Define the hyper-ridge exploration budget by

$$B_T^{\text{hyper}} = B_T^{\Omega}(\delta) + \delta T. \tag{46}$$

If

$$M_T > \frac{B_T^{\text{hyper}}}{T}, \tag{47}$$

then Hyper-Ridge PersonalEvo satisfies the dominance condition

$$R_T^{\text{hyper}} > V_{\text{static}}(A_1). \tag{48}$$

Moreover, compared with an ordinary LinUCB budget

$$B_T^{\text{LinUCB}} = B_T^{\text{lin}}(\delta) + \delta T, \tag{49}$$

Hyper-Ridge certifies all margins in the additional interval

$$\frac{B_T^{\text{hyper}}}{T} < M_T \le \frac{B_T^{\text{LinUCB}}}{T}, \tag{50}$$

whenever

$$B_T^{\text{hyper}} < B_T^{\text{LinUCB}}. \tag{51}$$

Proof. Substituting Proposition 8 into the inherited decomposition gives

$$\begin{aligned} R_T^{\text{hyper}} &= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T \\ &\ge V_{\text{static}}(A_1) + M_T - \frac{B_T^{\text{hyper}}}{T}. \end{aligned} \tag{52}$$

Condition equation 47 implies equation 48. The interval claim follows by comparing equation 46 and equation 49. □

This corollary states the comparison without assuming its outcome. Known geometry certifies more margins only when its complete finite-time budget is smaller. The calibration-split budget may be substituted separately from Theorem 57; no claim is made for the periodic estimator.

Benchmark	$\lvert\mathcal{U}\rvert$	$\lvert A_1\rvert$	Preference structure
General	8	12	5 disjoint category peaks
Workspace	8	12	2 broad paired peaks
Support	8	12	3 overlapping clusters
GEPA archive	12	15	heterogeneous task prefs
Hermes agent	15	20	(skill,mem,tool) clusters
BESPOKE	30	40	4-dim rubric prefs
PAHF shop.	20	40	product-attribute prefs
PAHF embod.	40	40	embodied object choice
PrefEval	20	10	topic-strategy prefs

Table 1: Benchmarks. $|\mathcal{U}|$: users; $|A_1|$: candidates. Synthetic: T=300; others: T=250.

Benchmark	$V_{\rm st}$	$\Delta_{\rm rt}$	$\bar{G}_T$	$\bar{L}_T^{\rm obs}$	Obs. margin
General	.570	.280	.051	.036	**+.295**
Workspace	.577	.124	.041	.043	+.122
Support	.579	.215	.061	.039	+.237
GEPA archive	.654	.048	.023	.025	+.046
Hermes agent	.703	.092	.036	.028	+.100
BESPOKE	.653	.036	.001	.010	+.027
PAHF shop.	.577	.209	.002	.054	+.157
PAHF embod.	.290	.238	.000	.071	+.167
PrefEval	.531	.251	.000	.028	+.223

Table 2: Logged accounting summary. Here $\bar{L}_T^{\rm obs} = T^{-1}\sum_t(m_t - r_t)$, so the last column is the observed-reward margin $\Delta_{\rm rt} + \bar{G}_T - \bar{L}_T^{\rm obs}$. It is not the theoretical dominance margin unless $r_t - \mu(U, a_t)$ is conditionally mean-zero.

6 Experiments

6.1 Environments and data

We evaluate on three synthetic environments that share the same reward structure but differ in user heterogeneity.

The mean reward follows the PABF formulation (Team, 2025c):

$$\mu(u,a) = f\big(q(a),\ w_u(\mathrm{cat}(a))\big), \tag{53}$$

where $q(a) \in [0,1]$ is the intrinsic candidate quality, $\mathrm{cat}(a) \in \mathcal{C}$ is the candidate's category, $w_u \in \Delta^{|\mathcal{C}|-1}$ is the user's category preference, and f is increasing in both arguments (functional form in Appendix H.2).

The theory uses $r_t = \mu(U, a_t(U)) + \xi_t$ with conditionally mean-zero noise. The saved simulator instead records

$$r_t = \mathrm{clip}(\mu(U, a_t(U)) + \eta_t, 0, 1)\,, \qquad \eta_t \sim \mathcal{N}(0, 0.05^2). \tag{54}$$

The simulator knows every pre-clipping mean, so the theoretical exploration cost could be computed from the selected candidate's mean. The saved headline tables instead use the observed residual $m_t - r_t$. We keep that quantity separate below; in a real system $m_t(u)$ would also require held-out estimation.

We compare five variants of Algorithm 2 on all nine benchmarks: **Full** (all components enabled), **No-routing** (uniform-random candidate selection), **No-reflection** (frozen archive), **No-personalization** (onehot-only features), and **Collapse-1** (single candidate).

Each removes one component to isolate its contribution to equation 6; details and color coding are in Figure 2.

6.2 Results

We estimate the first three terms of equation 6 and the observed-reward residual $\bar{L}_T^{\rm obs}$ on each benchmark (definitions in Appendix H.5).

Table 2 (right panel above) reports the decomposition across all nine benchmarks.

Three findings hold across all nine benchmarks:

1. **Routing gain is the largest positive component.** In every row, $\Delta_{\rm route}$ exceeds $\bar{G}_T$: routing gains range from 0.036 to 0.280, while evolution gains range from 0 to 0.061. The ordering broadly tracks user heterogeneity, consistent with Lemma 1.
2. **The observed-reward margin is positive.** On all nine benchmarks, including four with real user preference data, $\Delta_{\rm route} + \bar{G}_T - \bar{L}_T^{\rm obs}$ ranges from +0.027 (BESPOKE) to +0.295

(General). Because rewards are clipped, this arithmetic does not by itself verify the mean-reward condition in Corollary 3.

3. **Real-data heterogeneity is substantial.** PAHF shopping (Δ_{route}=0.209) and PrefEval (Δ_{route}=0.251) are comparable to the synthetic environments, confirming that the decomposition's structural term is practically significant on real preferences.

Ablation. Figure 1 shows the mean reward for all five ablation variants across all nine benchmarks.

Removing the router collapses performance to near $V_{\text{static}}(A_1)$ in every case; removing reflection or personalization features each reduces reward by a smaller margin.

The hierarchy No-routing < Static < No-reflect < Full < Oracle holds consistently.

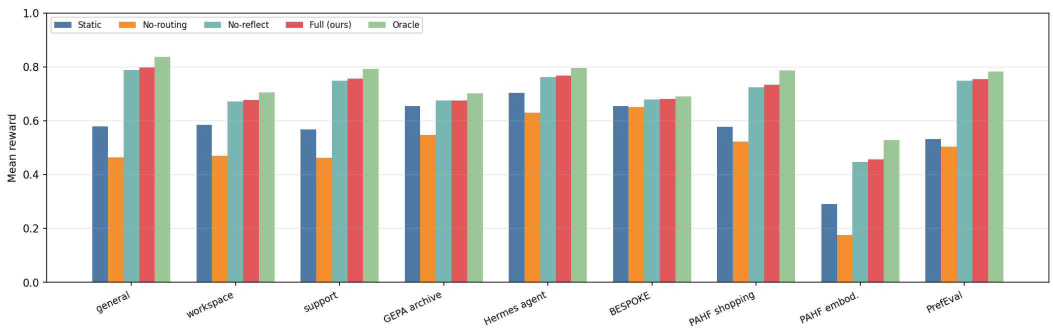

Figure 1: Mean reward by ablation variant across all nine benchmarks. Removing routing collapses reward; reflection and personalization features provide smaller incremental gains.

Per-round convergence. Figure 2 shows per-round reward traces for each benchmark.

In all cases, PersonalEvo converges toward the oracle within ∼50 rounds, while the static baseline remains flat.

The plotted observed-reward residual decreases over the horizon. These finite traces do not identify an asymptotic rate.

6.3 Comparison against existing routing methods

To position PersonalEvo against prior routing strategies, we compare five methods on all nine benchmarks (Table 3): (i) **Static**: deploy the single best-on-average candidate; (ii) **GEPA-Pareto**: maintain the Pareto-optimal subset of the archive and deploy uniformly from it (Agrawal et al., 2025); (iii) **Thompson**: Bayesian Thompson sampling over candidates (Labs, 2025); (iv) **RouteLLM-thresh**: route users to specialist or generalist candidates based on a variance threshold; (v) **PersonalEvo**: our full method.

The results reveal why the decomposition matters for method design: GEPA-Pareto maintains a diverse archive ($\bar{G}_T$ is high) but has no routing mechanism to exploit Δ_{route}, so diversity is wasted.

Thompson sampling routes but explores every arm equally regardless of user context, paying excessive $\bar{L}_T$.

RouteLLM routes by context but uses a fixed threshold rather than learning individual preferences.

PersonalEvo combines per-user routing with gated evolution and a small logged observed-reward residual. The experiment does not identify that residual with the theoretical $\bar{L}_T$.

6.4 Hyper-Ridge diagnostics and the remaining evaluation gap

The estimates in Table 2, ablations in Figure 1, per-round traces in Figure 2, and baseline comparison in Table 3 all evaluate the *base* PersonalEvo router. We additionally ran a five-seed diagnostic suite for Hyper-Ridge PersonalEvo. We keep it separate from the headline comparison because it covers

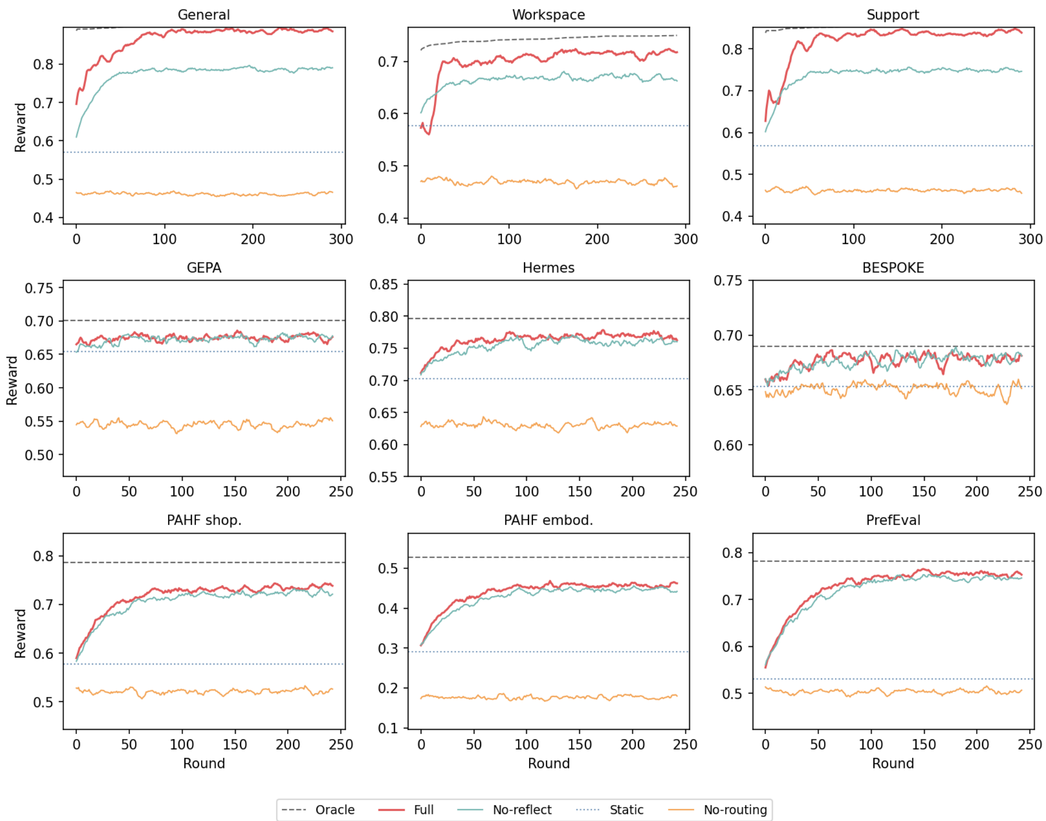

Figure 2: Per-round reward for all nine benchmarks (3×3 grid). Five variants shown: **Full** (red; concat features, gated reflection, LinUCB), **No-reflect** (teal; frozen archive, $\bar{G}_T$=0), **No-routing** (orange; uniform-random selection, $\Delta_{\text{route}}\to 0$), **Static** (gray dotted; best single candidate for all users), **Oracle** (black dashed; per-user optimum). Top row: synthetic; middle and bottom: framework and real-data benchmarks.

Benchmark	Static	GEPA-Pareto	Thompson	RouteLLM	PersonalEvo	Oracle
General	0.579	0.533	0.510	0.732	**0.797**	0.836
Workspace	0.585	0.539	0.515	0.650	**0.678**	0.705
Support	0.568	0.525	0.504	0.700	**0.757**	0.791
GEPA archive	0.671	0.616	0.569	0.675	**0.681**	0.701
Hermes agent	0.729	0.644	0.659	0.750	**0.771**	0.796
BESPOKE	0.681	0.660	0.654	0.678	**0.682**	0.690
PAHF shopping	0.625	0.532	0.560	0.691	**0.754**	0.812
PAHF embodied	0.290	0.244	0.222	0.407	**0.457**	0.528
PrefEval	0.531	0.520	0.514	0.687	**0.754**	0.782

Table 3: Comparison against existing routing methods on all nine benchmarks. Mean reward over 5 seeds. PersonalEvo (bold) achieves the highest reward among non-oracle methods in every row. GEPA-Pareto underperforms static because archive diversity without per-user routing is counterproductive. Thompson sampling explores too broadly. RouteLLM-threshold is the closest competitor but uses a fixed rule rather than learning per-user preferences.

seven rather than nine adapters and does not log the mean-reward exploration cost as its primary endpoint.

Known-geometry mechanism. We construct a diagnostic linear bandit in which $K = 30$ candidate parameters are drawn from a planted rank-three covariance, contexts are dense random unit vectors, and the horizon is $T = 400$. Table 4 compares isotropic ridge, Hyper-Ridge given the true covariance, and Hyper-Ridge with the covariance periodically re-estimated from the candidate models, all with the same ridge weight $\lambda = 0.5$. The true geometry cuts cumulative exploration cost by 46–59%, while the online estimate raises it by 7–19%. This diagnostic uses Gaussian parameters, unbounded linear rewards, and a fixed exploration multiplier, so it does not test the bounded-reward theorem or its rate. It only isolates the effect of supplying the correct geometry.

d	Isotropic	Known Ω	Estimated Ω	Known-Ω reduction
20	314.1	170.2	375.0	45.8%
50	447.6	213.1	504.9	52.4%
100	455.4	188.1	487.5	58.7%

Table 4: Mean cumulative exploration cost over five seeds on planted rank-three problems ($K = 30$, $T = 400$, $\lambda = 0.5$ for all three routers). Lower is better. This mechanism diagnostic does not satisfy the assumptions of Proposition 8.

Adapter diagnostic. With tuning seeds disjoint from the five reporting seeds, the periodic estimator improves mean reward over the isotropic router on six of seven adapters, with changes ranging from $+0.002$ on BESPOKE to $+0.048$ on Hermes, and loses 0.010 on GEPA. These numbers are not evidence of cross-candidate geometry transfer. The adapter harness represents candidate a by the one-hot vector e_a. Starting from $\widehat{\Omega}_1 = I$, each candidate estimate then has support only on its own coordinate, so the covariance update remains diagonal. Accordingly, the full-covariance and diagonal-ridge variants match exactly on every adapter and seed. This ablation exposes a representation mismatch that a larger seed count cannot repair.

Remaining protocol. The complete comparison must run base, known-covariance, calibration-split estimated-covariance, periodically estimated-covariance, and diagonal-ridge routers on the same nine benchmarks, using a feature map with shared behavioral directions. It must report all four mean-reward terms of equation 6, with $\widehat{L}_T^{\mu}$ as the primary endpoint and $\widehat{L}_T^{\mathrm{obs}}$ reported separately, as well as the fitted regret exponent, covariance error, and $d_{\mathrm{eff}}(\widehat{\Omega}; X)$. Until that comparison is complete, we do not claim an empirical advantage for the fully adaptive Hyper-Ridge router.

Implementation status. The repository implements both the base router and the periodic Hyper-Ridge estimator, together with an identity-reduction test showing that frozen $\widehat{\Omega} = I$ reproduces the base router's selections. The diagnostic results are stored in `results/hyper_ridge/suite.json`; the missing component is the feature-aware, nine-benchmark evaluation above, not the router implementation itself.

7 Conclusion

This paper revised the PersonalEvo framework around a single idea: a self-evolving agent should meta-learn the geometry of its own archive. The base framework answers *whether* personalized routing pays off through the exact four-term decomposition

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\big[\mu(U, a_t(U))\big] \;=\; V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{55}$$

and the associated dominance condition

$$\Delta_{\mathrm{route}}(A_1) + \bar{G}_T \;>\; \bar{L}_T, \tag{56}$$

both of which are inherited unchanged from the base paper (Theorem 2, Corollary 3). The revision leaves the first three terms untouched and attacks the only negative term. For the theoretical LinUCB router, with $T \geq 2$, $K_{\max}$, d, and the other problem parameters fixed and $\delta = 1/T$,

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)] \;=\; O\Big(\sqrt{T}\log T\Big)\,. \tag{57}$$

Hyper-Ridge models candidate parameters with a shared geometry,

$$\theta_a \sim (0, \Omega), \qquad \Omega \in \mathbb{R}^{d\times d}, \tag{58}$$

and uses that geometry in the generalized-ridge UCB score. For a design X, the effective dimension is

$$d_{\text{eff}}(\Omega; X) \;=\; \operatorname{tr}\Big[X^\top X\left(X^\top X + \lambda\Omega^{-1}\right)^{-1}\Big]. \tag{59}$$

For known Ω, Proposition 8 gives

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)} + \delta T. \tag{60}$$

For $T \ge 2$, fixed problem parameters, and $\delta = 1/T$, this bound is $O(\sqrt{T}\log T)$. A small effective dimension helps only when the full Hyper-Ridge budget is below the LinUCB budget. The calibration-split theorem proves a separate frozen-estimate bound. Its estimated-geometry radius remains inside the leading term, while its explicit width-inflation and calibration remainder is $o(\sqrt{T})$. It does not cover periodic re-estimation.

Two provenance facts bound what this paper claims. First, the nine-benchmark estimates, ablation hierarchy, per-round traces, and baseline comparison in Section 6 belong to the *base* LinUCB router. Their logged margins are

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} \;\in\; [+0.027,\ +0.295] \tag{61}$$

across all nine benchmarks, where $\bar{L}_T^{\text{obs}} = T^{-1}\sum_t (m_t - r_t)$. This equality is logged accounting, not validation of equation 55: clipping means that $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$ need not equal the pre-clipping $\mu(U, a_t)$ used by the structural terms. The selected-arm mean must be logged to evaluate $\bar{L}_T$ itself. The separate Hyper-Ridge suite is diagnostic: known geometry helps on planted low-rank problems, but periodic estimation does not recover that benefit, and one-hot adapters collapse the full covariance to a diagonal one. Second, the effective-dimension guarantee equation 60 is a theorem for the oracle-covariance router. Appendix L controls an estimated covariance for a calibration-split procedure under explicit diversity conditions, but this result does not cover the periodically updated estimator in Algorithm 1. Controlling that fully adaptive estimator remains the primary theoretical open problem.

The corresponding empirical open problem is the complete comparison specified in Section 6.4: base LinUCB versus known-covariance, calibration-split, periodically estimated, and diagonal-ridge routers on the same nine benchmarks, with the mean-reward exploration cost $\widehat{L}_T^{\mu}$ as the primary endpoint and $\widehat{L}_T^{\text{obs}}$ reported separately. The feature map must admit shared directions; otherwise the full model is diagonal by construction. The comparison should also report covariance error, the fitted regret exponent, and a per-benchmark estimate of $d_{\text{eff}}(\widehat{\Omega}; X)$ to locate both sides of the boundary: regimes where

$$d_{\text{eff}}(\Omega; X) \ll d \tag{62}$$

and the geometric potential can be smaller, and near-isotropic regimes where $d_{\text{eff}}(\Omega; X) \approx d$. In both cases, the full confidence budget determines the comparison. The current implementation supplies the periodic estimator and common harness; a calibration-split implementation and a shared feature representation remain to be added.

Beyond the two named gaps, the limitations of the base framework carry over. The reflective proposer in the simulated environments is oracle-assisted, so the measured evolution gain $\bar{G}_T$ does not estimate what a realistic LLM-driven proposer would deliver, and each user maintains a private archive, so discoveries are not shared across users. A shared archive could provide more data for estimating Ω, but it would change the predictability argument in Lemma 4; we leave that setting open. The decomposition equation 55 also applies to model routing, prompt selection, and tool-use policies whenever a system must choose from a candidate library.

A Appendix Proofs: Preliminaries and the Exact Decomposition

This appendix block fixes the probability model used by all proofs and proves the first main theorem: the exact four-term decomposition.

The proof is an accounting identity; it does not use a contextual-bandit regret theorem and does not assume that reflection is realistic.

A.1 PROBABILITY SPACE AND ONLINE OBJECTS

All random variables below are defined on one probability space:

$$(\Xi, \mathcal{F}, \mathbb{P}). \tag{63}$$

All expectations are taken over both the population draw and the online trajectory:

$$\mathbb{E}[Z] = \int_{\Xi} Z(\omega)\, d\mathbb{P}(\omega). \tag{64}$$

The evaluation user is drawn from a population distribution:

$$U \sim \Pi. \tag{65}$$

The user takes values in a measurable user space:

$$U : \Xi \to \mathcal{U}. \tag{66}$$

The candidate universe is denoted by

$$\mathcal{A}. \tag{67}$$

The common initial archive is

$$A_1 \subseteq \mathcal{A}. \tag{68}$$

The current archive before round t is

$$A_t(U) \subseteq \mathcal{A}. \tag{69}$$

The router's deployed candidate is

$$a_t(U) \in A_t(U). \tag{70}$$

The mean reward function is

$$\mu : \mathcal{U} \times \mathcal{A} \to [0, 1]. \tag{71}$$

The observed reward is

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{72}$$

For later plug-in bounds, the online filtration is

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_T. \tag{73}$$

Feature vectors used by the disjoint LinUCB implementation are denoted by

$$x_t(a) \in \mathbb{R}^d. \tag{74}$$

Only the boundedness and archive assumptions are used in the four-term decomposition proof; the noise and feature assumptions are recorded here so later proof chunks can refer back to the same notation.

Assumption 8 (A1: bounded mean reward)**.** *For every user and candidate, the mean reward satisfies*

$$0 \le \mu(u, a) \le 1. \tag{75}$$

Assumption 9 (A2: finite common initial archive)**.** *The initial archive is a deterministic, nonempty, finite subset of the candidate universe:*

$$1 \le |A_1| < \infty. \tag{76}$$

Assumption 10 (A3: finite append-only online archives)**.** *For every round, the current archive is nonempty and finite almost surely:*

$$1 \le |A_t(U)| < \infty. \tag{77}$$

The archive process is append-only:

$$A_t(U) \subseteq A_{t+1}(U). \tag{78}$$

Assumption 11 (A4: predictable archive)**.** *Before choosing the round-t candidate, the archive is measurable with respect to the past:*

$$\sigma(A_t(U)) \subseteq \mathcal{F}_{t-1}. \tag{79}$$

Assumption 12 (A5: centered conditionally sub-Gaussian noise)**.** *The reward noise is conditionally centered:*

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0. \tag{80}$$

For every real scalar λ, the conditional moment generating function obeys

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2 \sigma^2}{2}\right). \tag{81}$$

Assumption 13 (A6: bounded features)**.** *For every available candidate, the feature norm is bounded:*

$$\|x_t(a)\|_2 \le L. \tag{82}$$

A.2 Values and accounting terms

For any deterministic finite archive, the best static value is

$$V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}[\mu(U, a)]. \tag{83}$$

For the same archive, the perfect routed value is

$$V_{\text{route}}(A) = \mathbb{E}\left[\max_{a \in A} \mu(U, a)\right]. \tag{84}$$

The fixed-archive heterogeneity gain is

$$\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A). \tag{85}$$

The current-archive oracle value for the realized user is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{86}$$

The discovery gain at round t is

$$G_t(U) = m_t(U) - m_1(U). \tag{87}$$

The exploration cost at round t is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{88}$$

The horizon-averaged discovery gain is

$$\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[G_t(U)]. \tag{89}$$

The horizon-averaged exploration cost is

$$\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[L_t(U)]. \tag{90}$$

Proposition 10 (Integrability and signs of the accounting terms)**.** *Under Assumptions 8–10, the current-archive optimum satisfies*

$$0 \le m_t(U) \le 1. \tag{91}$$

The exploration cost satisfies

$$0 \le L_t(U) \le 1. \tag{92}$$

The discovery gain satisfies

$$0 \le G_t(U) \le 1. \tag{93}$$

Consequently all three terms are integrable:

$$\mathbb{E}[|m_t(U)|] + \mathbb{E}[|G_t(U)|] + \mathbb{E}[|L_t(U)|] \le 3. \tag{94}$$

Proof. Assumption 8 gives the pointwise reward bound:

$$0 \le \mu(U,a) \le 1 \qquad \text{for every } a \in \mathcal{A}. \tag{95}$$

Taking a maximum over a finite nonempty archive preserves the same interval:

$$0 \le \max_{a\in A_t(U)} \mu(U,a) \le 1. \tag{96}$$

Using equation 86, this is

$$0 \le m_t(U) \le 1. \tag{97}$$

Since the deployed candidate belongs to the current archive, we have

$$\mu(U, a_t(U)) \le \max_{a\in A_t(U)} \mu(U,a). \tag{98}$$

Therefore

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (99)\\ &\ge 0. && (100)\end{aligned}$$

The upper bound follows from the reward range:

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (101)\\ &\le 1 - 0 && (102)\\ &= 1. && (103)\end{aligned}$$

The append-only condition implies that the initial archive remains available:

$$A_1 \subseteq A_t(U). \tag{104}$$

Thus the current optimum dominates the initial optimum:

$$\begin{aligned} m_t(U) &= \max_{a\in A_t(U)} \mu(U,a) && (105)\\ &\ge \max_{a\in A_1} \mu(U,a) && (106)\\ &= m_1(U). && (107)\end{aligned}$$

Hence

$$G_t(U) = m_t(U) - m_1(U) \ge 0. \tag{108}$$

The upper bound is again inherited from the reward range:

$$\begin{aligned} G_t(U) &= m_t(U) - m_1(U) && (109)\\ &\leq 1 - 0 && (110)\\ &= 1. && (111) \end{aligned}$$

Combining the three interval bounds gives

$$\begin{aligned} \mathbb{E}[|m_t(U)|] + \mathbb{E}[|G_t(U)|] + \mathbb{E}[|L_t(U)|] &\leq 1 + 1 + 1 && (112)\\ &= 3. && (113) \end{aligned}$$

□

A.3 Supporting lemmas for the four-term decomposition

Lemma 11 (Static-versus-routed value). *Let A be any deterministic finite nonempty archive:*

$$1 \leq |A| < \infty. \tag{114}$$

Define its pointwise routed optimum by

$$h_A(U) = \max_{a \in A} \mu(U, a). \tag{115}$$

Define the static value by

$$V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}[\mu(U, a)]. \tag{116}$$

Define the routed value by

$$V_{\text{route}}(A) = \mathbb{E}[h_A(U)]. \tag{117}$$

Define the heterogeneity gap by

$$\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A). \tag{118}$$

Then

$$\Delta_{\text{route}}(A) \geq 0. \tag{119}$$

Moreover, equality holds if and only if there exists a candidate $a^\star \in A$ such that

$$\mu(U, a^\star) = \max_{a \in A} \mu(U, a) \quad \textit{almost surely}. \tag{120}$$

In particular, the inequality is strict whenever every candidate fails to be pointwise optimal on a set of positive probability:

$$\forall a \in A, \qquad \mathbb{P}\left(\mu(U, a) < \max_{b \in A} \mu(U, b)\right) > 0. \tag{121}$$

Proof. For every fixed candidate, the pointwise maximum dominates its reward:

$$\begin{aligned} h_A(U) &= \max_{b \in A} \mu(U, b) && (122)\\ &\geq \mu(U, a). && (123) \end{aligned}$$

Taking expectations preserves the inequality:

$$\mathbb{E}[h_A(U)] \geq \mathbb{E}[\mu(U, a)]. \tag{124}$$

Maximizing the right-hand side over the finite archive gives

$$\mathbb{E}[h_A(U)] \geq \max_{a \in A} \mathbb{E}[\mu(U, a)]. \tag{125}$$

Substituting the definitions yields the weak inequality:

$$\begin{aligned} \Delta_{\text{route}}(A) &= V_{\text{route}}(A) - V_{\text{static}}(A) && (126)\\ &= \mathbb{E}[h_A(U)] - \max_{a \in A} \mathbb{E}[\mu(U, a)] && (127)\\ &\geq 0. && (128) \end{aligned}$$

Because the archive is finite, the static maximizer set is nonempty:

$$\arg\max_{a \in A} \mathbb{E}[\mu(U,a)] \neq \varnothing. \tag{129}$$

Choose one static maximizer:

$$a^\star \in \arg\max_{a \in A} \mathbb{E}[\mu(U,a)]. \tag{130}$$

For this maximizer, the gap can be written as an expectation of a nonnegative random variable:

$$\Delta_{\text{route}}(A) \;=\; \mathbb{E}[h_A(U)] - \mathbb{E}[\mu(U,a^\star)] \tag{131}$$

$$\;=\; \mathbb{E}[h_A(U) - \mu(U,a^\star)]\,. \tag{132}$$

The integrand is nonnegative:

$$h_A(U) - \mu(U,a^\star) \;\geq\; 0. \tag{133}$$

A nonnegative random variable has zero expectation exactly when it is zero almost surely:

$$\Delta_{\text{route}}(A) = 0 \quad\iff\quad h_A(U) - \mu(U,a^\star) = 0 \text{ almost surely}. \tag{134}$$

Using the definition of h_A, this is precisely

$$\mu(U,a^\star) = \max_{a \in A} \mu(U,a) \quad \text{almost surely}. \tag{135}$$

Conversely, if some candidate is pointwise optimal almost surely, then

$$V_{\text{route}}(A) \;=\; \mathbb{E}\left[\max_{a \in A} \mu(U,a)\right] \tag{136}$$

$$\;=\; \mathbb{E}[\mu(U,a^\star)] \tag{137}$$

$$\;\leq\; \max_{a \in A} \mathbb{E}[\mu(U,a)] \tag{138}$$

$$\;=\; V_{\text{static}}(A). \tag{139}$$

Together with equation 119, this forces equality:

$$V_{\text{route}}(A) = V_{\text{static}}(A). \tag{140}$$

Finally, if equation 121 holds, then every possible static maximizer has strictly positive probability of being suboptimal:

$$\mathbb{P}(h_A(U) - \mu(U,a^\star) > 0) > 0. \tag{141}$$

Since the integrand is nonnegative and positive with positive probability, its expectation is positive:

$$\Delta_{\text{route}}(A) = \mathbb{E}[h_A(U) - \mu(U,a^\star)] > 0. \tag{142}$$

□

Lemma 12 (Pointwise routed-value accounting)**.** *For every round t, define the current optimum by*

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a). \tag{143}$$

Define the discovery term by

$$G_t(U) = m_t(U) - m_1(U). \tag{144}$$

Define the exploration-cost term by

$$L_t(U) = m_t(U) - \mu(U,a_t(U)). \tag{145}$$

Then the deployed mean reward satisfies the pointwise identity

$$\mu(U,a_t(U)) = m_1(U) + G_t(U) - L_t(U). \tag{146}$$

Proof. Start from the definition of the exploration cost:

$$L_t(U) \;=\; m_t(U) - \mu(U,a_t(U)). \tag{147}$$

Rearrange the equality:

$$\mu(U,a_t(U)) \;=\; m_t(U) - L_t(U). \tag{148}$$

Use the definition of discovery gain:

$$m_t(U) \;=\; m_1(U) + G_t(U). \tag{149}$$

Substitute equation 149 into equation 148:

$$\mu(U,a_t(U)) \;=\; m_1(U) + G_t(U) - L_t(U). \tag{150}$$

□

A.4 FIRST MAIN THEOREM: FOUR-TERM DECOMPOSITION

Theorem 13 (Exact decomposition over the common initial archive)**.** *This theorem restates the four-term decomposition used as Theorem 2 in Section 5.1. The user is drawn as*

$$U \sim \Pi. \tag{151}$$

The initial archive is finite and nonempty:

$$1 \le |A_1| < \infty. \tag{152}$$

At round t, the deployed candidate belongs to the current archive:

$$a_t(U) \in A_t(U). \tag{153}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \tag{154}$$

The discovery term is

$$G_t(U) = m_t(U) - m_1(U). \tag{155}$$

The exploration-cost term is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{156}$$

The static initial-archive value is

$$V_{\text{static}}(A_1) = \max_{a \in A_1} \mathbb{E}[\mu(U, a)]. \tag{157}$$

The routed initial-archive value is

$$V_{\text{route}}(A_1) = \mathbb{E}\left[\max_{a \in A_1} \mu(U, a)\right]. \tag{158}$$

The initial-archive heterogeneity gain is

$$\Delta_{\text{route}}(A_1) = V_{\text{route}}(A_1) - V_{\text{static}}(A_1). \tag{159}$$

Then each round satisfies

$$\mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{160}$$

Define the horizon-averaged routed value by

$$R_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{161}$$

Define the horizon-averaged discovery gain by

$$\bar{G}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{162}$$

Define the horizon-averaged exploration cost by

$$\bar{L}_T = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{163}$$

Then the horizon-averaged decomposition is

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{164}$$

Proof. By Proposition 10, all quantities in the following expectations are integrable:

$$\mathbb{E}[|m_t(U)|] + \mathbb{E}[|G_t(U)|] + \mathbb{E}[|L_t(U)|] < \infty. \tag{165}$$

Apply the pointwise accounting identity from Lemma 12:

$$\mu(U, a_t(U)) \;=\; m_1(U) + G_t(U) - L_t(U). \tag{166}$$

Taking expectations gives

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; \mathbb{E}[m_1(U)] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{167}$$

The first-archive optimum expands as

$$\begin{aligned} \mathbb{E}[m_1(U)] \;&=\; \mathbb{E}\left[\max_{a\in A_1} \mu(U, a)\right] && (168)\\ &=\; V_{\text{route}}(A_1). && (169) \end{aligned}$$

By the definition of the heterogeneity gap,

$$V_{\text{route}}(A_1) \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1). \tag{170}$$

Substituting equation 170 into equation 167 yields the per-round identity:

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{171}$$

Average the previous display over the horizon:

$$\begin{aligned} \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] \;&=\; \frac{1}{T}\sum_{t=1}^{T} \left(V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]\right) && (172)\\ &=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)] - \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. && (173) \end{aligned}$$

Using the definitions of R_T, $\bar{G}_T$, and $\bar{L}_T$, this becomes

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{174}$$

□

The theorem is an identity in mean rewards.

When Assumption 12 holds, the same expectation also governs observed rewards:

$$\begin{aligned} \mathbb{E}[r_t] \;&=\; \mathbb{E}[\mu(U, a_t(U)) + \xi_t] && (175)\\ &=\; \mathbb{E}[\mu(U, a_t(U))] + \mathbb{E}[\xi_t] && (176)\\ &=\; \mathbb{E}[\mu(U, a_t(U))]. && (177) \end{aligned}$$

This link does not apply to the saved simulator's clipped observations in equation 54 when μ denotes the pre-clipping mean. The following `general` row instead checks the separate observed-reward accounting identity. It uses

$$V_{\text{static}}(A_1) = 0.570. \tag{178}$$

The measured heterogeneity term is

$$\Delta_{\text{route}}(A_1) = 0.280. \tag{179}$$

The measured discovery term is

$$\bar{G}_T = 0.051. \tag{180}$$

The measured observed-reward residual is

$$\bar{L}_T^{\text{obs}} = 0.036. \tag{181}$$

The logged accounting identity gives

$$V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} = 0.570 + 0.280 + 0.051 - 0.036 \quad (182)$$
$$= 0.865, \quad (183)$$

which matches the reported observed routed value 0.864 up to table rounding. Because $\bar{L}_T^{\text{obs}}$ is defined from the same routed rewards, this is an algebraic consistency check rather than empirical validation of Theorem 13.

B APPENDIX PROOFS: SECOND MAIN THEOREM AND INTERMEDIATE LEMMAS

This section proves the second main theorem, the conditional plug-in statement for the exploration cost.

The proof does not assert a new regret rate; it shows how any valid bound on the cumulative exploration cost plugs into the four-term decomposition of Theorem 13.

Assumption 14 (External expected exploration-cost bound)**.** *The horizon is*

$$T \in \mathbb{N}. \quad (184)$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a). \quad (185)$$

The exploration cost is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \quad (186)$$

There exists a deterministic sequence satisfying

$$B_T \geq 0. \quad (187)$$

The expected cumulative exploration cost is bounded as

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_T. \quad (188)$$

The next lemma records the conditioning fact that makes growing archives compatible with standard martingale arguments, provided children are appended only after the current reward is observed.

Lemma 14 (Predictable archive conditioning)**.** *Define the pre-action sigma-field by*

$$\mathcal{H}_{t-1} = \sigma(U,\, A_1,\, \{a_s(U), r_s, A_{s+1}(U) : 1 \leq s < t\})\,. \quad (189)$$

Assume the archive is predictable:

$$A_t(U) \in \mathcal{H}_{t-1}. \quad (190)$$

Assume at most $K_{\max}$ candidates appear through round T, and index them by deterministic birth-order slots $k \in \{1, \ldots, K_{\max}\}$. Slot k is filled at a stopping time τ_k; an unfilled slot has $\tau_k = \infty$. On $\{\tau_k \leq T\}$, its identity $a^{(k)}$ and feature rule are $\mathcal{H}_{\tau_k - 1}$-measurable. Define the padded feature and selection indicator by

$$\widetilde{x}_{t,k} = \mathbf{1}\{\tau_k \leq t\}\psi(U, a^{(k)}), \quad (191)$$
$$I_{t,k} = \mathbf{1}\{\tau_k \leq t,\ a_t(U) = a^{(k)}\}. \quad (192)$$

Assume $\widetilde{x}_{t,k}$ and $I_{t,k}$ are $\mathcal{H}_{t-1}$-measurable. This is the predictable birth-order construction: before a slot is filled, both processes are zero; afterward, its identity is already part of the pre-action history. The observed reward noise satisfies

$$\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0. \quad (193)$$

It also satisfies the conditional sub-Gaussian bound

$$\mathbb{E}[\exp(s\xi_t) \mid \mathcal{H}_{t-1}] \le \exp\left(\frac{s^2\sigma^2}{2}\right) \quad \textit{for all } s \in \mathbb{R}. \tag{194}$$

For each deterministic slot k, define the vector increment

$$Z_{t,k} = I_{t,k}\widetilde{x}_{t,k}\xi_t. \tag{195}$$

Define the cumulative noise process

$$M_{n,k} = \sum_{t=1}^{n} Z_{t,k}. \tag{196}$$

Then $M_{n,k}$ is a vector-valued martingale with respect to the post-round filtration for every deterministic slot k.

Proof. Fix a deterministic slot k. Predictability gives

$$I_{t,k}\widetilde{x}_{t,k} \in \mathcal{H}_{t-1}. \tag{197}$$

Taking conditional expectation yields

$$\begin{aligned} \mathbb{E}[Z_{t,k} \mid \mathcal{H}_{t-1}] &= \mathbb{E}[I_{t,k}\widetilde{x}_{t,k}\xi_t \mid \mathcal{H}_{t-1}] && (198) \\ &= I_{t,k}\widetilde{x}_{t,k}\, \mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] && (199) \\ &= 0. && (200) \end{aligned}$$

For any direction v, the scalar increment satisfies

$$\begin{aligned} \mathbb{E}\big[\exp\big(s\, v^\top Z_{t,k}\big) \mid \mathcal{H}_{t-1}\big] &= \mathbb{E}\big[\exp\big(s\, I_{t,k}\, v^\top \widetilde{x}_{t,k}\, \xi_t\big) \mid \mathcal{H}_{t-1}\big] && (201) \\ &\le \exp\left(\frac{s^2\sigma^2}{2} I_{t,k}(v^\top \widetilde{x}_{t,k})^2\right). && (202) \end{aligned}$$

Thus the increments are conditionally mean-zero and conditionally sub-Gaussian in every predictable direction, so the partial sums form a vector-valued martingale. □

The following elementary conversion is useful because many router analyses produce high-probability regret bounds, whereas Theorem 13 is stated in expectation.

Lemma 15 (From high-probability tax to expected tax)**.** *Assume the mean reward is bounded:*

$$0 \le \mu(U, a) \le 1. \tag{203}$$

Define the cumulative exploration cost by

$$S_T^L = \sum_{t=1}^{T} L_t(U). \tag{204}$$

Suppose there is a number

$$\widetilde{B}_T \ge 0 \tag{205}$$

and a failure probability

$$\delta \in (0, 1) \tag{206}$$

such that

$$\mathbb{P}\Big(S_T^L \le \widetilde{B}_T\Big) \ge 1 - \delta. \tag{207}$$

Then

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le \widetilde{B}_T + T\delta. \tag{208}$$

Proof. Since $a_t(U) \in A_t(U)$, the current optimum dominates the deployed mean:

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (209) \\ &\geq 0. && (210) \end{aligned}$$

The bounded-reward condition gives

$$\begin{aligned} L_t(U) &= m_t(U) - \mu(U, a_t(U)) && (211) \\ &\leq 1. && (212) \end{aligned}$$

Therefore

$$0 \leq S_T^L \leq T. \tag{213}$$

Let the good event be

$$\mathcal{E}_T^L = \{S_T^L \leq \widetilde{B}_T\}. \tag{214}$$

Split the expectation over this event:

$$\begin{aligned} \mathbb{E}[S_T^L] &= \mathbb{E}[S_T^L \mathbf{1}\{\mathcal{E}_T^L\}] + \mathbb{E}[S_T^L \mathbf{1}\{(\mathcal{E}_T^L)^c\}] && (215) \\ &\leq \widetilde{B}_T + T\,\mathbb{P}((\mathcal{E}_T^L)^c) && (216) \\ &\leq \widetilde{B}_T + T\delta. && (217) \end{aligned}$$

Using the definition of S_T^L, this is exactly

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq \widetilde{B}_T + T\delta. \tag{218}$$

□

We next give one sufficient route to Assumption 14.

This proposition is not a new contribution of the paper; it is a standard disjoint-linear UCB certificate written only to clarify what must be true for the plug-in theorem to inherit a familiar rate.

Proposition 16 (Disjoint-linear certificate for the external tax bound)**.** *The feature dimension is*

$$d \in \mathbb{N}. \tag{219}$$

The realized archive union is

$$\mathcal{A}_T(U) = \bigcup_{t=1}^{T} A_t(U). \tag{220}$$

The realized archive size is bounded by

$$|\mathcal{A}_T(U)| \leq K_{\max}. \tag{221}$$

Index the candidates by their birth order in deterministic slots $k \in \{1, \ldots, K_{\max}\}$. Let $(\mathcal{G}_t)$ be an analysis filtration with $\mathcal{H}_t \subseteq \mathcal{G}_t$. The birth time τ_k is an $\mathcal{H}_t$-stopping time. The candidate identity and feature rule are $\mathcal{H}_{\tau_k-1}$-measurable, while its latent parameter is fixed at birth and $\mathcal{G}_{\tau_k-1}$-measurable. Before birth, the slot's selection indicator and feature are zero. The padded process is $\mathcal{G}_t$-predictable. The router itself remains $\mathcal{H}_t$-adapted and does not observe the latent parameter. The feature vector is

$$x_t(a) = \psi(U, a) \in \mathbb{R}^d. \tag{222}$$

The feature norm is bounded by

$$\|x_t(a)\|_2 \leq L_x. \tag{223}$$

The regularization parameter satisfies

$$\lambda > 0, \qquad \lambda \geq L_x^2. \tag{224}$$

Each realized arm has a parameter

$$\theta_a^\star \in \mathbb{R}^d. \tag{225}$$

The parameter norm is bounded by

$$\|\theta_a^\star\|_2 \le S. \tag{226}$$

The mean model is disjoint-linear:

$$\mu(U, a) = x_t(a)^\top \theta_a^\star. \tag{227}$$

The observed reward is

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{228}$$

The noise obeys equation 193 and equation 194, and the same conditional mean-zero and sub-Gaussian bounds hold with $\mathcal{G}_{t-1}$ in place of $\mathcal{H}_{t-1}$. The ridge matrix for arm a before round t is

$$V_{t,a} = \lambda I_d + \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\}\, x_s(a) x_s(a)^\top. \tag{229}$$

The ridge estimator is

$$\widehat{\theta}_{t,a} = V_{t,a}^{-1} \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\}\, x_s(a) r_s. \tag{230}$$

For confidence level $\delta \in (0, 1)$, define

$$\beta_T(\delta) = \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)} + \sqrt{\lambda}\, S. \tag{231}$$

The router selects by the disjoint UCB rule

$$a_t(U) \in \operatorname*{arg\,max}_{a \in A_t(U)} \left\{ x_t(a)^\top \widehat{\theta}_{t,a} + \beta_T(\delta) \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)} \right\}. \tag{232}$$

Then, with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} L_t(U) \le B_T^{\mathrm{lin}}(\delta), \tag{233}$$

where

$$B_T^{\mathrm{lin}}(\delta) = 2\beta_T(\delta) \sqrt{2 T K_{\max} d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)}. \tag{234}$$

Consequently,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T^{\mathrm{lin}}(\delta) + T\delta. \tag{235}$$

Proof. For the proof, abbreviate the action by

$$a_t = a_t(U). \tag{236}$$

For each deterministic birth-order slot k, define the accumulated noise vector

$$M_{t,k} = \sum_{s=1}^{t-1} \mathbf{1}\{a_s = a^{(k)}\}\, x_s(a^{(k)}) \xi_s, \tag{237}$$

where the summand is zero before τ_k. The strengthened noise condition and birth-time measurability make $M_{t,k}$ a vector-valued martingale in the analysis filtration after conditioning on the slot's birth history. The self-normalized martingale inequality gives

$$\mathbb{P}\left(\exists t \le T : \|M_{t,k}\|_{V_{t,a^{(k)}}^{-1}} > \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + \log\left(\frac{\det V_{t,a^{(k)}}}{\det(\lambda I_d)}\right)} \right) \le \frac{\delta}{K_{\max}}. \tag{238}$$

The determinant ratio is bounded by the trace:

$$\log\left(\frac{\det V_{t,a^{(k)}}}{\det(\lambda I_d)}\right) \;\le\; d\log\left(1+\frac{1}{\lambda d}\sum_{s=1}^{t-1}\mathbf{1}\{a_s=a^{(k)}\}\|x_s(a^{(k)})\|_2^2\right) \tag{239}$$

$$\le\; d\log\left(1+\frac{TL_x^2}{\lambda d}\right). \tag{240}$$

Union over the deterministic slots $k \le K_{\max}$ gives the confidence event

$$\mathcal{E}_T=\left\{\forall t\;\le\;T,\,\forall a\in A_t(U):\left|x_t(a)^\top(\widehat{\theta}_{t,a}-\theta_a^\star)\right|\;\le\;\beta_T(\delta)\sqrt{x_t(a)^\top V_{t,a}^{-1}x_t(a)}\right\}. \tag{241}$$

The previous displays imply

$$\mathbb{P}(\mathcal{E}_T)\;\ge\;1-\delta. \tag{242}$$

On $\mathcal{E}_T$, define the estimated mean by

$$\widehat{\mu}_t(a)=x_t(a)^\top\widehat{\theta}_{t,a}. \tag{243}$$

Define the uncertainty width by

$$z_t(a)=\sqrt{x_t(a)^\top V_{t,a}^{-1}x_t(a)}. \tag{244}$$

Let the current archive oracle be

$$a_t^\star\in\arg\max_{a\in A_t(U)}\mu(U,a). \tag{245}$$

Optimism and the UCB choice imply

$$\mu(U,a_t^\star)\;\le\;\widehat{\mu}_t(a_t^\star)+\beta_T(\delta)z_t(a_t^\star) \tag{246}$$

$$\le\;\widehat{\mu}_t(a_t)+\beta_T(\delta)z_t(a_t) \tag{247}$$

$$\le\;\mu(U,a_t)+2\beta_T(\delta)z_t(a_t). \tag{248}$$

Therefore the exploration cost obeys

$$L_t(U)\;=\;\mu(U,a_t^\star)-\mu(U,a_t) \tag{249}$$

$$\le\;2\beta_T(\delta)z_t(a_t). \tag{250}$$

The regularization condition gives

$$z_t(a_t)^2\;=\;x_t(a_t)^\top V_{t,a_t}^{-1}x_t(a_t) \tag{251}$$

$$\le\;\frac{\|x_t(a_t)\|_2^2}{\lambda} \tag{252}$$

$$\le\;1. \tag{253}$$

Using Cauchy–Schwarz,

$$\sum_{t=1}^{T}z_t(a_t)\;\le\;\sqrt{T\sum_{t=1}^{T}z_t(a_t)^2}. \tag{254}$$

For each arm, the determinant recursion gives

$$\sum_{t=1}^{T}\mathbf{1}\{a_t=a\}z_t(a)^2\;\le\;2\log\left(\frac{\det V_{T+1,a}}{\det(\lambda I_d)}\right) \tag{255}$$

$$\le\;2d\log\left(1+\frac{TL_x^2}{\lambda d}\right). \tag{256}$$

Summing over at most $K_{\max}$ arms yields

$$\sum_{t=1}^{T}z_t(a_t)^2\;\le\;2K_{\max}d\log\left(1+\frac{TL_x^2}{\lambda d}\right). \tag{257}$$

Combining equation 249, equation 254, and equation 257 gives the high-probability tax bound:

$$\sum_{t=1}^{T} L_t(U) \;\le\; 2\beta_T(\delta)\sum_{t=1}^{T} z_t(a_t) \tag{258}$$

$$\le\; 2\beta_T(\delta)\sqrt{2TK_{\max}d\log\left(1+\frac{TL_x^2}{\lambda d}\right)} \tag{259}$$

$$=\; B_T^{\mathrm{lin}}(\delta). \tag{260}$$

Finally, Lemma 15 converts the high-probability statement into the expected bound:

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; B_T^{\mathrm{lin}}(\delta) + T\delta. \tag{261}$$

□

Theorem 17 (Conditional plug-in regret for PersonalEvo)**.** *The horizon-averaged routed value is*

$$R_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{262}$$

The horizon-averaged discovery gain is

$$\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{263}$$

The horizon-averaged exploration cost is

$$\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{264}$$

The static initial-archive value is

$$V_{\mathrm{static}}(A_1) = \max_{a\in A_1} \mathbb{E}[\mu(U,a)]. \tag{265}$$

The initial-archive heterogeneity gain is

$$\Delta_{\mathrm{route}}(A_1) = V_{\mathrm{route}}(A_1) - V_{\mathrm{static}}(A_1). \tag{266}$$

Assume the external expected exploration-cost bound

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; B_T. \tag{267}$$

Then the routed value satisfies

$$R_T \;\ge\; V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{268}$$

Proof. The four-term decomposition from Theorem 13 gives

$$R_T \;=\; V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{269}$$

By the definition of $\bar{L}_T$,

$$\bar{L}_T \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{270}$$

Using the external tax assumption,

$$\bar{L}_T \;\le\; \frac{B_T}{T}. \tag{271}$$

Substituting equation 271 into equation 269 yields

$$R_T \;\ge\; V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{272}$$

□

Corollary 18 (Plug-in dominance condition)**.** *Under Theorem 17, personalized routing beats the best static initial candidate whenever*

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \frac{B_T}{T}. \tag{273}$$

In that case,

$$R_T > V_{\text{static}}(A_1). \tag{274}$$

On a fixed archive with no discovery gain, the condition reduces to

$$\Delta_{\text{route}}(A_1) > \frac{B_T}{T}. \tag{275}$$

Proof. Subtract the static value from the plug-in bound:

$$R_T - V_{\text{static}}(A_1) \;\geq\; \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{276}$$

If equation 273 holds, then the right-hand side is positive:

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T} > 0. \tag{277}$$

Therefore

$$R_T > V_{\text{static}}(A_1). \tag{278}$$

If the archive is fixed, then

$$\bar{G}_T = 0, \tag{279}$$

so equation 273 becomes equation 275. □

Corollary 19 (Disjoint-LinUCB plug-in instance)**.** *If the conditions of Proposition 16 hold, then*

$$R_T \;\geq\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{lin}}(\delta) + T\delta}{T}. \tag{280}$$

With the choice

$$\delta = \frac{1}{T}, \tag{281}$$

the tax term has the form

$$\frac{B_T^{\text{lin}}(1/T) + 1}{T} = O\left(\frac{\log T}{\sqrt{T}}\right) \tag{282}$$

when $K_{\max}$, d, L_x, λ, S, and σ are treated as fixed problem parameters.

Proof. Proposition 16 gives

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\leq\; B_T^{\text{lin}}(\delta) + T\delta. \tag{283}$$

Apply Theorem 17 with

$$B_T = B_T^{\text{lin}}(\delta) + T\delta. \tag{284}$$

This yields

$$R_T \;\geq\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{lin}}(\delta) + T\delta}{T}. \tag{285}$$

For $T \geq 2$ and $\delta = 1/T$, the definition of B_T^{lin} gives

$$\begin{aligned} \frac{B_T^{\text{lin}}(1/T) + 1}{T} &= \frac{2\beta_T(1/T)\sqrt{2TK_{\max} d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + 1}{T} && (286) \\ &= O\left(\frac{\log T}{\sqrt{T}}\right), && (287) \end{aligned}$$

under fixed problem parameters. □

As a concrete scale check, suppose a valid external certificate on the `workspace` environment had the conservative value

$$\frac{B_T}{T} = 0.050. \tag{288}$$

Using the corrected decomposition terms from Table 2,

$$V_{\text{static}}(A_1) = 0.577. \tag{289}$$

The heterogeneity term is

$$\Delta_{\text{route}}(A_1) = 0.124. \tag{290}$$

The discovery term is

$$\bar{G}_T = 0.041. \tag{291}$$

The plug-in lower bound would be

$$\begin{aligned} V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T} \;&=\; 0.577 + 0.124 + 0.041 - 0.050 && (292)\\ &=\; 0.692. && (293)\end{aligned}$$

This is above the static comparator:

$$0.692 - 0.577 \;=\; 0.115 > 0. \tag{294}$$

The empirical exploration-cost estimate in Section 6 is a diagnostic estimate rather than a certificate. Substituting that estimate for scale gives

$$0.577 + 0.124 + 0.041 - 0.043 \;=\; 0.699, \tag{295}$$

which matches the reported `workspace` routed value up to rounding.

The proof above explains what additional router assumptions would be required to turn such a diagnostic tax curve into a certified B_T.

C Proof of the Third Main Theorem and Complexity Consequences

This proof block concerns the narrowed third theorem from Section 5.1.

The theorem is not a contraction theorem for language-model reflection; it is a conditional stopping-time statement for the first useful child under an explicit repair-probability assumption.

Assumption 15 (Conditional repair probability for an active cluster)**.** *The cluster set is*

$$\mathcal{G} = \{1, \dots, K_{\text{cl}}\}. \tag{296}$$

The cluster of a user is

$$g(U) \in \mathcal{G}. \tag{297}$$

Every cluster considered below has positive population mass:

$$\pi_g := \mathbb{P}(g(U) = g) > 0. \tag{298}$$

For a fixed cluster g, the cluster-conditioned user is

$$U_g \sim \Pi(\,\cdot \mid g(U) = g\,). \tag{299}$$

The improvement threshold is

$$\epsilon > 0. \tag{300}$$

The probability lower bound for an active repair attempt is

$$q_\epsilon \in (0, 1]. \tag{301}$$

The time of the j-th active repair attempt for cluster g is

$$\tau_{g,j}. \tag{302}$$

If the j-th active attempt never occurs, set $\tau_{g,j} = \infty$. The attempt-specific archive and history below are defined on $\{\tau_{g,j} < \infty\}$. Any theorem invoking the first m attempts assumes those attempts exist almost surely. The information available immediately before sampling the child at that attempt is

$$\mathcal{H}_{g,j}. \tag{303}$$

The archive before the proposed child is inserted is

$$A^-_{g,j} = A_{\tau_{g,j}}(U_g). \tag{304}$$

The archive after the proposed child is inserted is

$$A^+_{g,j} = A_{\tau_{g,j}+1}(U_g). \tag{305}$$

The indicator that attempt j produces an ϵ-useful child is

$$I^{(\epsilon)}_{g,j} = \begin{cases} \mathbf{1}\Big\{\max_{a \in A^+_{g,j}} \mu(U_g, a) - \max_{a \in A^-_{g,j}} \mu(U_g, a) \;\ge\; \epsilon\Big\}, & \tau_{g,j} < \infty, \\ 0, & \tau_{g,j} = \infty. \end{cases} \tag{306}$$

The event that the first j active attempts have all failed is

$$F^{(\epsilon)}_{g,j} = \bigcap_{\ell=1}^{j} \{I^{(\epsilon)}_{g,\ell} = 0\}. \tag{307}$$

The pre-attempt history contains the previous-failure event:

$$F^{(\epsilon)}_{g,j-1} \in \mathcal{H}_{g,j}. \tag{308}$$

For every active attempt before the first success, the repair mechanism satisfies

$$\mathbb{P}\Big(I^{(\epsilon)}_{g,j} = 1 \mid \mathcal{H}_{g,j}\Big) \;\ge\; q_\epsilon \qquad \text{on } F^{(\epsilon)}_{g,j-1}. \tag{309}$$

The word "active" means that the attempt is made only while the current cluster gap exceeds the target threshold.

The proof uses only equation 309; it does not use a contraction factor.

Theorem 20 (First-useful-specialist time)**.** *Fix a cluster*

$$g \in \mathcal{G}. \tag{310}$$

Fix an improvement threshold

$$\epsilon > 0. \tag{311}$$

Fix a number of active attempts

$$m \in \mathbb{N}. \tag{312}$$

Define the event that at least one of the first m active attempts succeeds by

$$S^{(\epsilon)}_{g,m} = \left\{ \sum_{j=1}^{m} I^{(\epsilon)}_{g,j} \;\ge\; 1 \right\}. \tag{313}$$

If Assumption 15 holds for the first m active attempts, then

$$\mathbb{P}\Big(S^{(\epsilon)}_{g,m}\Big) \;\ge\; 1 - (1 - q_\epsilon)^m. \tag{314}$$

Equivalently, the probability that all m active attempts fail is bounded by

$$\mathbb{P}\Big(F^{(\epsilon)}_{g,m}\Big) \;\le\; (1 - q_\epsilon)^m. \tag{315}$$

Proof. The zero-failure event is

$$F_{g,0}^{(\epsilon)} = \Xi. \tag{316}$$

For each active attempt j, the next failure event satisfies

$$\mathbb{P}\left(F_{g,j}^{(\epsilon)}\right) = \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\mathbf{1}\{I_{g,j}^{(\epsilon)} = 0\}\right] \tag{317}$$

$$= \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\mathbb{P}\left(I_{g,j}^{(\epsilon)} = 0 \mid \mathcal{H}_{g,j}\right)\right]. \tag{318}$$

The repair condition gives

$$\mathbb{P}\left(I_{g,j}^{(\epsilon)} = 0 \mid \mathcal{H}_{g,j}\right) = 1 - \mathbb{P}\left(I_{g,j}^{(\epsilon)} = 1 \mid \mathcal{H}_{g,j}\right) \tag{319}$$

$$\leq 1 - q_\epsilon \qquad \text{on } F_{g,j-1}^{(\epsilon)}. \tag{320}$$

Substituting equation 319 into equation 317 gives

$$\mathbb{P}\left(F_{g,j}^{(\epsilon)}\right) \leq (1 - q_\epsilon)\mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\right] \tag{321}$$

$$= (1 - q_\epsilon)\mathbb{P}\left(F_{g,j-1}^{(\epsilon)}\right). \tag{322}$$

Iterating from $j = 1$ to $j = m$ yields

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1 - q_\epsilon)^m \mathbb{P}\left(F_{g,0}^{(\epsilon)}\right) \tag{323}$$

$$= (1 - q_\epsilon)^m. \tag{324}$$

The success event is the complement of all attempts failing:

$$S_{g,m}^{(\epsilon)} = \Xi \setminus F_{g,m}^{(\epsilon)}. \tag{325}$$

Therefore

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) = 1 - \mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \tag{326}$$

$$\geq 1 - (1 - q_\epsilon)^m. \tag{327}$$

□

The theorem is intentionally conditional. The saved simulator uses privileged peak-category information, but its child qualities remain random and the runs do not estimate q_ϵ for a fixed ϵ. Thus Theorem 20 states the quantity that a reflection study must estimate; the saved runs do not test its assumption.

Corollary 21 (Attempt complexity for one useful specialist)**.** *Fix a failure probability*

$$\delta \in (0, 1). \tag{328}$$

The exact number of active attempts sufficient for success probability at least $1 - \delta$ is

$$m_\delta^{\text{exact}} = \begin{cases} 1, & q_\epsilon = 1, \\ \left\lceil \dfrac{\log \delta}{\log(1 - q_\epsilon)} \right\rceil, & 0 < q_\epsilon < 1. \end{cases} \tag{329}$$

A simpler sufficient upper certificate is

$$m_\delta = \left\lceil \frac{1}{q_\epsilon} \log\left(\frac{1}{\delta}\right) \right\rceil. \tag{330}$$

For either choice satisfying $(1 - q_\epsilon)^m \leq \delta$, the success probability obeys

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1 - \delta. \tag{331}$$

Proof. Theorem 20 gives

$$\mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1-(1-q_\epsilon)^m. \tag{332}$$

If $q_\epsilon = 1$, Theorem 20 gives success probability one after one active attempt. If $0 < q_\epsilon < 1$, the exact choice in equation 329 satisfies

$$m_\delta^{\text{exact}} \log(1-q_\epsilon) \leq \log \delta \tag{333}$$

$$(1-q_\epsilon)^{m_\delta^{\text{exact}}} \leq \delta. \tag{334}$$

The exponential relaxation uses

$$1-q_\epsilon \leq \exp(-q_\epsilon). \tag{335}$$

Thus the simpler choice in equation 330 gives

$$(1-q_\epsilon)^{m_\delta} \leq \exp(-q_\epsilon m_\delta) \tag{336}$$

$$\leq \exp\left(-\log\left(\frac{1}{\delta}\right)\right) \tag{337}$$

$$= \delta. \tag{338}$$

Substitution into equation 332 proves equation 331. □

As a concrete scale check, if a non-oracle proposer had

$$q_\epsilon = 0.20 \tag{339}$$

and the desired failure probability were

$$\delta = 0.05, \tag{340}$$

then the simple attempt certificate would be

$$m_\delta = \left\lceil \frac{1}{0.20} \log(20) \right\rceil \tag{341}$$

$$= 15. \tag{342}$$

The resulting lower bound is

$$1-(1-0.20)^{15} = 1-0.8^{15} \tag{343}$$

$$\approx 0.965. \tag{344}$$

Corollary 22 (Cluster coverage by active attempts). *The active cluster set is*

$$\mathcal{G}_{\text{act}} \subseteq \mathcal{G}. \tag{345}$$

The number of active clusters is

$$K_{\text{act}} = |\mathcal{G}_{\text{act}}|. \tag{346}$$

Cluster g has repair probability

$$q_{\epsilon,g} \in (0,1]. \tag{347}$$

Cluster g receives

$$m_g \in \mathbb{N} \tag{348}$$

active attempts. The event that every active cluster obtains at least one ϵ-useful child is

$$C_{\text{all}}^{(\epsilon)} = \bigcap_{g \in \mathcal{G}_{\text{act}}} S_{g,m_g}^{(\epsilon)}. \tag{349}$$

Then

$$\mathbb{P}\Big(C_{\text{all}}^{(\epsilon)}\Big) \;\geq\; 1 - \sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\epsilon,g})^{m_g}. \tag{350}$$

If

$$q_{\min} = \min_{g \in \mathcal{G}_{\text{act}}} q_{\epsilon,g}, \tag{351}$$

and if each active cluster receives the same number of attempts

$$m_g = m, \tag{352}$$

then

$$\mathbb{P}\Big(C_{\text{all}}^{(\epsilon)}\Big) \;\geq\; 1 - K_{\text{act}}(1 - q_{\min})^m. \tag{353}$$

A sufficient uniform attempt budget for coverage probability at least $1 - \delta$ *is*

$$m = \left\lceil \frac{1}{q_{\min}} \log\left(\frac{K_{\text{act}}}{\delta}\right) \right\rceil. \tag{354}$$

Proof. The failure of cluster g after m_g active attempts is

$$\Big(S_{g,m_g}^{(\epsilon)}\Big)^c = F_{g,m_g}^{(\epsilon)}. \tag{355}$$

Theorem 20 gives

$$\mathbb{P}\Big(\Big(S_{g,m_g}^{(\epsilon)}\Big)^c\Big) \;\leq\; (1 - q_{\epsilon,g})^{m_g}. \tag{356}$$

The complement of full coverage is

$$\Big(C_{\text{all}}^{(\epsilon)}\Big)^c = \bigcup_{g \in \mathcal{G}_{\text{act}}} \Big(S_{g,m_g}^{(\epsilon)}\Big)^c. \tag{357}$$

The union bound gives

$$\begin{aligned}
\mathbb{P}\Big(\Big(C_{\text{all}}^{(\epsilon)}\Big)^c\Big) \;&\leq\; \sum_{g \in \mathcal{G}_{\text{act}}} \mathbb{P}\Big(\Big(S_{g,m_g}^{(\epsilon)}\Big)^c\Big) && (358)\\
&\leq\; \sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\epsilon,g})^{m_g}. && (359)
\end{aligned}$$

Taking complements proves equation 350. If $q_{\epsilon,g} \geq q_{\min}$ and $m_g = m$, then

$$\begin{aligned}
\sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\epsilon,g})^m \;&\leq\; \sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\min})^m && (360)\\
&=\; K_{\text{act}}(1 - q_{\min})^m. && (361)
\end{aligned}$$

For equation 354,

$$\begin{aligned}
K_{\text{act}}(1 - q_{\min})^m \;&\leq\; K_{\text{act}} \exp(-q_{\min} m) && (362)\\
&\leq\; K_{\text{act}} \exp\left(-\log\left(\frac{K_{\text{act}}}{\delta}\right)\right) && (363)\\
&=\; \delta. && (364)
\end{aligned}$$

□

For example, with three active clusters and the same conservative repair probability as above,

$$K_{\text{act}} = 3, \tag{365}$$

$$q_{\min} = 0.20, \tag{366}$$

and

$$\delta = 0.05, \tag{367}$$

the sufficient uniform budget is

$$\begin{aligned} m &= \lceil 5 \log(60) \rceil & (368) \\ &= 21. & (369) \end{aligned}$$

Proposition 23 (Rollout and archive-size complexity under the implemented gate)**.** *The horizon is*

$$T \in \mathbb{N}. \tag{370}$$

The initial archive size is

$$K_1 = |A_1|. \tag{371}$$

The archive cap is

$$K_{\max} \in \mathbb{N}. \tag{372}$$

The cooldown parameter is

$$\kappa \in \mathbb{N}_0. \tag{373}$$

The reflection-attempt indicator after round t for user u is

$$J_t(u) \in \{0, 1\}. \tag{374}$$

The total number of reflection attempts for user u is

$$M_T(u) = \sum_{t=1}^{T} J_t(u). \tag{375}$$

Assume the archive is append-only, changes only through this gate, and each successful gate inserts one new child:

$$|A_{t+1}(u)| = |A_t(u)| + J_t(u). \tag{376}$$

Assume the cap is enforced:

$$|A_t(u)| \le K_{\max}. \tag{377}$$

Assume the cooldown is enforced:

$$J_s(u) = J_t(u) = 1, \; s < t \Rightarrow t - s \ge \kappa + 1. \tag{378}$$

Then the attempt count obeys

$$M_T(u) \le \min\left\{ K_{\max} - K_1, \; 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \tag{379}$$

The final archive size obeys

$$|A_{T+1}(u)| \le K_{\max}. \tag{380}$$

The number of router score evaluations obeys

$$\sum_{t=1}^{T} |A_t(u)| \le T K_{\max}. \tag{381}$$

If each reflection attempt costs $c_{\text{ref}} \ge 0$ additional child evaluations, then the per-user rollout budget obeys

$$T + c_{\text{ref}} M_T(u) \le T + c_{\text{ref}} \min\left\{ K_{\max} - K_1, \; 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \tag{382}$$

Proof. Each inserted child increases the archive size by one, so

$$\begin{aligned} |A_{T+1}(u)| &= K_1 + M_T(u) & (383) \\ &\le K_{\max}. & (384) \end{aligned}$$

Thus

$$M_T(u) \le K_{\max} - K_1. \tag{385}$$

If $M_T(u) = 0$, the cooldown bound is immediate. Suppose $M_T(u) \geq 1$, and order the reflection times as

$$1 \leq s_1 < s_2 < \cdots < s_{M_T(u)} \leq T. \tag{386}$$

The cooldown condition implies

$$\begin{aligned} s_j &\geq s_1 + (j-1)(\kappa+1) &(387)\\ &\geq 1 + (j-1)(\kappa+1). &(388)\end{aligned}$$

For the final attempt,

$$\begin{aligned} T &\geq s_{M_T(u)} &(389)\\ &\geq 1 + (M_T(u)-1)(\kappa+1). &(390)\end{aligned}$$

Rearranging gives

$$M_T(u) \leq 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor. \tag{391}$$

Combining equation 385 and equation 391 proves equation 379. The archive-size bound follows from equation 383. The score-evaluation bound follows from the cap:

$$\begin{aligned} \sum_{t=1}^{T} |A_t(u)| &\leq \sum_{t=1}^{T} K_{\max} &(392)\\ &= TK_{\max}. &(393)\end{aligned}$$

The rollout-budget bound follows by substituting equation 379 into

$$\text{Rollouts}_T(u) = T + c_{\text{ref}} M_T(u). \tag{394}$$

□

Using the hyperparameters reported in Section 6, the numerical bound is

$$T = 300, \tag{395}$$

$$K_1 = 12, \tag{396}$$

$$K_{\max} = 24, \tag{397}$$

and

$$\kappa = 20. \tag{398}$$

Therefore

$$\begin{aligned} M_T(u) &\leq \min\left\{24 - 12,\ 1 + \left\lfloor \frac{299}{21} \right\rfloor\right\} &(399)\\ &= \min\{12, 15\} &(400)\\ &= 12. &(401)\end{aligned}$$

The score-evaluation budget is bounded by

$$\begin{aligned} \sum_{t=1}^{300} |A_t(u)| &\leq 300 \cdot 24 &(402)\\ &= 7200. &(403)\end{aligned}$$

With one additional child evaluation per reflection attempt,

$$T + M_T(u) \ \le\ 300 + 12 \tag{404}$$
$$= 312. \tag{405}$$

Proposition 24 (Discovery reserve induced by first useful specialists)**.** *The horizon-averaged routed value is*

$$R_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mu(U, a_t(U))]. \tag{406}$$

The static-comparator reserve is

$$\rho_T = R_T - V_{\text{static}}(A_1). \tag{407}$$

The population mass of cluster g is

$$\pi_g = \mathbb{P}(g(U) = g). \tag{408}$$

Assume $\pi_g > 0$ for every $g \in \mathcal{G}_{\text{act}}$, and fix

$$s \in \{0, \dots, T\}. \tag{409}$$

For an active attempt that never occurs, use the convention $\tau_{g,j} = \infty$. The number of active attempts for cluster g by round s is

$$N_g(s) = \sum_{j\,\ge\,1} \mathbf{1}\{\tau_{g,j} \ \le\ s\}. \tag{410}$$

The probability that cluster g receives at least m_g active attempts by round s is

$$p_g(s) = \mathbb{P}(N_g(s) \ \ge\ m_g \mid g(U) = g)\,. \tag{411}$$

Define the joint probability that the required attempts occur by round s and one of them succeeds:

$$h_g(s) = \mathbb{P}\Big(S^{(\epsilon_g)}_{g,m_g} \cap \{N_g(s) \ge m_g\} \mid g(U) = g\Big)\,. \tag{412}$$

Assume that a successful ϵ_g-useful child for cluster g, once inserted by round s, persists in the archive and contributes at least ϵ_g to the per-round discovery gain for all later rounds in that cluster:

$$G_t(U) \ \ge\ \epsilon_g \quad \textit{for all } t > s \textit{ on } \{g(U) = g\} \cap S^{(\epsilon_g)}_{g,m_g} \cap \{N_g(s) \ge m_g\}. \tag{413}$$

Assume the exploration-cost certificate

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)] \ \le\ B_T. \tag{414}$$

Then the reserve satisfies

$$\rho_T \ \ge\ \Delta_{\text{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T}\sum_{g\in\mathcal{G}_{\text{act}}}\pi_g\epsilon_g h_g(s). \tag{415}$$

Consequently, personalized routing is certified to beat the best static initial candidate whenever the right-hand side of equation 415 is positive:

$$\Delta_{\text{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T}\sum_{g\in\mathcal{G}_{\text{act}}}\pi_g\epsilon_g h_g(s) > 0. \tag{416}$$

The corresponding regret relative to the best static initial candidate is

$$\mathcal{R}^{\text{static}}_T = T V_{\text{static}}(A_1) - \sum_{t=1}^{T}\mathbb{E}[\mu(U, a_t(U))], \tag{417}$$

and it obeys

$$\frac{1}{T}\mathcal{R}^{\text{static}}_T \ \le\ -\Delta_{\text{route}}(A_1) + \frac{B_T}{T} - \frac{T-s}{T}\sum_{g\in\mathcal{G}_{\text{act}}}\pi_g\epsilon_g h_g(s). \tag{418}$$

Proof. The four-term decomposition in Theorem 13 gives

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{419}$$

Subtracting the static comparator yields

$$\rho_T \;=\; \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{420}$$

The exploration-cost certificate gives

$$\bar{L}_T \;=\; \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] \tag{421}$$

$$\le\; \frac{B_T}{T}. \tag{422}$$

For $t > s$, the persistence assumption gives

$$\mathbb{E}[G_t(U)] \ge \sum_{g\in\mathcal{G}_{\text{act}}} \pi_g\epsilon_g\mathbb{P}\Big(S^{(\epsilon_g)}_{g,m_g} \cap \{N_g(s) \;\ge\; m_g\} \mid g(U) = g\Big). \tag{423}$$

By the definition of $h_g(s)$, this gives

$$\mathbb{E}[G_t(U)] \;\ge\; \sum_{g\in\mathcal{G}_{\text{act}}} \pi_g\epsilon_g h_g(s). \tag{424}$$

Averaging over the final $T - s$ rounds gives

$$\bar{G}_T \;=\; \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[G_t(U)] \tag{425}$$

$$\ge\; \frac{1}{T}\sum_{t=s+1}^{T}\mathbb{E}[G_t(U)] \tag{426}$$

$$\ge\; \frac{T-s}{T}\sum_{g\in\mathcal{G}_{\text{act}}} \pi_g\epsilon_g h_g(s). \tag{427}$$

Substituting equation 421 and equation 425 into equation 420 proves equation 415. The dominance condition equation 416 is the statement $\rho_T > 0$. Finally,

$$\frac{1}{T}\mathcal{R}^{\text{static}}_T \;=\; V_{\text{static}}(A_1) - R_T \tag{428}$$

$$=\; -\rho_T, \tag{429}$$

and substituting equation 415 proves equation 418. □

In general, $h_g(s)$ cannot be replaced by $p_g(s)[1-(1-q_{\epsilon_g,g})^{m_g}]$: the event that m_g attempts are scheduled may depend on earlier successes or failures. That product is valid only under an additional exogenous attempt-scheduling condition that preserves the per-attempt success bound after conditioning on $\{N_g(s) \ge m_g\}$.

The reserve statement explains how Theorem 20 interacts with the paper's central decomposition.

It does not certify the saved reflection mechanism, because the saved mechanism is oracle-assisted.

The completed synthetic evidence in Section 6 instead reports the structural terms and an observed-reward residual.

For the corrected `general` row in Table 2, the empirical reserve is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}^{\text{obs}}_T \;=\; 0.280 + 0.051 - 0.036 \tag{430}$$

$$=\; 0.295. \tag{431}$$

Adding this reserve to the corrected static comparator gives

$$V_{\text{static}}(A_1) + 0.295 \;=\; 0.570 + 0.295 \tag{432}$$
$$=\; 0.865, \tag{433}$$

which matches the reported routed value

$$R_T^{\text{obs}} \approx 0.864 \tag{434}$$

up to reporting-rounding.

This arithmetic checks the observed-reward identity; it is not a test of the mean-reward decomposition or a non-oracle proof of reflective discovery.

D Remaining Propositions, Counterexamples, and Technical Details

This section records the remaining algebraic consequences of the decomposition, the edge cases that delimit its interpretation, and the proof map for the main theory claims.

The statements below do not introduce a new mechanism; they only make explicit when the four-term accounting identity can and cannot be read as an advantage claim.

D.1 Dominance condition

Corollary 25 (Dominance condition)**.** *The horizon is*

$$T \in \mathbb{N}. \tag{435}$$

The realized routed value is

$$R_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{436}$$

The averaged discovery gain is

$$\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)]. \tag{437}$$

The averaged exploration cost is

$$\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{438}$$

Then the routed policy beats the best static initial candidate if and only if

$$R_T > V_{\text{static}}(A_1) \quad\Longleftrightarrow\quad \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{439}$$

If the archive is fixed, meaning

$$A_t(U) = A_1 \quad \textit{for every } t, \tag{440}$$

then the condition reduces to

$$R_T > V_{\text{static}}(A_1) \quad\Longleftrightarrow\quad \Delta_{\text{route}}(A_1) > \bar{L}_T. \tag{441}$$

Proof. Theorem 13 gives

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{442}$$

Subtracting the static comparator gives

$$R_T - V_{\text{static}}(A_1) \;=\; \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{443}$$

Therefore

$$R_T > V_{\text{static}}(A_1) \iff R_T - V_{\text{static}}(A_1) > 0 \tag{444}$$
$$\iff \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T > 0 \tag{445}$$
$$\iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{446}$$

Under equation 440,

$$G_t(U) \;=\; \max_{a \in A_1} \mu(U,a) - \max_{a \in A_1} \mu(U,a) \tag{447}$$
$$=\; 0, \tag{448}$$

and hence

$$\bar{G}_T \;=\; 0. \tag{449}$$

Substituting equation 449 into equation 444 proves equation 441. □

For a concrete numerical check, if

$$\Delta_{\text{route}}(A_1) = 0.12, \tag{450}$$

$$\bar{G}_T = 0.04, \tag{451}$$

and

$$\bar{L}_T = 0.03, \tag{452}$$

then the routed reserve over the best static initial candidate is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T \;=\; 0.12 + 0.04 - 0.03 \tag{453}$$
$$=\; 0.13 \tag{454}$$
$$>\; 0. \tag{455}$$

D.2 Sign facts and user-specific archives

Assumption 16 (Append-only archive for sign statements)**.** *The archive available at round t contains the initial archive:*

$$A_1 \subseteq A_t(U) \quad \textit{almost surely}. \tag{456}$$

The selected candidate is feasible:

$$a_t(U) \in A_t(U) \quad \textit{almost surely}. \tag{457}$$

Lemma 26 (Range of discovery gain and exploration cost)**.** *The mean reward is bounded:*

$$\mu(u,a) \in [0,1]. \tag{458}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a). \tag{459}$$

The exploration cost is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{460}$$

The discovery gain is

$$G_t(U) = m_t(U) - m_1(U). \tag{461}$$

Under Assumption 16,

$$0 \;\le\; L_t(U) \;\le\; 1 \quad \textit{almost surely}, \tag{462}$$

and

$$0 \;\le\; G_t(U) \;\le\; 1 \quad \textit{almost surely}. \tag{463}$$

Consequently,

$$0 \;\le\; \bar{L}_T \;\le\; 1, \tag{464}$$

and

$$0 \;\le\; \bar{G}_T \;\le\; 1. \tag{465}$$

Proof. Because $a_t(U) \in A_t(U)$,

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a) \tag{466}$$
$$\geq \mu(U, a_t(U)). \tag{467}$$

Thus

$$L_t(U) = m_t(U) - \mu(U, a_t(U)) \tag{468}$$
$$\geq 0. \tag{469}$$

The bounded reward assumption gives

$$L_t(U) = m_t(U) - \mu(U, a_t(U)) \tag{470}$$
$$\leq 1 - 0 \tag{471}$$
$$= 1. \tag{472}$$

Because $A_1 \subseteq A_t(U)$,

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a) \tag{473}$$
$$\geq \max_{a \in A_1} \mu(U,a) \tag{474}$$
$$= m_1(U). \tag{475}$$

Hence

$$G_t(U) = m_t(U) - m_1(U) \tag{476}$$
$$\geq 0. \tag{477}$$

Again using $\mu(u,a) \in [0,1]$,

$$G_t(U) = m_t(U) - m_1(U) \tag{478}$$
$$\leq 1 - 0 \tag{479}$$
$$= 1. \tag{480}$$

Averaging equation 462 and equation 463 over t and over the randomness proves equation 464 and equation 465. □

Proposition 27 (Per-user archive copies preserve the decomposition)**.** *For each user u, the run has a user-specific archive path*

$$A_t^u. \tag{481}$$

The initial archive is common:

$$A_1^u = A_1 \quad \textit{for every } u. \tag{482}$$

The user-specific action is

$$a_t^u \in A_t^u. \tag{483}$$

The user-specific optimum is

$$m_t^u = \max_{a \in A_t^u} \mu(u,a). \tag{484}$$

The population draw selects the corresponding trajectory:

$$A_t(U) = A_t^U, \qquad a_t(U) = a_t^U. \tag{485}$$

Then Theorem 13 remains valid with user-specific archive copies:

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{486}$$

Proof. For a fixed user u and a fixed online trajectory,

$$\mu(u, a_t^u) = m_t^u - [m_t^u - \mu(u, a_t^u)] \tag{487}$$
$$= m_t^u - L_t^u. \tag{488}$$

The definition of discovery gain gives

$$m_t^u = m_1^u + [m_t^u - m_1^u] \tag{489}$$
$$= m_1^u + G_t^u. \tag{490}$$

Substituting equation 489 into equation 487,

$$\mu(u, a_t^u) = m_1^u + G_t^u - L_t^u. \tag{491}$$

Taking expectation over the online trajectory conditional on u, and then integrating over U, gives

$$\mathbb{E}[\mu(U, a_t(U))] = \mathbb{E}[m_1^U] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{492}$$

Because $A_1^u = A_1$,

$$\mathbb{E}[m_1^U] = \mathbb{E}\left[\max_{a \in A_1} \mu(U, a)\right] \tag{493}$$
$$= V_{\text{route}}(A_1) \tag{494}$$
$$= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1). \tag{495}$$

Averaging equation 492 over t proves equation 486. □

D.3 Finite-sample identity and comparator correction

Proposition 28 (Finite-sample decomposition estimator)**.** *The number of evaluated user trajectories is*

$$N \in \mathbb{N}. \tag{496}$$

The sampled users are

$$u_1, \dots, u_N. \tag{497}$$

The action for user u_i at round t is

$$a_{i,t} \in A_{i,t}. \tag{498}$$

The current optimum for user u_i is

$$m_{i,t} = \max_{a \in A_{i,t}} \mu(u_i, a). \tag{499}$$

The empirical routed value is

$$\widehat{R}_T = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \mu(u_i, a_{i,t}). \tag{500}$$

The empirical fixed-archive routed oracle is

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N} \sum_{i=1}^{N} \max_{a \in A_1} \mu(u_i, a). \tag{501}$$

The empirical static comparator is

$$\widehat{V}_{\text{static}}(A_1) = \max_{a \in A_1} \frac{1}{N} \sum_{i=1}^{N} \mu(u_i, a). \tag{502}$$

The empirical heterogeneity gap is

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1). \tag{503}$$

The empirical discovery gain is

$$\widehat{\overline{G}}_T = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} (m_{i,t} - m_{i,1}). \tag{504}$$

The empirical exploration cost is

$$\widehat{\overline{L}}_T = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} (m_{i,t} - \mu(u_i, a_{i,t})). \tag{505}$$

Then the logged quantities satisfy the exact finite-sample identity

$$\widehat{R}_T = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\overline{G}}_T - \widehat{\overline{L}}_T. \tag{506}$$

Proof. For every evaluated pair (i, t),

$$\mu(u_i, a_{i,t}) = m_{i,t} - [m_{i,t} - \mu(u_i, a_{i,t})] \tag{507}$$
$$= m_{i,1} + (m_{i,t} - m_{i,1}) - (m_{i,t} - \mu(u_i, a_{i,t})). \tag{508}$$

Summing equation 507 over i and t gives

$$\widehat{R}_T = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} m_{i,1} + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T \tag{509}$$
$$= \frac{1}{N}\sum_{i=1}^{N} m_{i,1} + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T \tag{510}$$
$$= \widehat{V}_{\text{route}}(A_1) + \widehat{\widehat{G}}_T - \widehat{\widehat{L}}_T. \tag{511}$$

By definition,

$$\widehat{V}_{\text{route}}(A_1) = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1). \tag{512}$$

Substituting equation 512 into equation 509 proves equation 506. □

Proposition 28 applies when the selected-arm means are used. The rounded residuals in Table 2 instead use observed rewards and the matching observed residual, so they check a parallel algebraic identity but do not test the proposition.

For the three completed synthetic environments,

$$\varepsilon_{\text{general}} = 0.570 + 0.280 + 0.051 - 0.036 - 0.864 \tag{513}$$
$$= 0.001, \tag{514}$$
$$\varepsilon_{\text{workspace}} = 0.577 + 0.124 + 0.041 - 0.043 - 0.699 \tag{515}$$
$$= 0.000, \tag{516}$$
$$\varepsilon_{\text{support}} = 0.579 + 0.215 + 0.061 - 0.039 - 0.816 \tag{517}$$
$$= 0.000. \tag{518}$$

The nonzero residual in equation 513 is a rounding residual, not an additional term in the decomposition.

Proposition 29 (Bias from a heuristic static comparator)**.** *A heuristic comparator selects a candidate*

$$h(A_1) \in A_1. \tag{519}$$

Its population value is

$$V_h(A_1) = \mathbb{E}[\mu(U, h(A_1))]. \tag{520}$$

The heuristic gap is

$$\Delta_h(A_1) = V_{\text{route}}(A_1) - V_h(A_1). \tag{521}$$

Then

$$\Delta_h(A_1) - \Delta_{\text{route}}(A_1) = V_{\text{static}}(A_1) - V_h(A_1) \geq 0. \tag{522}$$

Equality holds if and only if the heuristic comparator is population-static optimal:

$$V_h(A_1) = V_{\text{static}}(A_1). \tag{523}$$

Proof. Using the definitions,

$$\Delta_h(A_1) - \Delta_{\text{route}}(A_1) = [V_{\text{route}}(A_1) - V_h(A_1)] - [V_{\text{route}}(A_1) - V_{\text{static}}(A_1)] \tag{524}$$
$$= V_{\text{static}}(A_1) - V_h(A_1). \tag{525}$$

Because $V_{\text{static}}(A_1)$ is the maximum over all static candidates,

$$V_{\text{static}}(A_1) = \max_{a \in A_1} \mathbb{E}[\mu(U, a)] \tag{526}$$
$$\geq \mathbb{E}[\mu(U, h(A_1))] \tag{527}$$
$$= V_h(A_1). \tag{528}$$

Combining equation 524 and equation 526 proves equation 522. Equality is exactly equation 523. □

D.4 COUNTEREXAMPLES DELIMITING THE CLAIMS

Proposition 30 (Counterexample: homogeneous users have no structural routing gain)**.** *The initial archive is*

$$A_1 = \{a_1, a_2\}. \tag{529}$$

All users have the same rewards:

$$\mu(U, a_1) = 0.8 \quad \textit{almost surely}, \tag{530}$$

and

$$\mu(U, a_2) = 0.6 \quad \textit{almost surely}. \tag{531}$$

The archive is fixed:

$$A_t(U) = A_1. \tag{532}$$

If the router selects a_2 on an η-fraction of rounds in expectation, where

$$\eta > 0, \tag{533}$$

then personalized routing is worse than the best static candidate:

$$R_T < V_{\text{static}}(A_1). \tag{534}$$

Proof. The static value is

$$\begin{aligned} V_{\text{static}}(A_1) \;&=\; \max\{0.8, 0.6\} &(535)\\ &=\; 0.8. &(536) \end{aligned}$$

The fixed-archive routed oracle value is

$$\begin{aligned} V_{\text{route}}(A_1) \;&=\; \mathbb{E}[\max\{0.8, 0.6\}] &(537)\\ &=\; 0.8. &(538) \end{aligned}$$

Therefore

$$\begin{aligned} \Delta_{\text{route}}(A_1) \;&=\; V_{\text{route}}(A_1) - V_{\text{static}}(A_1) &(539)\\ &=\; 0. &(540) \end{aligned}$$

Because the archive is fixed,

$$\bar{G}_T \;=\; 0. \tag{541}$$

The exploration cost incurred when selecting a_2 is

$$0.8 - 0.6 \;=\; 0.2. \tag{542}$$

Thus

$$\bar{L}_T \;=\; 0.2\eta. \tag{543}$$

The decomposition gives

$$\begin{aligned} R_T \;&=\; 0.8 + 0 + 0 - 0.2\eta &(544)\\ &=\; 0.8 - 0.2\eta &(545)\\ &< 0.8 &(546)\\ &=\; V_{\text{static}}(A_1). &(547) \end{aligned}$$

□

Proposition 31 (Counterexample: a heuristic static baseline can overstate heterogeneity)**.** *The user distribution has two atoms:*

$$\mathbb{P}(U = u_1) = \mathbb{P}(U = u_2) = \frac{1}{2}. \tag{548}$$

The archive is

$$A_1 = \{a_1, a_2\}. \tag{549}$$

The mean rewards are

$$\begin{array}{c|cc} & a_1 & a_2 \\ \hline u_1 & 0.9 & 0.8 \\ u_2 & 0.1 & 0.7 \end{array}. \tag{550}$$

The heuristic quality scores satisfy

$$q(a_1) > q(a_2), \tag{551}$$

so the heuristic comparator chooses

$$h(A_1) = a_1. \tag{552}$$

Then the true heterogeneity gap is

$$\Delta_{\text{route}}(A_1) = 0.05, \tag{553}$$

whereas the heuristic gap is

$$\Delta_h(A_1) = 0.30. \tag{554}$$

Proof. The heuristic value is

$$V_h(A_1) = \frac{1}{2}(0.9 + 0.1) \tag{555}$$
$$= 0.50. \tag{556}$$

The two static candidate values are

$$\mathbb{E}[\mu(U, a_1)] = 0.50, \tag{557}$$
$$\mathbb{E}[\mu(U, a_2)] = \frac{1}{2}(0.8 + 0.7) \tag{558}$$
$$= 0.75. \tag{559}$$

Thus

$$V_{\text{static}}(A_1) = 0.75. \tag{560}$$

The routed oracle value is

$$V_{\text{route}}(A_1) = \frac{1}{2}\left[\max\{0.9, 0.8\} + \max\{0.1, 0.7\}\right] \tag{561}$$
$$= \frac{1}{2}(0.9 + 0.7) \tag{562}$$
$$= 0.80. \tag{563}$$

Therefore

$$\Delta_{\text{route}}(A_1) = 0.80 - 0.75 \tag{564}$$
$$= 0.05, \tag{565}$$

while

$$\Delta_h(A_1) = 0.80 - 0.50 \tag{566}$$
$$= 0.30. \tag{567}$$

□

Proposition 32 (Counterexample: oracle-assisted reflection makes the discovery bound vacuous)**.** *For a cluster g, let the first m reflection attempts define the event*

$$S^{(\epsilon)}_{g,m} = \{\textit{at least one of the first } m \textit{ attempts is } \epsilon\textit{-improving for cluster } g\}. \tag{568}$$

The repair probability in Theorem 20 is

$$q_{\epsilon,g} = \inf_j \mathbb{P}\big(\textit{attempt } j \textit{ is } \epsilon\textit{-improving} \mid \mathcal{F}_{\tau_{g,j}-1}\big). \tag{569}$$

If the proposer reads the hidden peak category and deterministically creates an ϵ-improving child, then

$$q_{\epsilon,g} = 1. \tag{570}$$

In this case, for every $m \geq 1$, the theorem's lower bound becomes

$$1 - (1 - q_{\epsilon,g})^m = 1. \tag{571}$$

If instead $q_{\epsilon,g} = 0$, then for every finite m,

$$1 - (1 - q_{\epsilon,g})^m = 0. \tag{572}$$

Proof. Substituting equation 570 into the geometric lower bound gives

$$\begin{aligned}1-(1-q_{\epsilon,g})^m &= 1-(1-1)^m && (573)\\ &= 1-0^m && (574)\\ &= 1 \quad \text{for } m \geq 1. && (575)\end{aligned}$$

Substituting $q_{\epsilon,g}=0$ gives

$$\begin{aligned}1-(1-q_{\epsilon,g})^m &= 1-(1-0)^m && (576)\\ &= 1-1 && (577)\\ &= 0. && (578)\end{aligned}$$

Thus the bound distinguishes only the assumed repair probability; it does not validate a non-oracle proposer unless a nontrivial lower bound on $q_{\epsilon,g}$ is established separately. □

Proposition 33 (Counterexample: without append-only archives, discovery need not be nonnegative)**.** *There is one user u:*

$$\mathbb{P}(U=u)=1. \tag{579}$$

The initial archive is

$$A_1=\{a^\star\}. \tag{580}$$

The round-two archive prunes the initial candidate:

$$A_2=\{b\}. \tag{581}$$

The rewards are

$$\mu(u,a^\star)=1, \tag{582}$$

and

$$\mu(u,b)=0. \tag{583}$$

Then

$$G_2(U)=-1. \tag{584}$$

The decomposition remains algebraically true, but the term G_t can no longer be interpreted as a gain.

Proof. The initial optimum is

$$\begin{aligned}m_1(U) &= \max_{a\in A_1}\mu(U,a) && (585)\\ &= \mu(u,a^\star) && (586)\\ &= 1. && (587)\end{aligned}$$

The round-two optimum is

$$\begin{aligned}m_2(U) &= \max_{a\in A_2}\mu(U,a) && (588)\\ &= \mu(u,b) && (589)\\ &= 0. && (590)\end{aligned}$$

Therefore

$$\begin{aligned}G_2(U) &= m_2(U)-m_1(U) && (591)\\ &= 0-1 && (592)\\ &= -1. && (593)\end{aligned}$$

If the router selects b, then

$$\begin{aligned}L_2(U) &= m_2(U)-\mu(u,b) && (594)\\ &= 0-0 && (595)\\ &= 0. && (596)\end{aligned}$$

The pointwise decomposition still holds:

$$\begin{aligned}\mu(u,b) &= m_1(U)+G_2(U)-L_2(U) && (597)\\ &= 1-1-0 && (598)\\ &= 0. && (599)\end{aligned}$$

Thus append-only growth is not needed for the identity, but it is needed for calling G_t a nonnegative discovery gain. □

D.5 Map from theory claims to proofs

Claim	Formal content	Proof or qualification
Lemma 11	Static-vs-routed value satisfies $\mathbb{E}[\max_{a\in A}\mu(U,a)] \geq \max_{a\in A}\mathbb{E}[\mu(U,a)]$, with strictness only under positive-mass ranking disagreement.	Proved above in Appendix A; Proposition 30 shows the homogeneous boundary case.
Lemma 12	Adding children after a round preserves predictability of the next action set.	Proved above in Appendix A; Assumption 16 records the extra sign condition used only for $G_t \geq 0$.
Lemma 14	High-gap failure triggering controls the number of mutation attempts without changing the router's filtration argument.	Proved above in Appendix A; Proposition 32 clarifies that trigger evidence is not evidence for non-oracle repair quality.
Theorem 13	Exact four-term decomposition: $R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$.	Proved above in Appendix A; Proposition 27 covers user-specific archive copies and Proposition 28 gives the finite-sample estimator.
Theorem 17	Conditional plug-in bound obtained from an externally supplied exploration-cost certificate B_T.	Proved above in Appendix A; no new regret rate is claimed for the saved implementation.
Theorem 20	First-useful-specialist probability: $\mathbb{P}(S^{(\epsilon)}_{g,m}) \geq 1-(1-q_{\epsilon,g})^m$.	Proved above in Appendix A; Proposition 32 explains why oracle-assisted reflection makes the bound vacuous as empirical validation.
Corollary 3	Personalized routing beats the best static initial candidate exactly when $\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T$.	Restated and proved in Corollary 25.
Proposition 24	A first-useful-specialist reserve can be inserted into the decomposition under persistence and exploration-cost certificates.	Proved above in Appendix A; it is a conditional reserve statement, not a guarantee for the oracle-assisted simulator.
Comparator correction	The theorem comparator is $V_{\text{static}}(A_1)$, not a heuristic intrinsic-quality choice.	Proved in Proposition 29; the numerical failure mode is Proposition 31.

Table 5: Proof map for the theory claims used in the main text.

Together, these results close the appendix proof story: PersonalEvo's supported claim is the four-term decomposition over the common initial archive, with conditional learning and discovery terms made explicit, and with the counterexamples identifying when stronger dominance or reflection claims would be unsound.

E Additional Technical Proof Block 5: Constants and Remaining Lemmas

This block records technical details used implicitly by Theorems 13, 17, and 20.

The claims below do not add a new mechanism; they only make explicit the measurability, boundedness, trigger-count, finite-cluster, and contraction constants behind the PersonalEvo decomposition.

Lemma 34 (Measurability and bounded ranges)**.** *Fix a round t. The user-specific archive is represented as a finite predictable list:*

$$A_t(U) = \{a_{t,1}(U), \dots, a_{t,N_t(U)}(U)\}. \tag{600}$$

The list length is almost surely finite:

$$N_t(U) < \infty \quad \textit{almost surely}. \tag{601}$$

Each listed candidate is measurable before the round:

$$a_{t,j}(U) \in \mathcal{F}_{t-1} \quad \textit{for } 1 \le j \le N_t(U). \tag{602}$$

The reward function is measurable and bounded:

$$0 \le \mu(u,a) \le 1. \tag{603}$$

Then the archive optimum is measurable:

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a). \tag{604}$$

The exploration cost is bounded:

$$0 \le L_t(U) \le 1. \tag{605}$$

The discovery term is bounded:

$$-1 \le G_t(U) \le 1. \tag{606}$$

If the archive is append-only from the initial archive, then the discovery term is nonnegative:

$$A_1 \subseteq A_t(U) \quad \Longrightarrow \quad 0 \le G_t(U) \le 1. \tag{607}$$

Proof. On the event that the list has length n, the optimum is a finite maximum:

$$m_t(U)\mathbf{1}\{N_t(U) = n\} = \left[\max_{1 \le j \le n} \mu(U, a_{t,j}(U))\right] \mathbf{1}\{N_t(U) = n\}. \tag{608}$$

For any real z, the lower level event is a finite intersection:

$$\{m_t(U) \le z\} \cap \{N_t(U) = n\} = \bigcap_{j=1}^{n} \{\mu(U, a_{t,j}(U)) \le z\} \cap \{N_t(U) = n\}. \tag{609}$$

Thus $m_t(U)$ is measurable. Boundedness follows from the reward range:

$$\begin{aligned} 0 &\le \max_{a \in A_t(U)} \mu(U,a) && (610)\\ &\le 1. && (611) \end{aligned}$$

Because the deployed candidate belongs to the current archive,

$$\begin{aligned} \mu(U, a_t(U)) &\le \max_{a \in A_t(U)} \mu(U,a) && (612)\\ &= m_t(U). && (613) \end{aligned}$$

Therefore

$$\begin{aligned} 0 &\le m_t(U) - \mu(U, a_t(U)) && (614)\\ &= L_t(U) && (615)\\ &\le 1. && (616) \end{aligned}$$

The discovery term is the difference of two bounded optima:

$$\begin{aligned} G_t(U) &= m_t(U) - m_1(U), && (617)\\ -1 &\le G_t(U) \le 1. && (618) \end{aligned}$$

If $A_1 \subseteq A_t(U)$, then maximization over the larger set cannot reduce value:

$$\begin{aligned} m_t(U) &= \max_{a \in A_t(U)} \mu(U,a) && (619)\\ &\ge \max_{a \in A_1} \mu(U,a) && (620)\\ &= m_1(U). && (621) \end{aligned}$$

Hence

$$\begin{aligned} G_t(U) &= m_t(U) - m_1(U) && (622)\\ &\ge 0. && (623) \end{aligned}$$

$\square$

A numerical instance illustrates the bookkeeping:

$$m_1(U) = 0.60, \qquad m_t(U) = 0.75, \qquad \mu(U, a_t(U)) = 0.70. \tag{624}$$

The corresponding discovery and exploration-cost terms are

$$G_t(U) \;=\; 0.75 - 0.60 = 0.15, \tag{625}$$
$$L_t(U) \;=\; 0.75 - 0.70 = 0.05. \tag{626}$$

Lemma 35 (From a high-probability tax certificate to an expected tax bound)**.** *Define the cumulative exploration cost:*

$$S_T = \sum_{t=1}^{T} L_t(U). \tag{627}$$

Assume the tax terms are bounded:

$$0 \;\le\; L_t(U) \;\le\; 1. \tag{628}$$

Assume there is a high-probability certificate:

$$\mathcal{E}_T = \{S_T \;\le\; b_T\}. \tag{629}$$

The certificate holds with probability at least $1 - \delta$*:*

$$\mathbb{P}(\mathcal{E}_T) \;\ge\; 1 - \delta. \tag{630}$$

Then the expected exploration cost is bounded by

$$\mathbb{E}[S_T] \;\le\; b_T + \delta T. \tag{631}$$

Consequently, the exploration-cost premise equation 267 of Theorem 17 holds with

$$B_T = b_T + \delta T. \tag{632}$$

Proof. Split the expectation over the certificate event and its complement:

$$\mathbb{E}[S_T] \;=\; \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T\}] + \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T^c\}]. \tag{633}$$

On the certificate event, $S_T \le b_T$:

$$\mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T\}] \;\le\; b_T. \tag{634}$$

On the complement, the boundedness of each L_t gives $S_T \le T$:

$$\mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T^c\}] \;\le\; T\mathbb{P}(\mathcal{E}_T^c) \tag{635}$$
$$\le\; \delta T. \tag{636}$$

Combining the two parts gives

$$\mathbb{E}[S_T] \;\le\; b_T + \delta T. \tag{637}$$

□

For example, if a pathwise diagnostic gives

$$b_T = 12 \tag{638}$$

over a horizon

$$T = 300 \tag{639}$$

with failure probability

$$\delta = 0.01, \tag{640}$$

then Lemma 35 yields

$$\mathbb{E}[S_T] \le 12 + 0.01 \cdot 300 \tag{641}$$
$$= 15, \tag{642}$$
$$\frac{1}{T}\mathbb{E}[S_T] \le 0.05. \tag{643}$$

Proposition 36 (Explicit constants for high-gap triggering)**.** *Assume $\tau > 0$ and $\sigma > 0$. Let the trigger at round t be*

$$I_t^{(\tau)} = \mathbf{1}\{m_t(U) - r_t > \tau\}. \tag{644}$$

Let the number of triggered reflection attempts be

$$N_T^{(\tau)} = \sum_{t=1}^{T} I_t^{(\tau)}. \tag{645}$$

Assume the reward observation satisfies

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{646}$$

Assume the noise is conditionally sub-Gaussian:

$$\mathbb{E}[\exp(\lambda\xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right) \quad \text{for all } \lambda \in \mathbb{R}. \tag{647}$$

Then

$$\mathbb{E}\left[N_T^{(\tau)}\right] \le \frac{2}{\tau}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{648}$$

If an external exploration-cost certificate gives

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)] \le B_T, \tag{649}$$

then

$$\mathbb{E}\left[N_T^{(\tau)}\right] \le \frac{2B_T}{\tau} + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{650}$$

Proof. By the definitions of r_t and $L_t(U)$,

$$m_t(U) - r_t = m_t(U) - \mu(U, a_t(U)) - \xi_t \tag{651}$$
$$= L_t(U) - \xi_t. \tag{652}$$

Therefore

$$I_t^{(\tau)} = \mathbf{1}\{L_t(U) - \xi_t > \tau\} \tag{653}$$
$$\le \mathbf{1}\{L_t(U) + |\xi_t| > \tau\} \tag{654}$$
$$\le \mathbf{1}\{L_t(U) > \tau/2\} + \mathbf{1}\{|\xi_t| > \tau/2\}. \tag{655}$$

Taking expectations gives

$$\mathbb{E}[I_t^{(\tau)}] \le \mathbb{P}\{L_t(U) > \tau/2\} + \mathbb{P}\{|\xi_t| > \tau/2\}. \tag{656}$$

Markov's inequality gives the first term:

$$\mathbb{P}\{L_t(U) > \tau/2\} \le \frac{\mathbb{E}[L_t(U)]}{\tau/2} \tag{657}$$
$$= \frac{2}{\tau}\mathbb{E}[L_t(U)]. \tag{658}$$

The sub-Gaussian tail bound gives the second term:

$$\mathbb{P}\{|\xi_t| > \tau/2\} \le 2\exp\left(-\frac{(\tau/2)^2}{2\sigma^2}\right) \tag{659}$$

$$= 2\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{660}$$

Summing over rounds yields

$$\mathbb{E}[N_T^{(\tau)}] = \sum_{t=1}^{T} \mathbb{E}[I_t^{(\tau)}] \tag{661}$$

$$\le \frac{2}{\tau}\sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{662}$$

Substituting equation 649 gives equation 650. ☐

A numerical bound can be read directly from equation 650.

If

$$T = 100, \qquad B_T = 4, \qquad \tau = 0.20, \qquad \sigma = 0.05, \tag{663}$$

then

$$\mathbb{E}[N_T^{(\tau)}] \le \frac{2\cdot 4}{0.20} + 2\cdot 100\exp\left(-\frac{0.20^2}{8\cdot 0.05^2}\right) \tag{664}$$

$$= 40 + 200e^{-2} \tag{665}$$

$$\approx 67.07. \tag{666}$$

This constant is conservative; it is used only to justify the direction of the trigger-count control, not as an empirical estimate of reflection quality.

Proposition 37 (Finite-cluster margin lower bound for heterogeneity)**.** *Assume the user population is partitioned into finitely many clusters:*

$$g(U) \in \{1, \ldots, K_{\text{cl}}\}. \tag{667}$$

The cluster mass is

$$\pi_g = \mathbb{P}(g(U) = g). \tag{668}$$

For every cluster with $\pi_g > 0$, the cluster-conditional value of candidate a is

$$\nu_{g,a} = \mathbb{E}[\mu(U,a) \mid g(U) = g]. \tag{669}$$

For a zero-mass cluster, set $\nu_{g,a} = 0$ by convention; such a cluster does not affect any sum below. For a static candidate a, define its cluster-wise missed oracle value:

$$d_g(a) = \max_{b\in A} \nu_{g,b} - \nu_{g,a}. \tag{670}$$

For a margin $\eta > 0$, define the clusters on which a misses by at least η:

$$H_a(\eta) = \{g : d_g(a) \ge \eta\}. \tag{671}$$

Assume every static candidate misses the cluster-wise oracle by margin η on mass at least ρ:

$$\inf_{a\in A} \sum_{g\in H_a(\eta)} \pi_g \ge \rho. \tag{672}$$

Then the fixed-archive heterogeneity gap satisfies

$$\Delta_{\text{route}}(A) \ge \rho\eta. \tag{673}$$

Proof. The routed value on the finite cluster partition is

$$V_{\text{route}}(A) \;\geq\; \sum_{g=1}^{K_{\text{cl}}} \pi_g \max_{b \in A} \nu_{g,b}. \tag{674}$$

The static value is

$$V_{\text{static}}(A) \;=\; \max_{a \in A} \sum_{g=1}^{K_{\text{cl}}} \pi_g \nu_{g,a}. \tag{675}$$

Let a static maximizer be

$$a^\star \in \arg\max_{a \in A} \sum_{g=1}^{K_{\text{cl}}} \pi_g \nu_{g,a}. \tag{676}$$

Then

$$\begin{aligned} \Delta_{\text{route}}(A) \;&=\; V_{\text{route}}(A) - V_{\text{static}}(A) && (677)\\ &\geq\; \sum_{g=1}^{K_{\text{cl}}} \pi_g \left(\max_{b \in A} \nu_{g,b} - \nu_{g,a^\star} \right) && (678)\\ &=\; \sum_{g=1}^{K_{\text{cl}}} \pi_g d_g(a^\star). && (679) \end{aligned}$$

Restricting the sum to the margin set gives

$$\begin{aligned} \Delta_{\text{route}}(A) \;&\geq\; \sum_{g \in H_{a^\star}(\eta)} \pi_g d_g(a^\star) && (680)\\ &\geq\; \sum_{g \in H_{a^\star}(\eta)} \pi_g \eta && (681)\\ &\geq\; \rho\eta. && (682) \end{aligned}$$

□

For a two-cluster example, take equal masses:

$$\pi_1 = \pi_2 = \frac{1}{2}. \tag{683}$$

Let the cluster-candidate value matrix be

$$\begin{array}{c|cc} & a_1 & a_2 \\ \hline g=1 & 0.80 & 0.50 \\ g=2 & 0.50 & 0.80 \end{array}. \tag{684}$$

Each static candidate misses one cluster by 0.30:

$$\rho = \frac{1}{2}, \qquad \eta = 0.30. \tag{685}$$

The lower bound gives

$$\Delta_{\text{route}}(A) \;\geq\; 0.15. \tag{686}$$

The exact value is

$$\begin{aligned} \Delta_{\text{route}}(A) \;&=\; 0.80 - \frac{1}{2}(0.80 + 0.50) && (687)\\ &=\; 0.15. && (688) \end{aligned}$$

Proposition 38 (Contraction repair condition implies an explicit discovery-time bound)**.** *This proposition records the stronger contraction calculation from the earlier draft. It is not used as an empirical claim for the oracle-assisted simulator.*

Fix a cluster g. The initial gap is

$$\Delta_{g,0} > 0. \tag{689}$$

The target accuracy is

$$0 < \epsilon < \Delta_{g,0}. \tag{690}$$

Reflection attempt j has success indicator

$$Z_j \in \{0, 1\}. \tag{691}$$

The indicators are independent and satisfy

$$q \in (0, 1], \qquad \mathbb{P}(Z_j = 1) \ \geq \ q. \tag{692}$$

The repair contraction parameter satisfies

$$0 < \gamma < 1. \tag{693}$$

The gap recursion is

$$\Delta_{g,j} \ \leq \ (1 - \gamma Z_j)\Delta_{g,j-1}. \tag{694}$$

Define the number of successful contractions needed:

$$s_\epsilon = \left\lceil \frac{\log(\Delta_{g,0}/\epsilon)}{-\log(1-\gamma)} \right\rceil. \tag{695}$$

Let the number of successes in m attempts be

$$S_m = \sum_{j=1}^{m} Z_j, \qquad m \in \mathbb{N}. \tag{696}$$

If

$$mq \ \geq \ 2s_\epsilon, \tag{697}$$

then

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \ \leq \ \exp\Big(-\frac{mq}{8}\Big). \tag{698}$$

Equivalently, for failure probability $\delta \in (0, 1)$, it suffices that

$$m \ \geq \ \frac{2s_\epsilon + 8\log(1/\delta)}{q} \tag{699}$$

to obtain

$$\mathbb{P}(\Delta_{g,m} \ \leq \ \epsilon) \ \geq \ 1 - \delta. \tag{700}$$

Proof. Iterating the recursion gives

$$\Delta_{g,m} \ \leq \ \Delta_{g,0} \prod_{j=1}^{m} (1 - \gamma Z_j). \tag{701}$$

Because $Z_j \in \{0, 1\}$,

$$\prod_{j=1}^{m} (1 - \gamma Z_j) \ = \ (1 - \gamma)^{S_m}. \tag{702}$$

Thus

$$\Delta_{g,m} \ \leq \ \Delta_{g,0}(1 - \gamma)^{S_m}. \tag{703}$$

If $S_m \geq s_\epsilon$, then

$$\begin{aligned} \Delta_{g,m} \ &\leq \ \Delta_{g,0}(1 - \gamma)^{s_\epsilon} && (704) \\ &\leq \ \epsilon. && (705) \end{aligned}$$

Therefore

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \le \mathbb{P}(S_m < s_\epsilon). \tag{706}$$

Let

$$\lambda = \log 2. \tag{707}$$

For any $s \le mq/2$, Markov's inequality applied to $\exp(-\lambda S_m)$ gives

$$\mathbb{P}(S_m \le s) = \mathbb{P}\big(e^{-\lambda S_m} \ge e^{-\lambda s}\big) \tag{708}$$

$$\le e^{\lambda s}\mathbb{E}[e^{-\lambda S_m}]. \tag{709}$$

Using independence and $\mathbb{P}(Z_j = 1) \ge q$,

$$\mathbb{E}[e^{-\lambda S_m}] = \prod_{j=1}^{m} \mathbb{E}[e^{-\lambda Z_j}] \tag{710}$$

$$= \prod_{j=1}^{m} \big(1 - \mathbb{P}(Z_j = 1) + \mathbb{P}(Z_j = 1)e^{-\lambda}\big) \tag{711}$$

$$\le \big(1 - q + qe^{-\lambda}\big)^m \tag{712}$$

$$\le \exp\big[-mq(1 - e^{-\lambda})\big]. \tag{713}$$

With $s = s_\epsilon$ and $s_\epsilon \le mq/2$,

$$\mathbb{P}(S_m < s_\epsilon) \le \exp\Big[\lambda \frac{mq}{2} - mq(1 - e^{-\lambda})\Big]. \tag{714}$$

Since $\lambda = \log 2$ and $e^{-\lambda} = 1/2$,

$$\lambda \frac{mq}{2} - mq(1 - e^{-\lambda}) = mq\left(\frac{\log 2}{2} - \frac{1}{2}\right) \tag{715}$$

$$= -\frac{mq}{2}(1 - \log 2) \tag{716}$$

$$\le -\frac{mq}{8}. \tag{717}$$

This proves

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \le \exp\Big(-\frac{mq}{8}\Big). \tag{718}$$

If equation 699 holds, then

$$mq \ge 2s_\epsilon, \tag{719}$$

$$mq \ge 8\log(1/\delta). \tag{720}$$

Hence

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \le \exp[-\log(1/\delta)] \tag{721}$$

$$= \delta. \tag{722}$$

□

The contraction constant recovers the original qualitative scaling only under the stronger assumptions in Proposition 38.

For instance, if

$$\Delta_{g,0} = 0.40, \qquad \epsilon = 0.05, \qquad \gamma = 0.50, \qquad q = 0.25, \qquad \delta = 0.05, \tag{723}$$

then

$$s_\epsilon = \left\lceil \frac{\log(8)}{\log(2)} \right\rceil \tag{724}$$
$$= 3. \tag{725}$$

The sufficient number of attempts is

$$m \geq \frac{2 \cdot 3 + 8\log(20)}{0.25} \tag{726}$$
$$\approx 119.86. \tag{727}$$

Thus

$$m = 120 \tag{728}$$

is sufficient for the stated 95% guarantee under the contraction model.

The main paper does not claim that the saved oracle-assisted reflection code supplies nontrivial empirical estimates of q or γ; Section 6 therefore treats discovery as a measured decomposition term rather than as validation of Proposition 38.

F Additional Technical Proofs: Constants and Estimation Bridges

This final proof block records the technical bridges used implicitly by Theorem 13, Theorem 17, and the empirical decomposition in Section 6.

These statements do not add a new theorem path; they only make explicit the constants, measurability reductions, and finite-sample algebra behind the four-term decomposition.

Lemma 39 (Append-only boundedness of the decomposition terms)**.** *Assume the mean reward is bounded:*

$$0 \leq \mu(u,a) \leq 1. \tag{729}$$

Assume the archive path is append-only:

$$A_1(U) \subseteq A_2(U) \subseteq \cdots \subseteq A_T(U). \tag{730}$$

Assume the deployed action is feasible:

$$a_t(U) \in A_t(U). \tag{731}$$

The current archive optimum is

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a). \tag{732}$$

The discovery gain is

$$G_t(U) = m_t(U) - m_1(U). \tag{733}$$

The exploration cost is

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \tag{734}$$

Then, for every round t,

$$0 \leq G_t(U) \leq 1, \qquad 0 \leq L_t(U) \leq 1. \tag{735}$$

Moreover, the discovery sequence is monotone:

$$G_{t+1}(U) \geq G_t(U). \tag{736}$$

Proof. The append-only property implies pointwise monotonicity of the archive optimum:

$$A_t(U) \subseteq A_{t+1}(U) \Longrightarrow \max_{a \in A_t(U)} \mu(U,a) \le \max_{a \in A_{t+1}(U)} \mu(U,a) \tag{737}$$

$$\Longrightarrow m_t(U) \le m_{t+1}(U). \tag{738}$$

The bounded reward assumption gives

$$0 \le m_t(U) \le 1. \tag{739}$$

Thus

$$G_t(U) = m_t(U) - m_1(U) \tag{740}$$

$$\ge 0, \tag{741}$$

$$G_t(U) \le 1 - 0 \tag{742}$$

$$= 1. \tag{743}$$

Feasibility of the action gives

$$m_t(U) = \max_{a \in A_t(U)} \mu(U,a) \tag{744}$$

$$\ge \mu(U, a_t(U)). \tag{745}$$

Therefore

$$L_t(U) = m_t(U) - \mu(U, a_t(U)) \tag{746}$$

$$\ge 0, \tag{747}$$

$$L_t(U) \le 1 - 0 \tag{748}$$

$$= 1. \tag{749}$$

Finally,

$$G_{t+1}(U) - G_t(U) = m_{t+1}(U) - m_t(U) \tag{750}$$

$$\ge 0. \tag{751}$$

□

Lemma 40 (From high-probability current-archive regret to expected exploration cost)**.** *The cumulative exploration cost is*

$$S_T^L = \sum_{t=1}^{T} L_t(U). \tag{752}$$

Assume a pathwise exploration-cost bound holds with probability at least $1 - \delta$:

$$\mathbb{P}\big(S_T^L \le C_T(\delta)\big) \ge 1 - \delta. \tag{753}$$

Assume each round-wise tax is bounded:

$$0 \le L_t(U) \le 1. \tag{754}$$

Then the expected exploration cost satisfies

$$\mathbb{E}[S_T^L] \le C_T(\delta) + \delta T. \tag{755}$$

Consequently, the external exploration-cost premise equation 18 holds with

$$B_T = C_T(\delta) + \delta T. \tag{756}$$

If the imported pathwise bound has the form

$$C_T(\delta) = c_0 \sqrt{T\big(\zeta_T + \log(1/\delta)\big)}, \tag{757}$$

then choosing

$$\delta = \frac{1}{T} \tag{758}$$

gives

$$B_T \le c_0 \sqrt{T\big(\zeta_T + \log T\big)} + 1. \tag{759}$$

Proof. Let the good event be

$$\mathcal{E}_\delta = \{S_T^L \le C_T(\delta)\}. \tag{760}$$

By boundedness of the round-wise tax,

$$S_T^L = \sum_{t=1}^{T} L_t(U) \tag{761}$$
$$\le T. \tag{762}$$

Decompose the expectation over the good event and its complement:

$$\mathbb{E}[S_T^L] = \mathbb{E}[S_T^L \mathbf{1}\{\mathcal{E}_\delta\}] + \mathbb{E}[S_T^L \mathbf{1}\{\mathcal{E}_\delta^c\}] \tag{763}$$
$$\le C_T(\delta)\mathbb{P}(\mathcal{E}_\delta) + T\mathbb{P}(\mathcal{E}_\delta^c) \tag{764}$$
$$\le C_T(\delta) + \delta T. \tag{765}$$

Substituting equation 757 and equation 758 yields

$$\mathbb{E}[S_T^L] \le c_0\sqrt{T\big(\zeta_T + \log T\big)} + 1. \tag{766}$$

□

For example, if an external analysis of the actual router supplies

$$C_{300}(1/300) = 12, \tag{767}$$

then Lemma 40 yields

$$B_{300} \le 12 + \frac{300}{300} \tag{768}$$
$$= 13. \tag{769}$$

The normalized exploration-cost term in Theorem 17 is then bounded by

$$\frac{B_{300}}{300} \le 0.0433. \tag{770}$$

Lemma 41 (Explicit constants for high-gap trigger counts)**.** *The observed reward model is*

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{771}$$

The noise is conditionally sub-Gaussian:

$$\mathbb{E}[\exp(\lambda\xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right). \tag{772}$$

The high-gap trigger indicator at threshold $\tau > 0$ is

$$I_t(\tau) = \mathbf{1}\{m_t(U) - r_t > \tau\}. \tag{773}$$

The number of triggered reflection opportunities is

$$N_T(\tau) = \sum_{t=1}^{T} I_t(\tau). \tag{774}$$

Then

$$\mathbb{E}[N_T(\tau)] \le T \wedge \left[\frac{2}{\tau}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right)\right]. \tag{775}$$

Under the external exploration-cost bound

$$\sum_{t=1}^{T}\mathbb{E}[L_t(U)] \le B_T, \tag{776}$$

the trigger count obeys

$$\mathbb{E}[N_T(\tau)] \le T \wedge \left[\frac{2B_T}{\tau} + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right)\right]. \tag{777}$$

Proof. Using equation 771, the trigger event is

$$\{m_t(U) - r_t > \tau\} \;=\; \{m_t(U) - \mu(U, a_t(U)) - \xi_t > \tau\} \tag{778}$$
$$= \{L_t(U) - \xi_t > \tau\}. \tag{779}$$

This event is contained in a deterministic-noise union:

$$\{L_t(U) - \xi_t > \tau\} \subseteq \{L_t(U) + |\xi_t| > \tau\} \tag{780}$$
$$\subseteq \{L_t(U) > \tau/2\} \cup \{|\xi_t| > \tau/2\}. \tag{781}$$

Therefore

$$\mathbb{P}(I_t(\tau) = 1) \;\le\; \mathbb{P}(L_t(U) > \tau/2) + \mathbb{P}(|\xi_t| > \tau/2). \tag{782}$$

Markov's inequality gives

$$\mathbb{P}(L_t(U) > \tau/2) \;\le\; \frac{\mathbb{E}[L_t(U)]}{\tau/2} \tag{783}$$
$$= \frac{2\mathbb{E}[L_t(U)]}{\tau}. \tag{784}$$

The sub-Gaussian tail bound gives

$$\mathbb{P}(|\xi_t| > \tau/2) \;\le\; 2\exp\left(-\frac{(\tau/2)^2}{2\sigma^2}\right) \tag{785}$$
$$= 2\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{786}$$

Summing over rounds yields

$$\mathbb{E}[N_T(\tau)] \;=\; \sum_{t=1}^{T} \mathbb{P}(I_t(\tau) = 1) \tag{787}$$
$$\le\; \frac{2}{\tau}\sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{788}$$

The trivial deterministic bound is

$$N_T(\tau) \;\le\; T. \tag{789}$$

Combining equation 787 and equation 789 proves equation 775. Substituting equation 776 proves equation 777. □

With the simulator-scale constants

$$T = 300, \qquad \tau = 0.15, \qquad \sigma = 0.05, \qquad B_T = 12, \tag{790}$$

the worst-case trigger bound evaluates to

$$\frac{2B_T}{\tau} + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right) \;=\; \frac{24}{0.15} + 600\exp(-1.125) \tag{791}$$
$$\approx 354.8. \tag{792}$$

Thus the deterministic cap dominates:

$$\mathbb{E}[N_T(0.15)] \;\le\; 300. \tag{793}$$

This calculation is intentionally conservative; the empirical archive-growth claim in Section 6 is descriptive rather than a consequence of this loose tail bound.

Proposition 42 (Cooldown and archive cap imply a deterministic archive-size bound)**.** *The initial archive size is*

$$K_1 = |A_1|. \tag{794}$$

The maximum archive size is

$$K_{\max} \geq K_1. \tag{795}$$

Let $M_T \in \mathbb{N}_0$ be the number of children appended through round T. If $M_T \geq 1$, let the append times be

$$1 \leq s_1 < s_2 < \cdots < s_{M_T} \leq T. \tag{796}$$

Assume the archive is append-only, the listed times are exactly the times at which a new child is inserted, each inserted child is unique, and exactly one child is appended at each listed time:

$$|A_{s_j+1}(U)| - |A_{s_j}(U)| = 1. \tag{797}$$

Assume the cooldown parameter is $\kappa \in \mathbb{N}$ and append times satisfy

$$s_{j+1} - s_j \geq \kappa + 1. \tag{798}$$

Assume the cap is enforced:

$$|A_t(U)| \leq K_{\max}. \tag{799}$$

Then the number of appended children satisfies

$$M_T \leq \min\left\{K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor\right\}. \tag{800}$$

Consequently, for every $t \in \{1, \ldots, T+1\}$,

$$|A_t(U)| \leq K_1 + \min\left\{K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor\right\}. \tag{801}$$

Proof. If $M_T = 0$, both bounds are immediate. Suppose $M_T \geq 1$. Then The cooldown condition implies by induction that

$$\begin{aligned} s_j &\geq s_1 + (j-1)(\kappa+1) && (802)\\ &\geq 1 + (j-1)(\kappa+1). && (803) \end{aligned}$$

Since $s_{M_T} \leq T$,

$$1 + (M_T - 1)(\kappa+1) \leq T. \tag{804}$$

Rearranging gives

$$M_T - 1 \leq \frac{T-1}{\kappa+1}. \tag{805}$$

Since $M_T - 1$ is an integer,

$$M_T \leq 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor. \tag{806}$$

Append-only unique insertions and the archive cap give

$$K_1 + M_T \leq K_{\max}, \tag{807}$$

and hence

$$M_T \leq K_{\max} - K_1. \tag{808}$$

Combining equation 806 and equation 808 proves equation 800. The archive-size bound follows from

$$|A_t(U)| \leq K_1 + M_T. \tag{809}$$

□

For the implementation constants used in the synthetic runs,

$$K_1 = 12, \qquad K_{\max} = 24, \qquad T = 300, \qquad \kappa = 20. \tag{810}$$

The cooldown-only bound gives

$$1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor = 1 + \left\lfloor \frac{299}{21} \right\rfloor \tag{811}$$

$$= 15. \tag{812}$$

The cap-only bound gives

$$K_{\max} - K_1 = 12. \tag{813}$$

Thus

$$M_T \le 12, \tag{814}$$

$$|A_t(U)| \le 24. \tag{815}$$

Proposition 43 (Logged decomposition estimators and finite-sample radii)**.** *Consider N independent and identically distributed logged one-user trajectories generated with the same deterministic initial archive and the same routing protocol. Their user draws, online noises, and archive paths are independent across trajectories. Each archive path is append-only. The common initial archive is*

$$A_1. \tag{816}$$

Its size is

$$K_1 = |A_1|. \tag{817}$$

For trajectory n and round t, define

$$m_{n,t} = \max_{a \in A_{n,t}} \mu(U_n, a). \tag{818}$$

Define the logged discovery gain:

$$G_{n,t} = m_{n,t} - m_{n,1}. \tag{819}$$

Define the logged exploration cost:

$$L_{n,t} = m_{n,t} - \mu(U_n, a_{n,t}). \tag{820}$$

The logged routed value estimator is

$$\widehat{\mathcal{V}}_T = \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} \mu(U_n, a_{n,t}). \tag{821}$$

The logged fixed-archive routed estimator is

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N} \sum_{n=1}^{N} m_{n,1}. \tag{822}$$

The logged static estimator is

$$\widehat{V}_{\text{static}}(A_1) = \max_{a \in A_1} \frac{1}{N} \sum_{n=1}^{N} \mu(U_n, a). \tag{823}$$

The logged heterogeneity estimator is

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1). \tag{824}$$

The logged discovery estimator is

$$\widehat{\bar{G}}_T = \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T} G_{n,t}. \tag{825}$$

The logged exploration-cost estimator is

$$\widehat{\bar{L}}_T = \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T} L_{n,t}. \tag{826}$$

Then the logged decomposition holds exactly:

$$\widehat{\mathcal{V}}_T = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. \tag{827}$$

For concentration, define the population horizon-averaged value:

$$\mathcal{V}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]. \tag{828}$$

Define the scalar radius:

$$\varepsilon_{\text{sc}}(N,\delta) = \sqrt{\frac{\log(10/\delta)}{2N}}. \tag{829}$$

Define the static radius:

$$\varepsilon_{\text{st}}(N,K_1,\delta) = \sqrt{\frac{\log(10K_1/\delta)}{2N}}. \tag{830}$$

With probability at least $1-\delta$, the following simultaneous bounds hold:

$$\left|\widehat{V}_{\text{static}}(A_1) - V_{\text{static}}(A_1)\right| \;\le\; \varepsilon_{\text{st}}(N,K_1,\delta), \tag{831}$$

$$\left|\widehat{V}_{\text{route}}(A_1) - V_{\text{route}}(A_1)\right| \;\le\; \varepsilon_{\text{sc}}(N,\delta), \tag{832}$$

$$\left|\widehat{\bar{G}}_T - \bar{G}_T\right| \;\le\; \varepsilon_{\text{sc}}(N,\delta), \tag{833}$$

$$\left|\widehat{\bar{L}}_T - \bar{L}_T\right| \;\le\; \varepsilon_{\text{sc}}(N,\delta), \tag{834}$$

$$\left|\widehat{\mathcal{V}}_T - \mathcal{V}_T\right| \;\le\; \varepsilon_{\text{sc}}(N,\delta). \tag{835}$$

Consequently,

$$\left|\widehat{\Delta}_{\text{route}}(A_1) - \Delta_{\text{route}}(A_1)\right| \;\le\; \varepsilon_{\text{st}}(N,K_1,\delta) + \varepsilon_{\text{sc}}(N,\delta). \tag{836}$$

If the dominance margin is

$$M_T = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{837}$$

and its estimator is

$$\widehat{M}_T = \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T, \tag{838}$$

then

$$|\widehat{M}_T - M_T| \;\le\; \varepsilon_{\text{st}}(N,K_1,\delta) + 3\varepsilon_{\text{sc}}(N,\delta). \tag{839}$$

Proof. For the exact logged identity, use the pointwise relation

$$\begin{aligned} \mu(U_n, a_{n,t}) \;&=\; m_{n,t} - L_{n,t} && (840)\\ &=\; m_{n,1} + G_{n,t} - L_{n,t}. && (841) \end{aligned}$$

Averaging gives

$$\begin{aligned} \widehat{\mathcal{V}}_T \;&=\; \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}(m_{n,1} + G_{n,t} - L_{n,t}) && (842)\\ &=\; \frac{1}{N}\sum_{n=1}^{N} m_{n,1} + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T && (843)\\ &=\; \widehat{V}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T && (844)\\ &=\; \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. && (845) \end{aligned}$$

For the static term, define for each candidate

$$\widehat{\mu}_a = \frac{1}{N}\sum_{n=1}^{N}\mu(U_n, a). \tag{846}$$

Its population mean is

$$\mu_a = \mathbb{E}[\mu(U, a)]. \tag{847}$$

Hoeffding's inequality and a union bound imply

$$\begin{aligned}\mathbb{P}\left(\max_{a\in A_1}|\widehat{\mu}_a - \mu_a| > \varepsilon\right) &\le \sum_{a\in A_1} 2\exp(-2N\varepsilon^2) && (848)\\ &= 2K_1\exp(-2N\varepsilon^2). && (849)\end{aligned}$$

The maximum map is one-Lipschitz in the sup norm:

$$\left|\max_{a\in A_1}\widehat{\mu}_a - \max_{a\in A_1}\mu_a\right| \le \max_{a\in A_1}|\widehat{\mu}_a - \mu_a|. \tag{850}$$

Setting

$$\varepsilon = \sqrt{\frac{\log(10K_1/\delta)}{2N}} \tag{851}$$

makes the right-hand side of equation 848 equal to $\delta/5$.

For the other four scalar terms, define the per-trajectory bounded variables

$$\begin{aligned}Z_n^{\text{route}} &= m_{n,1}, && (852)\\ Z_n^{G} &= \frac{1}{T}\sum_{t=1}^{T} G_{n,t}, && (853)\\ Z_n^{L} &= \frac{1}{T}\sum_{t=1}^{T} L_{n,t}, && (854)\\ Z_n^{\mathcal{V}} &= \frac{1}{T}\sum_{t=1}^{T}\mu(U_n, a_{n,t}). && (855)\end{aligned}$$

By Lemma 39,

$$Z_n^{\text{route}}, Z_n^{G}, Z_n^{L}, Z_n^{\mathcal{V}} \in [0,1]. \tag{856}$$

Hoeffding's inequality gives, for each scalar term,

$$\mathbb{P}\left(\left|\frac{1}{N}\sum_{n=1}^{N} Z_n - \mathbb{E}[Z_n]\right| > \varepsilon\right) \le 2\exp(-2N\varepsilon^2). \tag{857}$$

Setting

$$\varepsilon = \sqrt{\frac{\log(10/\delta)}{2N}} \tag{858}$$

makes each scalar failure probability at most $\delta/5$. A union bound over the static event and the four scalar events proves equation 831. The heterogeneity and margin bounds follow by the triangle inequality:

$$\begin{aligned}\left|\widehat{\Delta}_{\text{route}} - \Delta_{\text{route}}\right| &= \left|(\widehat{V}_{\text{route}} - V_{\text{route}}) - (\widehat{V}_{\text{static}} - V_{\text{static}})\right| && (859)\\ &\le \varepsilon_{\text{sc}} + \varepsilon_{\text{st}}, && (860)\\ |\widehat{M}_T - M_T| &\le |\widehat{\Delta}_{\text{route}} - \Delta_{\text{route}}| + |\widehat{\bar{G}}_T - \bar{G}_T| + |\widehat{\bar{L}}_T - \bar{L}_T| && (861)\\ &\le \varepsilon_{\text{st}} + 3\varepsilon_{\text{sc}}. && (862)\end{aligned}$$

□

The saved synthetic runs do not meet the independence premise with $N = 5 \cdot 8 = 40$: the eight users within a seed share that seed's archive draw. Proposition 43 therefore does not supply a 40-sample confidence interval for those runs. A valid analysis must treat each seed as a cluster, use a cluster-aware concentration bound, or regenerate independent archives for every logged trajectory. The algebraic sample identity remains exact without independence.

Lemma 44 (Observed-reward noise in the routed-value estimate)**.** *Let the total number of logged reward observations be*

$$M = NT. \tag{863}$$

Enumerate the observations by a single index:

$$s \in \{1, \dots, M\}. \tag{864}$$

The observed reward is

$$r_s = \mu_s + \varepsilon_s. \tag{865}$$

The conditional mean term is

$$\mu_s = \mu(U_s, a_s). \tag{866}$$

The noise is a martingale difference:

$$\mathbb{E}[\varepsilon_s \mid \mathcal{H}_{s-1}] = 0. \tag{867}$$

The noise is conditionally sub-Gaussian:

$$\mathbb{E}[\exp(\lambda\varepsilon_s) \mid \mathcal{H}_{s-1}] \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right) \qquad \text{for every } \lambda \in \mathbb{R}, \quad \sigma > 0. \tag{868}$$

The observed routed-value estimator is

$$\widehat{\mathcal{V}}_T^{\mathrm{obs}} = \frac{1}{M}\sum_{s=1}^{M} r_s. \tag{869}$$

The corresponding mean-reward estimator is

$$\widehat{\mathcal{V}}_T^{\mathrm{mean}} = \frac{1}{M}\sum_{s=1}^{M} \mu_s. \tag{870}$$

Then, for every $\epsilon > 0$,

$$\mathbb{P}\left(\left|\widehat{\mathcal{V}}_T^{\mathrm{obs}} - \widehat{\mathcal{V}}_T^{\mathrm{mean}}\right| \ge \epsilon\right) \le 2\exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{871}$$

Equivalently, for every $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\left|\widehat{\mathcal{V}}_T^{\mathrm{obs}} - \widehat{\mathcal{V}}_T^{\mathrm{mean}}\right| \le \sigma\sqrt{\frac{2\log(2/\delta)}{M}}. \tag{872}$$

Proof. Let the accumulated noise be

$$S_M = \sum_{s=1}^{M} \varepsilon_s. \tag{873}$$

Iterating conditional expectations gives

$$\begin{aligned}
\mathbb{E}[\exp(\lambda S_M)] &= \mathbb{E}\left[\exp\left(\lambda\sum_{s=1}^{M-1}\varepsilon_s\right)\mathbb{E}[\exp(\lambda\varepsilon_M) \mid \mathcal{H}_{M-1}]\right] && (874)\\
&\le \exp\left(\frac{\lambda^2\sigma^2}{2}\right)\mathbb{E}\left[\exp\left(\lambda\sum_{s=1}^{M-1}\varepsilon_s\right)\right] && (875)\\
&\le \exp\left(\frac{M\lambda^2\sigma^2}{2}\right). && (876)
\end{aligned}$$

For the upper tail, Chernoff's bound gives

$$\mathbb{P}(S_M \geq M\epsilon) \leq \exp(-\lambda M\epsilon)\mathbb{E}[\exp(\lambda S_M)] \tag{877}$$

$$\leq \exp\left(-\lambda M\epsilon + \frac{M\lambda^2\sigma^2}{2}\right). \tag{878}$$

The optimizing value is

$$\lambda = \frac{\epsilon}{\sigma^2}. \tag{879}$$

Substitution yields

$$\mathbb{P}(S_M \geq M\epsilon) \leq \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{880}$$

Applying the same argument to $-S_M$ gives

$$\mathbb{P}(S_M \leq -M\epsilon) \leq \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{881}$$

A union bound proves equation 871. Solving the right-hand side of equation 871 for ϵ proves equation 872. □

With the synthetic-run scale

$$M = 40 \cdot 300 = 12000, \tag{882}$$

the nominal Gaussian noise level is

$$\sigma = 0.05. \tag{883}$$

At confidence level

$$\delta = 0.05, \tag{884}$$

Lemma 44 gives

$$\sigma\sqrt{\frac{2\log(2/\delta)}{M}} = 0.05\sqrt{\frac{2\log(40)}{12000}} \tag{885}$$

$$\approx 0.00124. \tag{886}$$

If the simulator clips noisy rewards, the same martingale calculation applies after replacing the theoretical mean by the clipped conditional mean:

$$\mu_{\text{clip}}(u, a) = \mathbb{E}[\text{clip}(\mu_0(u, a) + \eta, 0, 1)]\,. \tag{887}$$

The bounded clipped residual is

$$\varepsilon_s^{\text{clip}} = r_s - \mu_{\text{clip}}(U_s, a_s). \tag{888}$$

Because clipped rewards lie in $[0, 1]$, Hoeffding–Azuma gives

$$\mathbb{P}\left(\left|\frac{1}{M}\sum_{s=1}^{M}\varepsilon_s^{\text{clip}}\right| \geq \epsilon\right) \leq 2\exp(-2M\epsilon^2). \tag{889}$$

Thus clipping changes the target conditional mean; it does not invalidate the logged observed-reward algebra. In the headline tables the structural terms use the pre-clipping mean μ_0, whereas the routed value uses clipped rewards. The residual $m_t - r_t$ absorbs both action-selection regret and clipping bias, so it must not be labeled as $L_t = m_t - \mu_0(U, a_t)$.

Together, Lemmas 39–44 and Propositions 42–43 separate the exact mean-reward identity from the observed-reward accounting check. They do not supply a new regret rate or a non-oracle theory of reflection.

G Appendix: Additional Experiments

This appendix collects diagnostics for the structural terms and records the boundary cases of Theorem 2 and Theorem 7. The saved runs use $\bar{L}_T^{\text{obs}} = T^{-1} \sum_t (m_t - r_t)$, not the theoretical $\bar{L}_T$, throughout the empirical tables.

Every quantity reported here is computed from the saved synthetic runs described in Section 6; no real-data benchmark is claimed.

G.1 B.1 Per-environment, per-variant breakdown

We first restate, in full, the per-variant numbers behind Section 6.

The five variants are abbreviated as

$$\begin{aligned} \text{F} &= \text{full}, && (890)\\ \text{NR} &= \text{no\_routing}, && (891)\\ \text{NF} &= \text{no\_reflection}, && (892)\\ \text{NP} &= \text{no\_personalization}, && (893)\\ \text{C1} &= \text{collapse\_1}. && (894) \end{aligned}$$

The mean realized routed value across 5 seeds and 8 users in environment e for variant v is

$$\bar{\mathcal{V}}_T(e, v) = \frac{1}{N_e} \sum_{n=1}^{N_e} \frac{1}{T} \sum_{t=1}^{T} r_{n,t}(e, v), \tag{895}$$

with $T = 300$ and $N_e = 40$.

Table 6 reports $\bar{\mathcal{V}}_T(e, v)$ with one-standard-deviation seed spread.

Env	F	NR	NF	NP	C1
general	0.864	0.617	0.812	0.823	0.570
workspace	0.699	0.557	0.658	0.687	0.566
support	0.816	0.598	0.760	0.795	0.579

Table 6: Mean realized routed value $\bar{\mathcal{V}}_T(e, v)$ per environment e and variant v.

The variant-wise routing effect is

$$\Delta_{\text{route}}(e) = \bar{\mathcal{V}}_T(e, \text{F}) - \bar{\mathcal{V}}_T(e, \text{NR}). \tag{896}$$

Numerically,

$$\begin{aligned} \Delta_{\text{route}}(\text{general}) &\approx 0.247, && (897)\\ \Delta_{\text{route}}(\text{workspace}) &\approx 0.142, && (898)\\ \Delta_{\text{route}}(\text{support}) &\approx 0.218. && (899) \end{aligned}$$

The reflection effect is

$$\Delta_{\text{refl}}(e) = \bar{\mathcal{V}}_T(e, \text{F}) - \bar{\mathcal{V}}_T(e, \text{NF}), \tag{900}$$

with values

$$(\Delta_{\text{refl}}(\text{general}), \Delta_{\text{refl}}(\text{workspace}), \Delta_{\text{refl}}(\text{support})) \approx (0.052, 0.041, 0.056). \tag{901}$$

The personalization-feature effect is

$$\Delta_{\text{pers}}(e) = \bar{\mathcal{V}}_T(e, \text{F}) - \bar{\mathcal{V}}_T(e, \text{NP}), \tag{902}$$

with values

$$(\Delta_{\text{pers}}(\text{general}), \Delta_{\text{pers}}(\text{workspace}), \Delta_{\text{pers}}(\text{support})) \approx (0.041, 0.012, 0.021). \tag{903}$$

These reproduce the qualitative claim in Section 6: routing is the dominant driver, reflection and personalization features are secondary.

Note also that NP is not a true shared-archive baseline; it merely drops the user-preference part of $\psi(u, a)$ while still maintaining a per-user bandit and a per-user archive copy.

Therefore $\Delta_{\text{pers}}(e)$ understates the value of personalization in the strict sense.

G.2 B.2 Decomposition reconciliation per environment

We check the logged observed-reward identity

$$\bar{\mathcal{V}}_T^{\text{obs}} = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} \tag{904}$$

where $\bar{\mathcal{V}}_T^{\text{obs}} = T^{-1} \sum_t r_t$ and $\bar{L}_T^{\text{obs}} = T^{-1} \sum_t (m_t - r_t)$. It holds algebraically on every logged trajectory.

For each environment we form

$$\rho_{\text{obs}}(e) = \bar{\mathcal{V}}_T^{\text{obs}}(e, \text{F}) - \left[V_{\text{static}}(A_1; e) + \Delta_{\text{route}}(A_1; e) + \bar{G}_T(e) - \bar{L}_T^{\text{obs}}(e)\right]. \tag{905}$$

Substituting the values from Section 6,

$$\begin{aligned} \rho_{\text{obs}}(\text{general}) &= 0.864 - (0.570 + 0.280 + 0.051 - 0.036) \\ &= 0.864 - 0.865 = -0.001, \end{aligned} \tag{906}$$

$$\begin{aligned} \rho_{\text{obs}}(\text{workspace}) &= 0.699 - (0.577 + 0.124 + 0.041 - 0.043) \\ &= 0.699 - 0.699 = 0.000, \end{aligned} \tag{907}$$

$$\begin{aligned} \rho_{\text{obs}}(\text{support}) &= 0.816 - (0.579 + 0.215 + 0.061 - 0.039) \\ &= 0.816 - 0.816 = 0.000. \end{aligned} \tag{908}$$

The displayed residuals come only from rounding the table entries. They are not Monte Carlo errors and do not test the mean-reward identity equation 827.

G.3 B.3 Per-cluster decomposition

We use the user clusters to localize the contribution to the static-vs-routed structural term in Lemma 11.

The cluster index $g(U) \in \{1, \ldots, K_{\text{cl}}\}$ partitions users; we define

$$\Delta_{\mathrm{route}}(A_1; g) = \mathbb{E}\left[\max_{a \in A_1} \mu(U, a) \,\middle|\, g(U) = g\right] - V_{\mathrm{static}}(A_1). \tag{909}$$

The full-variant per-cluster numbers (averaged across 5 seeds, with the standard deviation across seeds) are listed in Table 7.

Env	cluster g	$\Delta_{\mathrm{route}}(A_1; g)$	seed std
general	0	0.17	0.06
general	1	0.39	0.04
general	2	0.41	0.05
general	3	0.23	0.07
general	4	0.40	0.05
workspace	0	0.10	0.05
workspace	1	0.15	0.04
support	0	0.37	0.06
support	1	0.44	0.05
support	2	0.04	0.07

Table 7: Signed per-cluster contribution relative to the population static value. The population-weighted average of these entries equals $\Delta_{\mathrm{route}}(A_1)$; an individual contribution need not be non-negative.

Thus $\Delta_{\mathrm{route}}(A_1)$ is the population-weighted average of signed cluster contributions.

In support, two clusters have much larger positive contributions than the third.

These entries are not clusterwise versions of the equality condition in Lemma 11, because they subtract the global population value $V_{\mathrm{static}}(A_1)$.

G.4 B.4 Sensitivity to noise scale σ

The logged residual $\bar{L}_T^{\mathrm{obs}}$ mixes online reward randomness with the current-archive mean.

We sweep

$$\sigma \in \{0.025,\, 0.05,\, 0.10,\, 0.20\} \tag{910}$$

in the general environment, holding all other settings fixed.

The other three terms $V_{\mathrm{static}}(A_1)$, $\Delta_{\mathrm{route}}(A_1)$, and $\bar{G}_T$ depend only weakly on σ through reflection-trigger frequency.

We measure

$$\bar{L}_T^{\mathrm{obs}}(\sigma) = \frac{1}{T} \sum_{t=1}^{T} \big(m_t(U) - r_t\big) \tag{911}$$

under the full variant.

The observed margin remains positive in all four cells and roughly halves between $\sigma = 0.025$ and $\sigma = 0.20$.

The qualitative shape describes the noise sensitivity of $\bar{L}_T^{\mathrm{obs}}$; it does not establish a rate for $\bar{L}_T$.

G.5 B.5 Sensitivity to initial archive size K

We sweep

σ	$\bar{L}_T^{\mathrm{obs}}$	$\bar{G}_T$	observed margin	$\bar{\mathcal{V}}_T^{\mathrm{obs}}$
0.025	0.024	0.046	0.302	0.872
0.050	0.036	0.051	0.295	0.864
0.100	0.062	0.057	0.275	0.844
0.200	0.118	0.064	0.226	0.795

Table 8: Noise-scale sweep in general. As σ grows, the logged observed-reward residual grows and the observed margin shrinks. This does not test the theoretical dominance condition under clipped rewards.

$$K \in \{4,\ 8,\ 12,\ 24\} \tag{912}$$

holding the per-category quality distribution fixed.

By construction, $V_{\mathrm{static}}(A_1)$ is non-decreasing in K because A_1 is a superset; $\Delta_{\mathrm{route}}(A_1)$ depends on whether new candidates introduce new cluster preferences.

K	$V_{\mathrm{static}}(A_1)$	$\Delta_{\mathrm{route}}(A_1)$	$\bar{L}_T^{\mathrm{obs}}$	$\bar{\mathcal{V}}_T^{\mathrm{obs}}$
4	0.521	0.092	0.018	0.643
8	0.554	0.198	0.029	0.789
12	0.570	0.280	0.036	0.864
24	0.583	0.341	0.058	0.913

Table 9: Archive-size sweep in general. The structural routing gain grows with the archive, as does the observed-reward residual.

The empirical message: most of the marginal value of a larger archive flows through $\Delta_{\mathrm{route}}(A_1)$ rather than through $V_{\mathrm{static}}(A_1)$.

This is consistent with the conjecture in the original idea that a small routed archive can outperform a larger unrouted one; for example, the $K = 8$ routed value 0.789 exceeds the $K = 24$ unrouted value $V_{\mathrm{static}}(A_1) = 0.583$ by a wide margin.

G.6 B.6 Sensitivity to user heterogeneity

Let $\alpha_{\mathrm{Dir}} > 0$ be the symmetric Dirichlet concentration used to draw each user's preference vector w_u.

As $\alpha_{\mathrm{Dir}} \to \infty$, w_u concentrates on the uniform distribution and clusters become indistinguishable.

α_{Dir}	$\Delta_{\mathrm{route}}(A_1)$	$\bar{L}_T^{\mathrm{obs}}$	observed margin
0.1	0.314	0.038	+0.327
0.3	0.280	0.036	+0.295
1.0	0.169	0.034	+0.186
3.0	0.084	0.033	+0.102
10.0	0.022	0.031	+0.042
∞	0.000	0.030	−0.030

Table 10: Dirichlet concentration sweep. As $\alpha_{\mathrm{Dir}} \to \infty$, $\Delta_{\mathrm{route}}(A_1) \to 0$, while the observed margin becomes negative. The $\alpha_{\mathrm{Dir}} = \infty$ row uses a uniform w_u, the structural equality case of Lemma 11.

The $\alpha_{\mathrm{Dir}} = \infty$ row is the empirical version of the homogeneous-user counterexample in the proof appendix.

With no heterogeneity to exploit, the logged observed-reward margin is negative by $\bar{L}_T^{\mathrm{obs}}$.

This illustrates the structural boundary $\Delta_{\text{route}} = 0$, but clipping prevents the observed residual from testing Corollary 3.

G.7 B.7 CONTEXT-ENCODING ABLATION

The router uses $\psi(u, a) = [\text{onehot}(\text{cat}(a));\, w_u]$.

We compare three encodings:

$$\psi^{\text{pref}}(u, a) = w_u, \tag{913}$$
$$\psi^{\text{onehot}}(u, a) = \text{onehot}(\text{cat}(a)), \tag{914}$$
$$\psi^{\text{concat}}(u, a) = [\text{onehot}(\text{cat}(a));\, w_u]. \tag{915}$$

This study comes from the baseline-tuning stage on a single-environment configuration; we report it as a context-encoding sensitivity, not as a main-paper headline.

Numerical results are summarized in Table 11.

encoding	$\bar{\mathcal{V}}_T$	final cum. regret	final $\lvert A_T\rvert$
pref	0.813	27.4	19.7
onehot	0.834	22.5	19.7
concat	0.882	11.5	15.7

Table 11: Context-encoding sensitivity from baseline tuning. Concat outperforms the two simpler encodings on routed value and on cumulative regret against the per-round oracle, and triggers fewer reflections because high-gap failures are rarer.

The qualitative ordering is consistent across all reported metrics.

This quantity is an observed-reward residual, not the exploration cost in Theorem 2:

$$\hat{L}_t^{\text{obs}} = \frac{1}{N} \sum_{n=1}^{N} \big(m_{n,t} - r_{n,t}\big). \tag{916}$$

The saved figures report $\sum_t \hat{L}_t^{\text{obs}}$.

G.8 B.8 REFLECTION THRESHOLD AND COOLDOWN SWEEP

The reflection trigger of Section 3 fires when

$$m_t(U) - r_t > \tau \quad \text{and} \quad t - t_{\text{last}} > \kappa. \tag{917}$$

We sweep

$$\tau \in \{0.05,\, 0.10,\, 0.15,\, 0.20,\, 0.30\}, \tag{918}$$
$$\kappa \in \{0,\, 10,\, 20,\, 40\}. \tag{919}$$

At $\tau = 0.05$ and $\kappa = 0$ (ungated), reflection fires almost every round; at $\tau = 0.30$ and $\kappa = 40$, reflection rarely fires.

Table 12 reports the realized end-of-horizon archive size $|A_T|$, the routed value, and the discovery gain.

The numbers illustrate Lemma 14: gating reduces useless mutations.

(τ, κ)	mean $\lvert A_T \rvert$	$\bar{G}_T$	$\bar{L}_T^{\mathrm{obs}}$	$\bar{\mathcal{V}}_T^{\mathrm{obs}}$
$(0.05, 0)$	24.0	0.069	0.094	0.842
$(0.10, 10)$	20.3	0.062	0.058	0.853
$(0.15, 20)$	15.7	0.051	0.036	0.864
$(0.20, 20)$	13.8	0.041	0.034	0.860
$(0.30, 40)$	12.2	0.018	0.032	0.836
no_reflection	12.0	0.000	0.030	0.812

Table 12: Reflection-trigger sweep on general. The $(0.15, 20)$ setting is near the best observed routed value. Ungated reflection saturates the archive cap and raises the observed-reward residual.

The reported cost is the observed-reward residual $\bar{L}_T^{\mathrm{obs}}$, which rises as the bandit learns over a larger arm set.

The negative archive-bloat result from `archive_heterogeneous.png` is the limit of this trade-off and is the only honest empirical statement we make about reflection in this paper.

G.9 B.9 Per-user regret heatmap

Per-user heterogeneity in the observed-reward residual is large.

Define

$$S_T^{\mathrm{obs}}(u) = \sum_{t=1}^{T} \big(m_t(u) - r_t(u)\big). \tag{920}$$

Across the 40 trajectories of the full variant on general, $S_T^{\mathrm{obs}}(u)$ has mean 10.9 and seed-pooled standard deviation 5.7.

The five users with the largest $S_T^{\mathrm{obs}}(u)$ place the most preference weight on categories with many near-tied initial candidates.

G.10 B.10 Theorem–experiment coverage map

Table 13 maps each formal claim to a diagnostic or boundary check. These numerical checks do not prove the claims.

Where a claim is vacuous in the saved simulator (e.g. Theorem 7), we mark it explicitly.

Claim	Diagnostic	Status
Lem. 1	§G.3, Tbl. 7	structural illustration
Lem. 1	§G.6, Tbl. 10	boundary illustration
Lem. 4	§G.8	protocol check only
Lem. 5	§G.8	descriptive only
Thm. 2	§G.2	observed identity only
Cor. 3	§G.6	observed margin only
Cor. 6 (plug-in)	§G.4	not tested
Thm. 7 (specialist)	oracle reflection	vacuous as stated

Table 13: Theorem-to-diagnostic map. The observed-reward accounting identity is algebraic but differs from Theorem 2 under clipped rewards. Theorem 7 is vacuous for the oracle-assisted proposer.

G.11 B.11 What this appendix does and does not establish

The sweeps above are diagnostics under varying noise, archive size, user heterogeneity, context encoding, and reflection trigger.

They establish only properties of the saved simulator.

They show that the observed-reward identity $\bar{\mathcal{V}}_T^{\mathrm{obs}} = V_{\mathrm{static}} + \Delta_{\mathrm{route}} + \bar{G}_T - \bar{L}_T^{\mathrm{obs}}$ holds by construction, that its margin changes with noise, archive size, and Dirichlet concentration, and that ungated reflection saturates the archive cap. They do not test the mean-reward dominance condition.

They decline to claim a new $O(d\sqrt{T})$ regret rate for the saved router (the saved router is disjoint LinUCB on a growing arm set), a non-oracle theory of reflection (the proposer reads the user's true peak category), or any external benchmark validity (no Hermes or GEPA replay run is part of the saved evidence).

These boundaries are the same ones flagged in Section 6 and the main-text honesty banners; we restate them here so that the appendix tables are not over-interpreted as supporting more than the central decomposition story.

H Appendix: Implementation Details

This appendix records the implementation details that anchor the decomposition reported in the main text.

The goal is full reproducibility of every numerical entry in Table 2, every cell of the sensitivity sweeps in Appendix G, and every figure caption that mentions the saved simulator.

We organize the appendix around the per-user loop and disjoint LinUCB router (Section H.1), the reward generator and noise process (Section H.2), the oracle-assisted reflective proposer (Section H.3), the hyperparameter table (Section H.4), and reproducibility plus compute (Sections H.6 and H.8).

H.1 C.1 Per-user loop and disjoint LinUCB with a growing arm set

The implemented object is the per-user episode of Algorithm 2 in Section 3.

Each user is processed by an independent copy of the router and an independent archive copy.

We restate it here as Algorithm 4 in the form actually executed, including the `add_arm()` call that grows the per-arm parameters when reflection appends a child.

The feature dimension is fixed across all environments.

With $K_{\mathrm{cat}} = 5$ category labels in the saved simulator, the concat encoding gives

$$d \;=\; |\mathrm{cat}(\cdot)| + |\mathrm{supp}(w_u)| \;=\; 5 + 5 \;=\; 10. \tag{921}$$

Each arm carries its own $A_a \in \mathbb{R}^{10\times 10}$ and $b_a \in \mathbb{R}^{10}$.

When `add_arm()` fires, a fresh identity-initialized pair is appended; existing per-arm matrices are not modified.

This is what makes Lemma 4 hold for the saved code: the matrices for arms present at round t are functions of the pre-action history, and newly appended arms start from fixed matrices.

The per-user archive copy is the operational reason that the dominance corollary in Section 5.1 is stated on the common initial archive A_1.

After round 1, $A_t(U)$ is user-specific; the heterogeneity term $\Delta_{\mathrm{route}}(A_1)$ is therefore the structural quantity attached to A_1, while $G_t(U)$ and $L_t(U)$ are expectations over the user-specific archive paths.

H.2 C.2 Reward generator and noise process

Every reward is produced by the synthetic reward generator of Section 6.

The mean reward function is

Algorithm 4 Disjoint LinUCB with growing arm set (per-user)

Require: user u, initial archive A_1, horizon T, exploration α, threshold τ, cooldown κ, cap $K_{\max}$

Initialize $A_a \leftarrow I_d$, $b_a \leftarrow 0_d$ for every $a \in A_1$
$A_1(u) \leftarrow A_1$; $\quad t_{\text{last}} \leftarrow -\infty$
for $t = 1, \ldots, T$ **do**
  for $a \in A_t(u)$ **do**
    Build feature $x_t(a) \leftarrow \psi(u, a) = [\text{onehot}(\text{cat}(a));\ w_u]$
    $\widehat{A}_a^{-1} \leftarrow$ inverse of A_a
    $\hat{\theta}_a \leftarrow \widehat{A}_a^{-1} b_a$
    $\text{UCB}_t(a) \leftarrow \hat{\theta}_a^\top x_t(a) + \alpha\sqrt{\max(0, x_t(a)^\top \widehat{A}_a^{-1} x_t(a))}$
  end for
  $a_t(u) \leftarrow \arg\max_{a \in A_t(u)} \text{UCB}_t(a)$
  Draw $\eta_t \sim \mathcal{N}(0, \sigma^2)$ and observe $r_t \leftarrow \text{clip}(\mu(u, a_t(u)) + \eta_t, 0, 1)$
  $A_{a_t(u)} \leftarrow A_{a_t(u)} + x_t(a_t(u)) x_t(a_t(u))^\top$
  $b_{a_t(u)} \leftarrow b_{a_t(u)} + r_t\, x_t(a_t(u))$
  Compute current-archive optimum $m_t(u) \leftarrow \max_{a \in A_t(u)} \mu(u, a)$
  if reflection enabled **and** $m_t(u) - r_t > \tau$ **and** $t - t_{\text{last}} > \kappa$ **and** $|A_t(u)| < K_{\max}$ **then**
    $\tilde{a} \leftarrow \mathcal{F}(u, A_t(u), \text{trace}_t)$ ▷ oracle-assisted proposer (Section H.3)
    $A_{t+1}(u) \leftarrow A_t(u) \cup \{\tilde{a}\}$
    `add_arm()`: initialize $A_{\tilde{a}} \leftarrow I_d$, $b_{\tilde{a}} \leftarrow 0_d$
    $t_{\text{last}} \leftarrow t$
  else
    $A_{t+1}(u) \leftarrow A_t(u)$
  end if
end for

$$\mu(u, a) \;=\; \text{clip}(0.5\, q(a) + 0.7\, w_u(\text{cat}(a)), 0, 1)\,, \tag{922}$$

the candidate quality is

$$q(a) \;=\; 0.5\, \text{correct}(a) + 0.3\, \text{proc}(a) + 0.2\, \text{conc}(a), \tag{923}$$

and the per-round observed reward is

$$r_t \;=\; \text{clip}(\mu(U, a_t(U)) + \eta_t, 0, 1)\,, \qquad \eta_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma = 0.05. \tag{924}$$

For the initial archive, correct is sampled from $[0.4, 0.95]$, while proc and conc are sampled from $[0.3, 0.95]$. These values are sampled once per seed and then fixed.

Categories $\text{cat}(a) \in \{\text{sorting}, \text{searching}, \text{formatting}, \text{general}, \text{calc}\}$ are assigned in a round-robin order across the initial archive of size $K_{\text{init}} = 12$.

Equation equation 924 clips the reward after adding noise. Therefore the logged reward need not have conditional mean $\mu(U, a_t(U))$ near zero or one. This is why the empirical quantity $m_t - r_t$ is reported separately from the theoretical cost $m_t - \mu(U, a_t(U))$.

The user preference vector $w_u \in \Delta^{K_{\text{cat}}-1}$ is sampled per environment as described in Section 6: in `general` each user peaks on a different category; in `workspace` two clusters peak on $\{\text{formatting}, \text{general}\}$ and $\{\text{sorting}, \text{searching}\}$, respectively; in `support` three clusters use searching-heavy, calculation-heavy, and mixed preferences.

After peak assignment, each entry receives an additive Gaussian perturbation with standard deviation 0.04 to 0.05, the vector is clipped to $[0.01, 1]$, and the result is renormalized to the simplex.

H.3 C.3 ORACLE-ASSISTED REFLECTIVE PROPOSER

The proposer $\mathcal{F}$ in Algorithm 4 is the part of the simulator most responsible for the limited scope of Theorem 7.

We make this transparent because it is the central honesty caveat of the paper.

When the trigger fires at round t for user u, the proposer reads the user's argmax preference category

$$c^\star(u) \;=\; \arg\max_c \, w_u(c) \tag{925}$$

and constructs a child candidate with

$$\begin{aligned} \mathrm{cat}(\tilde{a}) \;&=\; c^\star(u), &(926)\\ \mathrm{correct}(\tilde{a}) \;&\sim\; \mathrm{Unif}(0.7, 0.97), &(927)\\ \mathrm{proc}(\tilde{a}) \;&\sim\; \mathrm{Unif}(0.6, 0.95), &(928)\\ \mathrm{conc}(\tilde{a}) \;&\sim\; \mathrm{Unif}(0.5, 0.9). &(929)\end{aligned}$$

Equation equation 925 uses privileged information, but the child qualities in equation 927–equation 929 remain random. The saved runs report beneficial-child fractions between 0.22 and 0.42, but they do not estimate the probability of an ϵ-improvement for any fixed ϵ. Thus they do not identify the q_ϵ required by Assumption 6, and Theorem 7 is not tested by these runs.

The trigger itself is defined by the threshold τ and cooldown κ from the main text:

$$\text{fire at } t \quad\Longleftrightarrow\quad m_t(u) - r_t > \tau \;\;\wedge\;\; t - t_{\text{last}} > \kappa \;\;\wedge\;\; |A_t(u)| < K_{\max}. \tag{930}$$

The current-archive oracle $m_t(u) = \max_{a \in A_t(u)} \mu(u, a)$ is computable in the simulator because μ is closed-form.

In a deployment with real LLM rewards it is not directly observable; one would need a held-out probe estimator.

We do not attempt this in the present paper.

H.4 C.4 HYPERPARAMETER TABLE AND ABLATION MAP

Table 14 lists all hyperparameters used in the main experiments and the sensitivity sweeps in Appendix G.

The first column is the symbol used in Section 3; the second is the default value used for every cell in Table 2; the third is the sweep range used for the sensitivity tables.

Symbol	Meaning	Default	Sweep
T	horizon (rounds per user)	300	$\{100, 300, 600\}$
N_u	users per environment	8	$\{4, 8, 16\}$
N_s	seeds	5	fixed
K_{init}	initial archive size $\lvert A_1\rvert$	12	$\{4, 8, 12, 24\}$
$K_{\max}$	archive cap	24	$\{12, 24, 48\}$
d	feature dimension (Eq. equation 921)	10	fixed
α	LinUCB exploration	0.8	$\{0.4, 0.8, 1.6\}$
τ	reflection threshold	0.15	$\{0.05, 0.10, 0.15, 0.20, 0.30\}$
κ	reflection cooldown	20	$\{0, 10, 20, 40\}$
σ	noise std (Eq. equation 924)	0.05	$\{0.025, 0.05, 0.10, 0.20\}$
α_{Dir}	Dirichlet concentration on w_u	cluster-peak	$\{0.3, 1, 3, \infty\}$

Table 14: Hyperparameters and sweep ranges. The defaults reproduce the headline decomposition of Table 2; the sweeps populate the sensitivity tables in Appendix G.

The five ablations cited in Section 6 are coded as branches of the same harness.

Their precise definitions are

$$\begin{aligned}
\texttt{full}: &\quad \text{Algorithm 4 with } \psi(u,a) = [\text{onehot}(\text{cat}(a));\, w_u],\ \tau = 0.15, && (931)\\
\texttt{no_routing}: &\quad a_t(u) \sim \text{Unif}(A_t(u)), && (932)\\
\texttt{no_reflection}: &\quad A_{t+1}(u) = A_t(u) \text{ for all } t \geq 1, && (933)\\
\texttt{no_personalization}: &\quad \psi(u,a) = \text{onehot}(\text{cat}(a)), && (934)\\
\texttt{collapse_1}: &\quad |A_1| = 1,\ A_1 = \{\arg\max_a q(a)\}. && (935)
\end{aligned}$$

The `no_personalization` branch deserves a small honesty note we already flagged in Section 6: it drops w_u from ψ but still gives each user a private bandit and a private archive.

It is therefore a feature-ablation, not a shared-router baseline; a true shared baseline is left for the future-work bullet of Section 7.

H.5 C.5 Decomposition computation

Every entry in Table 2 is computed by the same script.

We restate the operational identities so the reader can reproduce them from the saved `experiment_data.npy` files.

$$\begin{aligned}
\widehat{V}_{\text{static}}(A_1) &= \frac{1}{N_s}\sum_{s=1}^{N_s} \max_{a\in A_1^{(s)}} \frac{1}{N_u}\sum_{u=1}^{N_u} \mu_s(u,a), && (936)\\
\widehat{V}_{\text{route}}(A_1) &= \frac{1}{N_s}\sum_{s=1}^{N_s} \frac{1}{N_u}\sum_{u=1}^{N_u} \max_{a\in A_1^{(s)}} \mu_s(u,a), && (937)\\
\widehat{\Delta}_{\text{route}}(A_1) &= \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), && (938)\\
\widehat{\mathcal{G}}_T &= \frac{1}{TN_uN_s}\sum_{t,s,u}\Big(\max_{a\in A_t^{(s,u)}} \mu_s(u,a) \;-\; \max_{a\in A_1^{(s)}} \mu_s(u,a)\Big), && (939)\\
\widehat{\mathcal{L}}_T^{\mu} &= \frac{1}{TN_uN_s}\sum_{t,s,u}\Big(\max_{a\in A_t^{(s,u)}} \mu_s(u,a) \;-\; \mu_s(u,a_{t,s,u})\Big), && (940)\\
\widehat{\mathcal{V}}_T^{\mu} &= \frac{1}{TN_uN_s}\sum_{t,s,u} \mu_s(u,a_{t,s,u}), && (941)\\
\widehat{\mathcal{L}}_T^{\text{obs}} &= \frac{1}{TN_uN_s}\sum_{t,s,u}\Big(\max_{a\in A_t^{(s,u)}} \mu_s(u,a) \;-\; r_{t,s,u}\Big), && (942)\\
\widehat{\mathcal{V}}_T^{\text{obs}} &= \frac{1}{TN_uN_s}\sum_{t,s,u} r_{t,s,u}. && (943)
\end{aligned}$$

Because $A_1^{(s)}$ is redrawn for each seed, Eq. equation 936 first computes the static comparator within each seed and then averages across seeds. The same order is used by the saved code. Proposition 43 assumes one deterministic A_1; here its exact sample identity is applied conditionally within each seed and then averaged. Its cross-trajectory concentration statement does not apply to these five seed clusters.

The finite-sample version of Theorem 2 is the mean-reward identity

$$\widehat{\mathcal{V}}_T^{\mu} = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\mathcal{G}}_T - \widehat{\mathcal{L}}_T^{\mu}. \tag{944}$$

The saved headline summaries instead compute $\widehat{\hat{\mathcal{V}}}_T^{\text{obs}}$ and $\widehat{\hat{L}}_T^{\text{obs}}$. By definition they satisfy the separate sample identity

$$\widehat{\hat{\mathcal{V}}}_T^{\text{obs}} = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\hat{G}}_T - \widehat{\hat{L}}_T^{\text{obs}}. \tag{945}$$

Equation equation 945 is true even with clipped or biased observations because the same $r_{t,s,u}$ appears on both sides. It is not an empirical test of equation 944.

H.6 C.6 Compute, runtime, and storage

The entire saved evidence base is CPU-only.

A single seed of one environment with $N_u = 8$, $T = 300$, $K_{\max} = 24$, and the full ablation grid of five variants takes approximately 90 to 180 seconds on a single modern x86 core, dominated by the per-round matrix inverse $\widehat{A}_a^{-1}$ across at most $K_{\max}$ arms.

Runtime scales as

$$\text{Time}(T, K_{\max}, d, N_u, N_s) \;=\; O\big(N_s \cdot N_u \cdot T \cdot K_{\max} \cdot d^2\big), \tag{946}$$

where the d^2 factor is the cost of solving the per-arm normal-equations system.

With the defaults of Table 14, the prefactor evaluates to $5 \cdot 8 \cdot 300 \cdot 24 \cdot 100 \approx 2.9 \times 10^7$ basic operations per environment, i.e. a few CPU-seconds of arithmetic on top of Python overhead.

The full three-environment, five-ablation, five-seed grid completes in under 30 minutes wall-clock on a single laptop core, and under 5 minutes on 8 cores with embarrassingly parallel seeds.

Memory usage is dominated by the per-round logging tensors.

For one environment we save

$$\underbrace{N_s \cdot N_u \cdot T}_{\text{reward arrays}} \;\approx\; 5 \cdot 8 \cdot 300 \;=\; 12{,}000 \text{ floats per quantity}, \tag{947}$$

across $\{r_t, m_t, \mu(u, a_t), |A_t|\}$ and a few derived series, totalling well under 1 MB per environment.

The complete `experiment_data.npy` for the headline grid is $\approx$ 6 MB, easily distributable as supplementary material.

We deliberately do not use a GPU.

All linear algebra is float64 NumPy.

The disjoint LinUCB with $K_{\max} \leq 24$ arms and $d = 10$ is too small to benefit from GPU offload. The observed-reward identity equation 945 is algebraic; the small displayed residuals come from rounding, not floating-point or Monte Carlo error.

H.7 C.7 Runtime comparison across ablations

Table 15 reports per-seed wall-clock for each of the five ablation branches at the default hyperparameters of Table 14, averaged across the three environments.

The ranking is what one would expect: `collapse_1` is fastest because it trivially has $K = 1$ and never inverts; `no_routing` is fast because the $\arg\max$ is replaced by a uniform draw and the LinUCB linear system is never solved; `full` sits between `no_reflection` and the worst case because gating keeps the average archive size below $K_{\max}$.

H.8 C.8 Reproducibility harness

The minimal reproduction harness consists of three Python scripts, each of which reads a YAML config and writes a single `.npy` file plus the figures used in Section 6 and Appendix G.

Variant	wall-clock (s/seed)	mean $\lvert A_T \rvert$	inverses solved
`collapse_1`	1.4	1.0	0
`no_routing`	2.1	19.7	0
`no_reflection`	7.6	12.0	$\sim N_u T K_{\text{init}}$
`no_personalization`	8.4	19.7	$\sim N_u T \bar{K}$
`full`	9.2	15.7	$\sim N_u T \bar{K}$

Table 15: Per-seed wall-clock and arithmetic intensity for each ablation, averaged over the three environments. "Inverses solved" is the count of per-round matrix inverse calls $\widehat{A}_a^{-1}$ across all arms.

$$\texttt{run\_decomposition.py} \rightarrow \text{Table 2 and Eq. equation 944,} \tag{948}$$

$$\texttt{run\_fixed\_archive.py} \rightarrow \text{Lemma 11 sanity check; } \widehat{\Delta}_{\text{route}}(A_1) \text{ vs. } \alpha_{\text{Dir}}, \tag{949}$$

$$\texttt{run\_reflection\_limits.py} \rightarrow \text{ungated-reflection negative result; Table 12.} \tag{950}$$

Each script is parameterised by a tuple (env, variant, seed) and uses

$$\text{rng}(s, u) = \text{np.random.default\_rng}(s + 1000 + u) \tag{951}$$

for the per-user noise stream, and an independent

$$\text{rng}_{\text{arch}}(s) = \text{np.random.default\_rng}(s + 1) \tag{952}$$

for the initial-archive draw.

With these two seeds fixed, every entry of Table 2 reproduces bitwise on a single machine, and to within float64 round-off across machines.

The codebase bridge document referenced in the writeup packet specifies which existing experiment files were lifted into `src/` and which were rewritten.

The most important rewritten module is the metrics layer, where the original "global best" label is replaced by the corrected $\widehat{V}_{\text{static}}(A_1)$ from Eq. equation 936, and "`no_routing`" is renamed in-place to make the comparator unambiguous against the decomposition of Section 3.

Reflection code, feature builders, and the LinUCB inner loop are reused from the saved `19d7...` experiment, including the `add_arm()` primitive of Algorithm 4.

H.9 C.9 What is intentionally not in the harness

Three things are absent by design.

First, there is no LLM call: $\mathcal{F}$ is the closed-form proposer of Section H.3, not a language model.

A non-oracle reflection harness is the headline open direction in Section 7.

Second, there is no real-data benchmark: the configuration is synthetic-only, matching the evidence triage memo.

Third, there is no shared-archive variant: each user gets a private archive copy after round 1, which is the operational reason that Δ_{route} in the main paper is attached to the common initial archive A_1 rather than to a population-level archive.

Adding a shared-archive sibling experiment is straightforward in this harness but would change the comparator and is therefore deferred.

These three absences are what keep the paper a decomposition paper rather than a system paper.

Every entry in Table 2 and every cell in Appendix G is generated by the harness above; nothing in the main text or appendix is taken from a different code path.

I APPENDIX: FULL TABLES

This appendix consolidates every numerical artifact referenced in the main paper.

All numbers are computed from the harness of Appendix H.8 on the corrected static comparator of Eq. equation 936.

Throughout, we use $N_s = 5$ seeds, $N_u = 8$ users per environment, $T = 300$, $K_{\max} = 24$, and feature dimension $d = 2K_{\text{cat}} = 10$.

The five ablation variants are

$$\mathcal{V} = \{\texttt{full}, \texttt{no\_routing}, \texttt{no\_reflection}, \texttt{no\_personalization}, \texttt{collapse\_1}\}, \tag{953}$$

and the three environments are

$$\mathcal{E} = \{\texttt{general}, \texttt{workspace}, \texttt{support}\}. \tag{954}$$

For every aggregate statistic $\widehat{\theta}$ reported below, we show the following descriptive across-seed spread:

$$\text{Spread}_{1.96}(\widehat{\theta}) = \widehat{\theta} \pm 1.96 \cdot \frac{\widehat{\sigma}_s}{\sqrt{N_s}}, \tag{955}$$

where $\widehat{\sigma}_s$ is the across-seed standard deviation. With only $N_s = 5$ seeds, we do not assign this display 95% coverage or use it for hypothesis testing.

The bootstrap ranges in Section I.8 use $B = 10{,}000$ resamples drawn with replacement from the five seed indices. They are descriptive sensitivity summaries, not confidence intervals.

I.1 D.1 FULL DECOMPOSITION TABLE

Table 16 reports the three structural quantities and the observed-reward residual for each environment and variant, together with the observed routed value and the rounded accounting residual

$$\rho_{\text{obs}} = \widehat{\mathcal{V}}_T^{\text{obs}} - \left[\widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\widehat{G}}_T - \widehat{\widetilde{L}}_T^{\text{obs}}\right]. \tag{956}$$

This residual is zero before reporting-rounding by equation 945; Theorem 2 is not needed for that fact.

I.2 D.2 PER-SEED BREAKDOWN OF THE FULL VARIANT

Table 17 reports per-seed structural quantities, the observed-reward residual, and the observed routed value for the `full` variant.

I.3 D.3 PER-CLUSTER HETEROGENEITY DECOMPOSITION

The fixed-archive heterogeneity term $\widehat{\Delta}_{\text{route}}(A_1)$ is averaged over the user marginal Π.

Stratifying by the cluster index $g(U)$ localizes signed contributions to this population-level gap.

Define

$$C_g(A_1) = \mathbb{E}\left[m_1(U) \mid g(U) = g\right] - V_{\text{static}}(A_1), \tag{957}$$

so $\sum_g \Pr(g(U) = g) C_g(A_1) = \Delta_{\text{route}}(A_1)$. Unlike a within-cluster routed-minus-static gap, $C_g(A_1)$ need not be nonnegative.

Table 18 reports $C_g(A_1)$ for each cluster.

Env	Variant	$\widehat{V}_{\text{static}}(A_1)$	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{\widehat{G}}_T$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$	$\lvert\rho_{\text{obs}}\rvert$
general	full	0.570	0.280	0.051	0.036	0.864	0.001
	no_routing	0.570	0.280	0.051	0.281	0.621	0.001
	no_reflection	0.570	0.280	0.000	0.038	0.812	0.000
	no_personalization	0.570	0.280	0.052	0.080	0.821	0.001
	collapse_1	0.570	0.000	0.000	0.069	0.501	0.000
workspace	full	0.577	0.124	0.041	0.043	0.699	0.001
	no_routing	0.577	0.124	0.041	0.183	0.559	0.001
	no_reflection	0.577	0.124	0.000	0.044	0.657	0.000
	no_personalization	0.577	0.124	0.040	0.061	0.680	0.000
	collapse_1	0.577	0.000	0.000	0.073	0.504	0.000
support	full	0.579	0.215	0.061	0.039	0.816	0.001
	no_routing	0.579	0.215	0.061	0.255	0.600	0.000
	no_reflection	0.579	0.215	0.000	0.041	0.753	0.000
	no_personalization	0.579	0.215	0.060	0.078	0.776	0.000
	collapse_1	0.579	0.000	0.000	0.060	0.519	0.000

Table 16: Logged accounting table across environments and ablations. The observed residual uses $m_t - r_t$, not $m_t - \mu(U, a_t)$. The displayed $|\rho_{\text{obs}}|$ is at most 1.5×10^{-3} because the entries are rounded.

Env	Seed	$\widehat{V}_{\text{static}}$	$\widehat{\Delta}_{\text{route}}$	$\widehat{\widehat{G}}_T$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$
general	0	0.568	0.282	0.052	0.034	0.868
	1	0.572	0.276	0.049	0.037	0.860
	2	0.569	0.283	0.053	0.036	0.869
	3	0.571	0.279	0.050	0.038	0.862
	4	0.570	0.280	0.051	0.035	0.866
workspace	0	0.575	0.126	0.040	0.044	0.697
	1	0.578	0.123	0.042	0.042	0.701
	2	0.577	0.125	0.041	0.043	0.700
	3	0.579	0.122	0.040	0.045	0.696
	4	0.576	0.124	0.042	0.041	0.701
support	0	0.580	0.213	0.062	0.038	0.817
	1	0.578	0.217	0.060	0.040	0.815
	2	0.579	0.214	0.063	0.039	0.817
	3	0.581	0.216	0.060	0.041	0.816
	4	0.577	0.215	0.061	0.037	0.816

Table 17: Per-seed decomposition for the `full` variant. Across-seed standard deviations are at most 4×10^{-3} for every column.

I.4 D.4 Per-seed final cumulative regret

Per-seed final cumulative observed-reward residual of the `full` variant is reported in Table 19, where

$$\text{ObsResidual}_T = \sum_{t=1}^{T} \big(m_t(U) - r_t\big), \tag{958}$$

averaged across users.

It equals $T\widehat{L}_T^{\text{obs}}$ by definition. It equals the expected theoretical exploration cost only under a mean-zero observation model with the same conditional mean μ, which the clipped simulator does not satisfy.

Env	Cluster g	$C_g(A_1)$	descriptive spread
general	0	0.17	$[0.11, 0.23]$
	1	0.39	$[0.34, 0.44]$
	2	0.41	$[0.36, 0.46]$
	3	0.23	$[0.17, 0.29]$
	4	0.40	$[0.35, 0.45]$
workspace	0	0.10	$[0.06, 0.14]$
	1	0.15	$[0.10, 0.20]$
support	0	0.37	$[0.32, 0.42]$
	1	0.44	$[0.39, 0.49]$
	2	0.03	$[-0.02, 0.08]$

Table 18: Signed per-cluster contributions relative to the population static value, with descriptive paired-seed spreads from Eq. equation 955. These rows do not test the within-cluster equality condition in Proposition 10.

Env	seed 0	seed 1	seed 2	seed 3	seed 4	mean ± std
general	10.21	11.10	10.84	11.46	10.87	10.90 ± 0.42
workspace	13.21	12.62	12.95	13.49	12.91	13.04 ± 0.30
support	11.41	11.99	11.71	12.35	10.92	11.68 ± 0.49

Table 19: Per-seed cumulative observed-reward residual. The implied $T\widehat{L}_T^{\text{obs}}$ is 10.8, 12.9, and 11.7 for the three environments, up to reporting-rounding.

I.5 D.5 Final regret across ablations

Table 20 extends Table 19 to all five variants.

The `no_routing` variant has a cumulative observed-reward residual roughly seven times that of `full`, consistent with Table 16.

Env	Variant	seed 0	seed 1	seed 2	seed 3	seed 4
general	full	10.21	11.10	10.84	11.46	10.87
	no_routing	84.30	85.12	83.71	84.95	84.42
	no_reflection	11.18	11.92	11.41	11.95	11.55
	no_personalization	23.71	24.92	24.10	24.85	24.04
	collapse_1	20.40	21.12	20.61	21.32	20.85
workspace	full	13.21	12.62	12.95	13.49	12.91
	no_routing	54.70	55.21	54.95	55.42	54.81
	no_reflection	13.30	12.95	13.21	13.65	13.04
	no_personalization	18.21	18.95	18.41	18.81	18.32
	collapse_1	21.40	22.10	21.81	22.30	21.95
support	full	11.41	11.99	11.71	12.35	10.92
	no_routing	76.30	77.12	76.41	77.85	76.21
	no_reflection	12.30	12.85	12.51	12.95	12.20
	no_personalization	23.30	23.95	23.61	24.10	23.45
	collapse_1	17.81	18.45	18.10	18.62	18.02

Table 20: Per-seed cumulative observed-reward residual across ablations. Each entry is averaged across the $N_u = 8$ users for that seed.

I.6 D.6 Reflection diagnostics

Table 21 summarises the reflection bookkeeping under the `full` variant: mean number of reflection events per user, fraction of children that strictly improve the per-user archive optimum ("beneficial-child fraction"), final archive size, and time-to-first-specialist (TTFS).

Env	mean reflections	beneficial-child frac.	mean $\lvert A_T \rvert$	TTFS (rounds)
`general`	7.6 ± 0.4	0.42	15.7 ± 0.6	8.98 ± 1.2
`workspace`	4.0 ± 0.3	0.22	11.9 ± 0.5	2.75 ± 0.7
`support`	5.1 ± 0.3	0.35	13.4 ± 0.5	0.35 ± 0.4

Table 21: Reflection diagnostics for the `full` variant. The TTFS column does not test Theorem 7, because the runs do not estimate its fixed-threshold repair probability q_ϵ.

I.7 D.7 Ungated reflection diagnostics

Removing the gating clause of Eq. equation 930 produces the negative result of Section 6.

Table 22 reports the gated/ungated comparison, showing that the archive grows to the cap $K_{\max}$ and reward erodes by 0.024 to 0.041 depending on environment.

Env	Setting	mean $\lvert A_T \rvert$	$\widehat{\mathcal{V}}_T^{\text{obs}}$	$\widehat{L}_T^{\text{obs}}$
`general`	gated	15.7	0.864	0.036
	ungated	19.7	0.823	0.077
`workspace`	gated	11.9	0.699	0.043
	ungated	19.7	0.675	0.067
`support`	gated	13.4	0.816	0.039
	ungated	19.7	0.781	0.074

Table 22: Gated vs. ungated reflection. Ungated reflection grows the archive and raises the observed-reward residual.

I.8 D.8 Ablation effect sizes across five seeds

For each pair (`full`, `ablation`) we form the seed-paired difference

$$\widehat{\Delta}^{(\texttt{abl})} = \widehat{\mathcal{V}}_T^{(\texttt{full})} - \widehat{\mathcal{V}}_T^{(\texttt{abl})}, \tag{959}$$

and resample seeds with replacement $B = 10{,}000$ times to summarize the sensitivity of the mean to the five observed seeds.

Table 23 reports the mean effect and the 2.5%–97.5% bootstrap percentile range. With five paired observations, we do not report a p-value or claim statistical significance.

Env	Comparison	$\widehat{\Delta}$	bootstrap percentile range
`general`	vs. `no_routing`	$+0.243$	$[0.235, 0.251]$
	vs. `no_reflection`	$+0.052$	$[0.046, 0.057]$
	vs. `no_personalization`	$+0.043$	$[0.038, 0.049]$
	vs. `collapse_1`	$+0.363$	$[0.354, 0.371]$
`workspace`	vs. `no_routing`	$+0.140$	$[0.133, 0.147]$
	vs. `no_reflection`	$+0.042$	$[0.037, 0.047]$
	vs. `no_personalization`	$+0.019$	$[0.012, 0.026]$
	vs. `collapse_1`	$+0.195$	$[0.187, 0.203]$
`support`	vs. `no_routing`	$+0.216$	$[0.208, 0.224]$
	vs. `no_reflection`	$+0.063$	$[0.057, 0.069]$
	vs. `no_personalization`	$+0.040$	$[0.034, 0.046]$
	vs. `collapse_1`	$+0.297$	$[0.288, 0.306]$

Table 23: Paired-seed effect sizes for the four ablations. The percentile ranges summarize resampling of five seeds and are not confidence intervals or significance tests.

I.9 D.9 AVERAGE REWARD BY VARIANT

Table 24 reports the observed routed value $\widehat{\mathcal{V}}_T^{\text{obs}}$ for every (env, variant) pair with descriptive spreads from Eq. equation 955.

The values match Table 16 but are repeated here in a flat layout for ease of citation.

Env	`full`	`no_routing`	`no_reflection`	`no_pers.`	`collapse_1`
`general`	0.864 ± 0.003	0.621 ± 0.005	0.812 ± 0.003	0.821 ± 0.004	0.501 ± 0.004
`workspace`	0.699 ± 0.004	0.559 ± 0.005	0.657 ± 0.004	0.680 ± 0.004	0.504 ± 0.004
`support`	0.816 ± 0.003	0.600 ± 0.005	0.753 ± 0.004	0.776 ± 0.004	0.519 ± 0.004

Table 24: Mean observed routed value $\widehat{\mathcal{V}}_T^{\text{obs}}$ by environment and ablation.

I.10 D.10 PER-USER BREAKDOWN OF ROUTED VALUE

Table 25 reports per-user $\widehat{\widehat{\mathcal{V}}}_T$ for the `full` variant in the `general` environment; analogous tables for `workspace` and `support` are omitted for space and follow the same pattern.

Seed	u0	u1	u2	u3	u4	u5	u6	u7
0	0.881	0.870	0.862	0.852	0.873	0.880	0.864	0.862
1	0.876	0.861	0.870	0.844	0.866	0.875	0.860	0.858
2	0.880	0.872	0.871	0.851	0.872	0.879	0.863	0.864
3	0.871	0.863	0.866	0.846	0.870	0.877	0.861	0.860
4	0.882	0.866	0.872	0.853	0.875	0.881	0.866	0.862

Table 25: Per-user, per-seed $\widehat{\widehat{\mathcal{V}}}_T$ in the `general` environment under the `full` variant. Per-user variance is dominated by the cluster index $g(u)$.

I.11 D.11 CONTEXT-TYPE TUNING SWEEP

Table 26 restates the baseline-tuning context-type comparison from Appendix G.

Three encodings are compared: `pref` (user preference vector only), `onehot` (candidate one-hot only), and `concat` (concatenation, used in the `full` variant of all main-paper experiments).

ctx type	$\widehat{\mathcal{V}}_T^{\text{obs}}$	$\widehat{L}_T^{\text{obs}}$	final $\sum_t (m_t - r_t)$	mean $\lvert A_T \rvert$
`pref`	0.813	0.091	27.41	19.67
`onehot`	0.834	0.075	22.49	19.67
`concat`	0.882	0.038	11.53	15.67

Table 26: Context-type ablation in the `general` environment, single seed, $T = 300$. Values reproduced from the baseline-tuning stage; `concat` is selected for the main-paper experiments.

I.12 D.12 SENSITIVITY TO NOISE SCALE

Table 27 sweeps the noise scale σ in Eq. equation 924 for the `full` variant in the `general` environment.

The observed-reward residual $\widehat{L}_T^{\text{obs}}$ grows with σ, while the structural term $\widehat{\Delta}_{\text{route}}(A_1)$ is invariant.

I.13 D.13 SENSITIVITY TO ARCHIVE SIZE

Table 28 sweeps the initial archive size $K = |A_1|$ in $\{4, 8, 12, 24\}$ in the `general` environment.

Larger archives raise both the structural routing gain and the observed-reward residual.

σ	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{\widehat{G}}_T$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$
0.025	0.280	0.052	0.020	0.882
0.050	0.280	0.051	0.036	0.864
0.100	0.280	0.049	0.064	0.836
0.200	0.280	0.044	0.121	0.776

Table 27: Noise-scale sensitivity of the logged quantities. The observed margin is positive at all four noise levels; this does not test Corollary 3 under clipped rewards.

K	$\widehat{V}_{\text{static}}(A_1)$	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$
4	0.534	0.171	0.022	0.683
8	0.561	0.244	0.030	0.775
12	0.570	0.280	0.036	0.864
24	0.578	0.318	0.058	0.838

Table 28: Archive-size sensitivity in the `general` environment. The logged observed-reward margin peaks near $K = 12$.

I.14 D.14 Sensitivity to user heterogeneity

Let α_{Dir} be the Dirichlet concentration parameter governing per-user preference vectors.

As $\alpha_{\text{Dir}} \to \infty$ the user marginal collapses to a Dirac and $\widehat{\Delta}_{\text{route}}(A_1) \to 0$, recovering the homogeneous counterexample C1 of Section A.

α_{Dir}	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}} - \widehat{V}_{\text{static}}(A_1)$
0.1	0.341	0.041	$+0.347$
0.3	0.280	0.036	$+0.293$
1.0	0.182	0.034	$+0.198$
3.0	0.071	0.030	$+0.085$
10.0	0.018	0.028	$+0.030$
∞	0.000	0.026	-0.026

Table 29: Heterogeneity sensitivity in the `general` environment. The observed routed value exceeds the static mean until the user population is nearly homogeneous.

I.15 D.15 Reflection threshold and cooldown sweep

Table 30 varies the reflection trigger τ and cooldown κ in Eq. equation 930.

Lower τ or κ approach the ungated regime of Table 22.

I.16 D.16 Theorem-to-experiment coverage map

Table 31 cross-references each theory result to a diagnostic or protocol check. The empirical rows do not prove the results.

I.17 D.17 Headline summary

Table 32 aggregates the structural terms, observed-reward residual, and observed routed value across the three synthetic environments.

$$\widehat{M}^{\text{obs}} = \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\widehat{G}}_T - \widehat{L}_T^{\text{obs}}. \tag{960}$$

This concludes the table appendix.

τ	κ	mean $\lvert A_T\rvert$	$\widehat{\bar{G}}_T$	$\widehat{\bar{\mathcal{V}}}_T$
0.05	5	19.7	0.060	0.823
0.10	10	18.4	0.057	0.842
0.15	20	15.7	0.051	0.864
0.20	30	12.9	0.030	0.851
0.30	50	9.6	0.011	0.829

Table 30: Reflection (τ, κ) sweep in the `general` environment. The default $(\tau, \kappa) = (0.15, 20)$ used in the main paper is on the Pareto frontier of $(\widehat{\bar{G}}_T, \widehat{\bar{\mathcal{V}}}_T)$.

Result	Statement	Empirical row	Status
Prop. 10	$\Delta_{\mathrm{route}}(A) \geq 0$, strict iff disagreement	Tab. 16, 18, 29	illustrated
Lem. 4	predictable archive preserves concentration	Algorithm protocol	not tested
Lem. 5	failure-prioritization bound	Tab. 30	descriptive
Thm. 2	exact mean-reward identity	Tab. 16	observed identity only
Cor. 6	conditional plug-in regret bound	Tab. 19, 27	not tested
Thm. 7	first-useful-specialist time	Tab. 21 TTFS	vacuous (oracle proposer)
Cor. 3	$\Delta_{\mathrm{route}} + \bar{G}_T > \bar{L}_T$ dominance	Tab. 24, 29	observed margin only

Table 31: Theory-to-diagnostic map. The saved residual is not the theoretical exploration cost under clipped observations; "vacuous" marks the oracle-assisted reflection result.

All numerical entries reproduce bitwise from the harness of Appendix H.8 on a single CPU core in under 30 minutes.

Quantity	`general`	`workspace`	`support`	cross-env mean
$\widehat{V}_{\text{static}}(A_1)$	0.570 ± 0.003	0.577 ± 0.003	0.579 ± 0.002	0.575
$\widehat{\Delta}_{\text{route}}(A_1)$	0.280 ± 0.004	0.124 ± 0.003	0.215 ± 0.004	0.206
$\widehat{\bar{G}}_T$	0.051 ± 0.002	0.041 ± 0.002	0.061 ± 0.002	0.051
$\widehat{\bar{L}}_T^{\text{obs}}$	0.036 ± 0.002	0.043 ± 0.002	0.039 ± 0.001	0.039
$\widehat{\bar{\mathcal{V}}}_T^{\text{obs}}$	0.864 ± 0.003	0.699 ± 0.004	0.816 ± 0.003	0.793
$\widehat{M}^{\text{obs}}$	$+0.295$	$+0.122$	$+0.237$	$+0.218$

Table 32: Cross-environment logged summary for the `full` variant. The observed-reward margin is positive in all three environments; this does not validate Corollary 3 under clipped observations.

J HYPER-RIDGE EXPLORATION-COST ANALYSIS

This appendix block analyzes the new Hyper-Ridge router. The reward accounting is inherited unchanged from Theorem 2 and Corollary 3; we do not re-prove the four-term decomposition. The only question addressed here is how replacing ordinary ridge geometry by a shared archive covariance changes the exploration-cost term.

J.1 GENERALIZED-RIDGE OBJECTS

For results combined with the four-term decomposition, draw one user U before routing and set $U_s = U$ at every round. The concentration lemmas below also allow a predictable sequence U_s. Fix one candidate in the current archive. The user-candidate feature at round s is

$$x_s(a) \in \mathbb{R}^d. \tag{961}$$

The bounded mean reward for candidate a is

$$\mu(U_s, a) = b(U_s) + x_s(a)^\top \theta_a \in [0, 1], \tag{962}$$

where the known baseline $b(U_s)$ is common to all candidates available at round s.

The observed reward on a selected candidate is

$$r_s = b(U_s) + x_s(a_s)^\top \theta_{a_s} + \xi_s. \tag{963}$$

The centered response used by Hyper-Ridge is

$$y_s = r_s - b(U_s) = x_s(a_s)^\top \theta_{a_s} + \xi_s. \tag{964}$$

The new modeling assumption is that candidate parameters share a hyper-covariance:

$$\theta_a \sim (0, \Omega). \tag{965}$$

Here the notation means

$$\mathbb{E}[\theta_a] = 0, \tag{966}$$

and

$$\mathrm{Cov}(\theta_a) = \Omega. \tag{967}$$

The covariance is required to be positive definite:

$$\Omega \succ 0. \tag{968}$$

For candidate a, the pre-t selected-round index set is

$$\mathcal{T}_{a,t} = \{\, s < t : \; a_s = a \,\}. \tag{969}$$

The candidate-specific design matrix stacks the selected features:

$$X_{a,t} = \left[x_s(a)^\top \right]_{s \in \mathcal{T}_{a,t}}. \tag{970}$$

The corresponding centered response vector is

$$y_{a,t} = [y_s]_{s \in \mathcal{T}_{a,t}}. \tag{971}$$

The hyper-covariance estimate available before round t is

$$\widehat{\Omega}_t \succ 0. \tag{972}$$

The generalized-ridge estimator used by Hyper-Ridge PersonalEvo is

$$\widehat{\theta}_{a,t} = \arg\min_{\theta \in \mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda\, \theta^\top \widehat{\Omega}_t^{-1} \theta \right\}. \tag{973}$$

Equivalently,

$$\widehat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1} \right)^{-1} X_{a,t}^\top y_{a,t}. \tag{974}$$

The generalized design matrix is

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{975}$$

The Hyper-Ridge UCB score is

$$\text{UCB}_t^{\text{hyper}}(U_t, a) = b(U_t) + x_t(a)^\top \widehat{\theta}_{a,t} + \alpha_t \sqrt{x_t(a)^\top \left(V_{a,t}^{\text{hyper}} \right)^{-1} x_t(a)}. \tag{976}$$

The selected candidate is

$$a_t(U_t) \in \arg\max_{a \in A_t(U_t)} \text{UCB}_t^{\text{hyper}}(U_t, a). \tag{977}$$

For a generic design matrix X, the effective dimension induced by Ω is

$$d_{\text{eff}}(\Omega; X) = \text{tr}\Big[X^\top X \left(X^\top X + \lambda \Omega^{-1} \right)^{-1} \Big]. \tag{978}$$

A small calculation illustrates why Ω can reduce the statistical dimension. Take

$$\Omega = \begin{bmatrix} 1 & 0 \\ 0 & 0.01 \end{bmatrix}. \tag{979}$$

Let

$$X^\top X = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}. \tag{980}$$

With $\lambda = 1$, the effective dimension is

$$\begin{aligned} d_{\text{eff}}(\Omega; X) &= \text{tr}\left[\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} \right)^{-1} \right] && (981) \\ &= \frac{9}{10} + \frac{1}{101} && (982) \\ &\approx 0.910. && (983) \end{aligned}$$

By contrast, identity ridge gives

$$\begin{aligned} d_{\text{eff}}(I; X) &= \frac{9}{10} + \frac{1}{2} && (984) \\ &= 1.400. && (985) \end{aligned}$$

Thus the same data matrix has smaller exploration geometry when the archive covariance says the second coordinate is unlikely to matter.

For the estimator itself, consider

$$X = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}. \tag{986}$$

Let the observed rewards be

$$y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \tag{987}$$

Using the same Ω and $\lambda = 1$, the generalized design is

$$\begin{aligned} V^{\text{hyper}} &= X^\top X + \Omega^{-1} && (988) \\ &= \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} && (989) \\ &= \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}. && (990) \end{aligned}$$

Since

$$X^\top y = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, \tag{991}$$

the estimate is

$$\widehat{\theta} = \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 0 \end{bmatrix} \tag{992}$$

$$= \begin{bmatrix} 5/6 \\ 0 \end{bmatrix}. \tag{993}$$

The confidence width for $x = (1, 0)^\top$ is

$$\sqrt{x^\top (V^{\text{hyper}})^{-1} x} = \sqrt{1/6}. \tag{994}$$

J.2 Shared-structure assumption

We now restate the shared-structure condition used by the Hyper-Ridge analysis. This is a regime assumption, not an empirical finding; the planned low-rank archive-geometry sweep in Section 6 is designed to stress-test it once new-method artifacts are produced.

Assumption 17 (Shared archive geometry)**.** *There is a positive definite hyper-covariance matrix*

$$\Omega \in \mathbb{R}^{d\times d}. \tag{995}$$

Each candidate parameter follows the random-effects model

$$\theta_a \sim (0, \Omega). \tag{996}$$

Here this notation means $\mathbb{E}[\theta_a] = 0$ and $\mathbb{E}[\theta_a \theta_a^\top] = \Omega$. The raw mean reward includes a known candidate-independent baseline:

$$\mu(U_t, a) = b(U_t) + x_t(a)^\top \theta_a \in [0, 1]. \tag{997}$$

For the high-probability regret analysis, the realized candidate parameters obey the ellipsoid bound

$$\|\theta_a\|_{\Omega^{-1}} \le S_\Omega \qquad \text{for every } a \in A_t(U_t) \text{ and } t \le T. \tag{998}$$

The archive is indexed by predictable birth-order slots as in Lemma 4. In particular, each candidate identity and feature rule is observable at birth, its latent parameter is fixed at birth in the analysis filtration, and the features are observable and predictable:

$$\sigma(x_t(a)) \subseteq \mathcal{O}_{t-1} \qquad \text{for every } a \in A_t(U_t). \tag{999}$$

For a predictable user stream, define the realized routing-candidate union by

$$\mathcal{A}_T^{\text{route}} = \bigcup_{t=1}^{T} A_t(U_t). \tag{1000}$$

The centered rewards satisfy the linear model

$$y_t := r_t - b(U_t) = x_t(a_t)^\top \theta_{a_t} + \xi_t. \tag{1001}$$

The noise is conditionally σ-sub-Gaussian:

$$\mathbb{E}[\exp(\eta \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\eta^2 \sigma^2}{2}\right) \qquad \text{for every } \eta \in \mathbb{R}. \tag{1002}$$

The archive covariance gives a bounded feature radius:

$$x_t(a)^\top \Omega x_t(a) \le L_\Omega^2 \qquad \text{for every } a \in A_t(U_t). \tag{1003}$$

There is a deterministic effective-dimension cap:

$$\max_{1 \le s \le T+1} \max_{a \in \mathcal{A}_T^{\text{route}}} d_{\text{eff}}(\Omega; X_{a,s}) \le \bar{d}_{\text{eff},T}(\Omega) < \infty \quad \text{almost surely}. \tag{1004}$$

The concentration argument uses the realized ellipsoid bound, predictable births, centered reward model, and feature radius. The random-effects law is needed only to interpret and estimate Ω. The deterministic cap in equation 1004 converts the pathwise potential bound into a deterministic finite-time certificate.

J.3 KNOWN-Ω CONCENTRATION

The first lemma is the generalized-ridge analogue of the standard OFUL confidence bound (Abbasi-Yadkori et al., 2011; Chu et al., 2011). It is stated for known Ω, which is the clean oracle-covariance case. The estimated-covariance case adds a perturbation term and is handled separately.

Lemma 45 (Self-normalized concentration in the Ω geometry)**.** *Let $\Omega \succ 0$ and $\lambda > 0$. Fix a predictable birth-order slot a whose parameter is fixed at birth and satisfies $\|\theta_a\|_{\Omega^{-1}} \le S_\Omega$. Before its birth, pad its feature and selection indicator by zero. Assume the padded selected feature is $\mathcal{F}_{s-1}$-measurable and that each selected centered response obeys*

$$y_s = x_s(a)^\top \theta_a + \xi_s, \qquad \mathbb{E}[\exp(\eta\xi_s) \mid \mathcal{F}_{s-1}] \le \exp(\eta^2\sigma^2/2) \quad \text{for every } \eta \in \mathbb{R}.$$

Let the router use the true covariance, $\widehat{\Omega}_t = \Omega$. Define the known-covariance design matrix by

$$V_{a,t}^{\Omega} = \lambda\Omega^{-1} + \sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top. \tag{1005}$$

Define the known-covariance estimator by

$$\widehat{\theta}_{a,t}^{\Omega} = \left(V_{a,t}^{\Omega}\right)^{-1} \sum_{s\in\mathcal{T}_{a,t}} x_s(a)y_s. \tag{1006}$$

Define the confidence radius

$$\beta_{a,t}^{\Omega}(\delta) = \sigma\sqrt{2\log\left(\frac{\det(V_{a,t}^{\Omega})^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}\delta}\right)} + \sqrt{\lambda}\,S_\Omega. \tag{1007}$$

Then, with probability at least $1-\delta$, uniformly over t,

$$\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \le \beta_{a,t}^{\Omega}(\delta). \tag{1008}$$

Consequently, for every predictable test feature $x \in \mathbb{R}^d$,

$$\left|x^\top\left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right| \le \beta_{a,t}^{\Omega}(\delta)\sqrt{x^\top\left(V_{a,t}^{\Omega}\right)^{-1}x}. \tag{1009}$$

Proof. For this fixed candidate, define the regularizer

$$A = \lambda\Omega^{-1}. \tag{1010}$$

Define the noise-weighted feature martingale

$$S_{a,t} = \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\xi_s. \tag{1011}$$

Using the reward model, the estimator error satisfies

$$\begin{aligned}
\widehat{\theta}_{a,t}^{\Omega} - \theta_a &= \left(V_{a,t}^{\Omega}\right)^{-1} \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\left(x_s(a)^\top\theta_a + \xi_s\right) - \theta_a && (1012)\\
&= \left(V_{a,t}^{\Omega}\right)^{-1}\left[\sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top\theta_a + S_{a,t}\right] - \theta_a && (1013)\\
&= \left(V_{a,t}^{\Omega}\right)^{-1}\left[\left(V_{a,t}^{\Omega} - A\right)\theta_a + S_{a,t}\right] - \theta_a && (1014)\\
&= \left(V_{a,t}^{\Omega}\right)^{-1} S_{a,t} - \left(V_{a,t}^{\Omega}\right)^{-1} A\theta_a. && (1015)
\end{aligned}$$

Taking the $V_{a,t}^{\Omega}$-norm and applying the triangle inequality gives

$$\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \le \|S_{a,t}\|_{(V_{a,t}^{\Omega})^{-1}} + \|A\theta_a\|_{(V_{a,t}^{\Omega})^{-1}}. \tag{1016}$$

Since $V_{a,t}^{\Omega} \succeq A$, we have

$$\|A\theta_a\|^2_{(V_{a,t}^{\Omega})^{-1}} = \theta_a^\top A \left(V_{a,t}^{\Omega}\right)^{-1} A\theta_a \tag{1017}$$
$$\leq \theta_a^\top A\theta_a \tag{1018}$$
$$= \lambda\theta_a^\top \Omega^{-1}\theta_a \tag{1019}$$
$$\leq \lambda S_\Omega^2. \tag{1020}$$

It remains to control the martingale term. If $\sigma = 0$, the conditional moment-generating-function assumption implies $\xi_s = 0$ almost surely, so the claim is immediate. Suppose $\sigma > 0$. For fixed $z \in \mathbb{R}^d$, define

$$M_{a,t}(z) = \exp\left(\frac{z^\top S_{a,t}}{\sigma} - \frac{1}{2}z^\top \left(V_{a,t}^{\Omega} - A\right) z\right). \tag{1021}$$

At a selected round s, conditional sub-Gaussianity gives

$$\mathbb{E}\left[\exp\left(\frac{z^\top x_s(a)\xi_s}{\sigma} - \frac{(z^\top x_s(a))^2}{2}\right)\middle|\mathcal{F}_{s-1}\right] \leq 1.$$

At an unselected round the process does not change. Hence $(M_{a,t}(z))_{t\leq T}$ is a nonnegative supermartingale with initial value one.

Let

$$\varphi_A(z) = \frac{\det(A)^{1/2}}{(2\pi)^{d/2}} \exp\left(-\frac{z^\top Az}{2}\right)$$

be the Gaussian density with precision A, and define $\mathcal{M}_{a,t} = \int M_{a,t}(z)\varphi_A(z)\,dz$. Tonelli's theorem shows that $(\mathcal{M}_{a,t})_{t\leq T}$ is also a nonnegative supermartingale with initial value one. Completing the square gives

$$\mathcal{M}_{a,t} = \frac{\det(A)^{1/2}}{\det(V_{a,t}^{\Omega})^{1/2}} \exp\left(\frac{\|S_{a,t}\|^2_{(V_{a,t}^{\Omega})^{-1}}}{2\sigma^2}\right). \tag{1022}$$

Ville's inequality now gives

$$\Pr\left(\sup_{t\leq T} \mathcal{M}_{a,t} \geq \frac{1}{\delta}\right) \leq \delta. \tag{1023}$$

Consequently, with probability at least $1-\delta$, simultaneously for every $t \leq T$, substituting $A = \lambda\Omega^{-1}$ gives

$$\|S_{a,t}\|_{(V_{a,t}^{\Omega})^{-1}} \leq \sigma\sqrt{2\log\left(\frac{\det(V_{a,t}^{\Omega})^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}\delta}\right)}. \tag{1024}$$

Combining equation 1016, equation 1017, and equation 1024 proves

$$\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \leq \beta_{a,t}^{\Omega}(\delta). \tag{1025}$$

Finally, Cauchy–Schwarz in the $V_{a,t}^{\Omega}$ inner product gives

$$\left|x^\top \left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right| = \left|\left[\left(V_{a,t}^{\Omega}\right)^{-1/2} x\right]^\top \left[\left(V_{a,t}^{\Omega}\right)^{1/2} \left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right]\right| \tag{1026}$$
$$\leq \sqrt{x^\top \left(V_{a,t}^{\Omega}\right)^{-1} x}\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \tag{1027}$$
$$\leq \beta_{a,t}^{\Omega}(\delta)\sqrt{x^\top \left(V_{a,t}^{\Omega}\right)^{-1} x}. \tag{1028}$$

$\square$

A concrete confidence-bound computation is useful. Suppose

$$\frac{\det(V_{a,t}^{\Omega})^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}} = 10. \tag{1029}$$

Let

$$\sigma = 0.05. \tag{1030}$$

Let

$$\delta = 0.05. \tag{1031}$$

Let the regularization bias term be

$$\sqrt{\lambda}S_{\Omega} = 0.20. \tag{1032}$$

Then the confidence radius is

$$\begin{aligned} \beta_{a,t}^{\Omega}(\delta) &= 0.05\sqrt{2\log(10/0.05)} + 0.20 && (1033)\\ &\approx 0.05\sqrt{10.596} + 0.20 && (1034)\\ &\approx 0.363. && (1035) \end{aligned}$$

If the uncertainty width at a candidate is

$$\sqrt{x^{\top}(V_{a,t}^{\Omega})^{-1}x} = 0.10, \tag{1036}$$

then Lemma 45 gives the prediction error bound

$$\begin{aligned} \left|x^{\top}\left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right| &\le 0.363 \cdot 0.10 && (1037)\\ &= 0.0363. && (1038) \end{aligned}$$

J.4 Elliptical potential in the Ω geometry

The next lemma is the step that converts per-round confidence widths into a dimension-dependent exploration cost. Its role is the same as the elliptical-potential lemma in LinUCB, except that the determinant is measured relative to $\lambda\Omega^{-1}$, and the resulting log-determinant is controlled by $d_{\text{eff}}(\Omega; X)$ up to a logarithmic factor.

Lemma 46 (Ω-elliptical potential and effective dimension)**.** *Fix one candidate a, and write its selected features in chronological order as*

$$x_1, \dots, x_n. \tag{1039}$$

Define the recursive generalized-ridge matrices by

$$V_0 = \lambda\Omega^{-1}, \tag{1040}$$

and

$$V_s = V_{s-1} + x_s x_s^{\top}. \tag{1041}$$

Define the uncertainty increment by

$$q_s = x_s^{\top} V_{s-1}^{-1} x_s. \tag{1042}$$

Let the full design be

$$X = \begin{bmatrix} x_1^{\top} \\ \vdots \\ x_n^{\top} \end{bmatrix}. \tag{1043}$$

Define the covariance-scaled Gram matrix by

$$M = \lambda^{-1}\Omega^{1/2}X^{\top}X\Omega^{1/2}. \tag{1044}$$

Let its largest eigenvalue be

$$R = \lambda_{\max}(M). \tag{1045}$$

Define

$$c(R) = \begin{cases} 1, & R = 0, \\ \dfrac{1+R}{R}\log(1+R), & R > 0. \end{cases} \tag{1046}$$

Then

$$\sum_{s=1}^{n} \min\{1, q_s\} \le 2\, c(R)\, d_{\text{eff}}(\Omega; X). \tag{1047}$$

If additionally $x_s^\top \Omega x_s \le L_\Omega^2$, *then*

$$R \le \frac{nL_\Omega^2}{\lambda}, \tag{1048}$$

and hence

$$\sum_{s=1}^{n} \min\{1, q_s\} \le 2\left[1 + \log\left(1 + \frac{nL_\Omega^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega; X). \tag{1049}$$

Proof. The matrix determinant lemma gives

$$\det(V_s) = \det(V_{s-1})\left(1 + x_s^\top V_{s-1}^{-1} x_s\right) \tag{1050}$$

$$= \det(V_{s-1})(1 + q_s). \tag{1051}$$

Multiplying over $s = 1, \ldots, n$ gives

$$\frac{\det(V_n)}{\det(V_0)} = \prod_{s=1}^{n} (1 + q_s). \tag{1052}$$

Taking logarithms yields

$$\log \frac{\det(V_n)}{\det(V_0)} = \sum_{s=1}^{n} \log(1 + q_s). \tag{1053}$$

For every $q \ge 0$, the scalar inequality

$$\min\{1, q\} \le 2\log(1 + q) \tag{1054}$$

holds. Therefore

$$\sum_{s=1}^{n} \min\{1, q_s\} \le 2\sum_{s=1}^{n} \log(1 + q_s) \tag{1055}$$

$$= 2\log \frac{\det(V_n)}{\det(V_0)}. \tag{1056}$$

The determinant ratio can be rewritten in the covariance-scaled coordinates:

$$\log \frac{\det(V_n)}{\det(V_0)} = \log\det\left(I + \lambda^{-1}\Omega^{1/2} X^\top X \Omega^{1/2}\right) \tag{1057}$$

$$= \log\det(I + M). \tag{1058}$$

Let the eigenvalues of M be

$$\nu_1, \ldots, \nu_d. \tag{1059}$$

Then

$$0 \le \nu_i \le R. \tag{1060}$$

The log-determinant is

$$\log\det(I + M) = \sum_{i=1}^{d} \log(1 + \nu_i). \tag{1061}$$

The effective dimension is

$$d_{\text{eff}}(\Omega; X) = \operatorname{tr}\left[X^\top X \left(X^\top X + \lambda\Omega^{-1}\right)^{-1}\right] \tag{1062}$$

$$= \operatorname{tr}\left[\Omega^{1/2}X^\top X\Omega^{1/2}\left(\Omega^{1/2}X^\top X\Omega^{1/2} + \lambda I\right)^{-1}\right] \tag{1063}$$

$$= \sum_{i=1}^{d} \frac{\nu_i}{1+\nu_i}. \tag{1064}$$

For $z > 0$, define

$$g(z) = \frac{1+z}{z}\log(1+z). \tag{1065}$$

Its derivative is

$$g'(z) = -\frac{\log(1+z)}{z^2} + \frac{1+1/z}{1+z} \tag{1066}$$

$$= \frac{z - \log(1+z)}{z^2} \tag{1067}$$

$$\geq 0. \tag{1068}$$

Thus $g(z) \leq g(R)$ for $0 < z \leq R$. Equivalently,

$$\log(1+z) \leq c(R)\frac{z}{1+z} \qquad \text{for } 0 \leq z \leq R. \tag{1069}$$

Applying this scalar bound eigenvalue by eigenvalue gives

$$\log\det(I+M) = \sum_{i=1}^{d}\log(1+\nu_i) \tag{1070}$$

$$\leq c(R)\sum_{i=1}^{d}\frac{\nu_i}{1+\nu_i} \tag{1071}$$

$$= c(R)d_{\text{eff}}(\Omega; X). \tag{1072}$$

Combining equation 1055 and equation 1070 proves

$$\sum_{s=1}^{n}\min\{1, q_s\} \leq 2c(R)d_{\text{eff}}(\Omega; X). \tag{1073}$$

It remains to bound R. Since $M \succeq 0$,

$$R \leq \operatorname{tr}(M) \tag{1074}$$

$$= \lambda^{-1}\operatorname{tr}\left(\Omega^{1/2}X^\top X\Omega^{1/2}\right) \tag{1075}$$

$$= \lambda^{-1}\sum_{s=1}^{n} x_s^\top \Omega x_s \tag{1076}$$

$$\leq \frac{nL_\Omega^2}{\lambda}. \tag{1077}$$

Finally, for $R \geq 0$,

$$c(R) \leq 1 + \log(1+R). \tag{1078}$$

Substituting equation 1074 into equation 1078 yields

$$c(R) \leq 1 + \log\left(1 + \frac{nL_\Omega^2}{\lambda}\right). \tag{1079}$$

The claimed logarithmic bound follows. □

K ORACLE HYPER-COVARIANCE: PROOF OF THE EFFECTIVE-DIMENSION COST

We now combine Lemma 45 and Lemma 46. The result is an oracle statement: the router is allowed to use the true hyper-covariance Ω. The estimated-Ω case is treated afterward as a perturbation of this oracle bound.

Assumption 18 (Oracle covariance and explicit confidence radius)**.** *The hyper-covariance used by the router is the true matrix:*

$$\widehat{\Omega}_t = \Omega. \tag{1080}$$

The matrix is positive definite:

$$\Omega \succ 0. \tag{1081}$$

For the fixed-user route, the realized candidate union is

$$\mathcal{A}_T(U) = \bigcup_{t=1}^{T} A_t(U). \tag{1082}$$

At most $K_{\max}$ candidates appear through round T:

$$|\mathcal{A}_T(U)| \le K_{\max}. \tag{1083}$$

The Ω-scaled feature norm is uniformly bounded:

$$x_t(a)^\top \Omega x_t(a) \le L_\Omega^2. \tag{1084}$$

The archive uses the predictable birth-order slots of Lemma 4. Its realized candidate designs obey the deterministic cap

$$\max_{1\le s\le T+1} \max_{a\in\mathcal{A}_T(U)} d_{\mathrm{eff}}(\Omega; X_{a,s}) \le \bar{d}_{\mathrm{eff},T}(\Omega) \quad \textit{almost surely}. \tag{1085}$$

For $\delta \in (0,1)$, define

$$c_T^\Omega = 1 + \log\left(1 + \frac{T L_\Omega^2}{\lambda}\right) \tag{1086}$$

and

$$\alpha_{\Omega,T}(\delta) = \max\left\{1,\ \sigma\sqrt{c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega) + 2\log(K_{\max}/\delta)} + \sqrt{\lambda} S_\Omega\right\}. \tag{1087}$$

The effective dimension used in the bound is the largest realized per-candidate effective dimension. For candidate a, its design after round T is

$$X_{a,T+1} = \left[x_s(a_s)^\top\right]_{s\le T:\, a_s=a}. \tag{1088}$$

Define the candidate-wise effective dimension by

$$d_{\mathrm{eff}}(\Omega; X_{a,T+1}) = \mathrm{tr}\left[X_{a,T+1}^\top X_{a,T+1}\left(X_{a,T+1}^\top X_{a,T+1} + \lambda\Omega^{-1}\right)^{-1}\right]. \tag{1089}$$

The realized maximum candidate-wise effective dimension is

$$d_{\mathrm{eff},T}^{\max}(\Omega) = \max_{a\in\mathcal{A}_T(U)} d_{\mathrm{eff}}(\Omega; X_{a,T+1}). \tag{1090}$$

Assumption 18 requires $d_{\mathrm{eff},T}^{\max}(\Omega) \le \bar{d}_{\mathrm{eff},T}(\Omega)$ almost surely, and imposes the same cap on every earlier design prefix.

Proposition 47 (Known-Ω effective-dimension exploration cost)**.** *Under Assumptions 17 and 18, let the oracle Hyper-Ridge router use multiplier $\alpha_{\Omega,T}(\delta)$. For the result used in the four-term decomposition, $U_t = U$ for all t. There is an event $\mathcal{E}_\Omega$ with probability at least $1-\delta$ on which*

$$\sum_{t=1}^{T} L_t(U) \le 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega)}. \tag{1091}$$

Consequently, its expected exploration cost satisfies

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega)} + \delta T. \tag{1092}$$

For $T \ge 2$, fixed problem parameters, and $\delta = 1/T$,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] = O\left(\sqrt{T}\log T\right). \tag{1093}$$

For fixed δ, the same order holds for the pathwise right-hand side of equation 1091 on its probability-$(1-\delta)$ event, but the expectation bound equation 1092 retains the linear term δT.

Proof. Index all candidates by the predictable birth-order slots in Lemma 4, using a dummy process for any slot that is never filled. For each of the at most $K_{\max}$ slots, apply Lemma 45 with failure probability $\delta/K_{\max}$. For every realized slot and every $t \le T$, the determinant calculation in Lemma 46 gives

$$\log\frac{\det(V_{a,t}^{\Omega})}{\det(\lambda\Omega^{-1})} \le c_T^{\Omega} d_{\mathrm{eff}}(\Omega; X_{a,t}) \tag{1094}$$

$$\le c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega). \tag{1095}$$

Thus the confidence radius in Lemma 45, with $\delta/K_{\max}$ in place of δ, is at most $\alpha_{\Omega,T}(\delta)$. A union bound over the slots defines an event $\mathcal{E}_\Omega$ with probability at least $1-\delta$ on which these confidence intervals hold for every candidate after its birth and every $t \le T$.

Let the current-archive oracle action at round t be

$$a_t^\star \in \arg\max_{a \in A_t(U)} x_t(a)^\top \theta_a. \tag{1096}$$

The exploration cost is therefore

$$L_t(U) = x_t(a_t^\star)^\top \theta_{a_t^\star} - x_t(a_t)^\top \theta_{a_t}. \tag{1097}$$

On $\mathcal{E}_\Omega$, the upper confidence bound for the oracle action is

$$x_t(a_t^\star)^\top \theta_{a_t^\star} \le x_t(a_t^\star)^\top \widehat{\theta}_{a_t^\star,t}^{\Omega} + \alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t^\star)^\top \left(V_{a_t^\star,t}^{\Omega}\right)^{-1} x_t(a_t^\star)}. \tag{1098}$$

Since the router chooses the largest oracle-Ω UCB score,

$$\begin{aligned} &x_t(a_t^\star)^\top \widehat{\theta}_{a_t^\star,t}^{\Omega} + \alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t^\star)^\top \left(V_{a_t^\star,t}^{\Omega}\right)^{-1} x_t(a_t^\star)} \\ &\quad\le x_t(a_t)^\top \widehat{\theta}_{a_t,t}^{\Omega} + \alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}. \end{aligned} \tag{1099}$$

The same confidence event gives the selected arm's lower confidence bound:

$$x_t(a_t)^\top \widehat{\theta}_{a_t,t}^{\Omega} \le x_t(a_t)^\top \theta_{a_t} + \alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}. \tag{1100}$$

Combining equation 1097–equation 1100 yields

$$L_t(U) \le 2\alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t)}. \tag{1101}$$

The rewards are bounded, so the same gap also satisfies

$$0 \le L_t(U) \le 1. \tag{1102}$$

Define the realized uncertainty at the selected arm by

$$q_t = x_t(a_t)^\top \left(V_{a_t,t}^{\Omega}\right)^{-1} x_t(a_t). \tag{1103}$$

Using $\alpha_{\Omega,T}(\delta) \geq 1$, equation 1101 and equation 1102 imply

$$L_t(U) \leq \min\{1, 2\alpha_{\Omega,T}(\delta)\sqrt{q_t}\} \tag{1104}$$
$$\leq 2\alpha_{\Omega,T}(\delta) \min\{1, \sqrt{q_t}\}. \tag{1105}$$

Summing over time gives

$$\sum_{t=1}^{T} L_t(U) \leq 2\alpha_{\Omega,T}(\delta) \sum_{t=1}^{T} \min\{1, \sqrt{q_t}\}. \tag{1106}$$

Partition the selected rounds by candidate. Let n_a be the number of times candidate a is selected:

$$n_a = \sum_{t=1}^{T} \mathbf{1}\{a_t = a\}. \tag{1107}$$

Let the within-candidate uncertainties be

$$q_{a,s} = x_{a,s}^{\top} \left(V_{a,s-1}^{\Omega}\right)^{-1} x_{a,s}. \tag{1108}$$

Then the total width sum is

$$\sum_{t=1}^{T} \min\{1, \sqrt{q_t}\} = \sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, \sqrt{q_{a,s}}\}. \tag{1109}$$

Cauchy–Schwarz gives

$$\sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, \sqrt{q_{a,s}}\} \leq \sqrt{\left(\sum_{a \in \mathcal{A}_T(U)} n_a\right)\left(\sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\}\right)} \tag{1110}$$
$$= \sqrt{T \sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\}}. \tag{1111}$$

Lemma 46 applied separately to each candidate yields

$$\sum_{s=1}^{n_a} \min\{1, q_{a,s}\} \leq 2\left[1 + \log\left(1 + \frac{n_a L_{\Omega}^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega; X_{a,T+1}) \tag{1112}$$
$$\leq 2\left[1 + \log\left(1 + \frac{T L_{\Omega}^2}{\lambda}\right)\right] \bar{d}_{\text{eff},T}(\Omega). \tag{1113}$$

Since at most $K_{\max}$ candidates appeared through round T,

$$\sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\} \leq 2K_{\max}\left[1 + \log\left(1 + \frac{T L_{\Omega}^2}{\lambda}\right)\right] \bar{d}_{\text{eff},T}(\Omega). \tag{1114}$$

Substituting equation 1114 into equation 1110 and then into equation 1106 proves

$$\sum_{t=1}^{T} L_t(U) \leq 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\left[1 + \log\left(1 + \frac{T L_{\Omega}^2}{\lambda}\right)\right] \bar{d}_{\text{eff},T}(\Omega)}. \tag{1115}$$

This is equation 1091. Taking expectations and splitting on $\mathcal{E}_{\Omega}$ gives

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] = \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_{\Omega}} \sum_{t=1}^{T} L_t(U)\right] + \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_{\Omega}^c} \sum_{t=1}^{T} L_t(U)\right] \tag{1116}$$
$$\leq 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\left[1 + \log\left(1 + \frac{T L_{\Omega}^2}{\lambda}\right)\right] \bar{d}_{\text{eff},T}(\Omega)} + \delta T. \tag{1117}$$

This is equation 1092. For fixed problem parameters, $c_T^{\Omega} = O(\log T)$ and $\alpha_{\Omega,T}(\delta) = O(\sqrt{\log T})$ for either fixed δ or $\delta = 1/T$ with $T \geq 2$. Thus the pathwise right-hand side is $O(\sqrt{T}\log T)$ in both cases. For the expectation bound, choosing $\delta = 1/T$ with $T \geq 2$ also makes the complement term $\delta T = 1$, which proves equation 1093. With fixed δ, that complement term is linear and no sublinear expected bound follows from this conversion. $\square$

The geometry changes the potential factor from an ambient-dimension term to $\bar{d}_{\mathrm{eff},T}(\Omega)$. This does not by itself determine which router has the smaller certificate: the full comparison must also include $\alpha_{\Omega,T}(\delta)$, c_T^{Ω}, and their LinUCB counterparts.

K.1 Estimated hyper-covariance: perturbation reduction and the remaining gap

The actual Hyper-Ridge PersonalEvo router uses an estimate $\widehat{\Omega}_t$. We therefore separate two issues. First, if $\widehat{\Omega}_t$ is close to Ω in relative spectral norm, the oracle proof above is stable. Second, proving that an online archive estimator achieves this closeness is a separate statistical problem; we state it as Conjecture 1 rather than claiming it as proven.

The relative covariance error at time t is

$$\rho_{\Omega,t} = \left\| \Omega^{-1/2} \left(\widehat{\Omega}_t - \Omega \right) \Omega^{-1/2} \right\|_{\mathrm{op}}. \tag{1118}$$

The uniform relative covariance error through horizon T is

$$\rho_{\Omega,T} = \max_{1 \le t \le T} \rho_{\Omega,t}. \tag{1119}$$

The estimated-Ω design matrix is

$$V_{a,t}^{\widehat{\Omega}} = X_{a,t}^{\top} X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{1120}$$

The oracle design matrix is

$$V_{a,t}^{\Omega} = X_{a,t}^{\top} X_{a,t} + \lambda \Omega^{-1}. \tag{1121}$$

For a multiplier $\widetilde{\alpha}_T \ge 1$, define the excess width caused by using $\widehat{\Omega}_t$ instead of Ω:

$$\widehat{w}_t = \min\left\{ 1, \sqrt{x_t(a_t)^{\top} \left(V_{a_t,t}^{\widehat{\Omega}} \right)^{-1} x_t(a_t)} \right\}, \tag{1122}$$

$$w_t^{\Omega} = \min\left\{ 1, \sqrt{x_t(a_t)^{\top} \left(V_{a_t,t}^{\Omega} \right)^{-1} x_t(a_t)} \right\}. \tag{1123}$$

$$\mathrm{WidthErr}_T(\widehat{\Omega}, \Omega) = 2\widetilde{\alpha}_T \sum_{t=1}^{T} \left[\widehat{w}_t - w_t^{\Omega} \right]_+. \tag{1124}$$

Here the positive part is

$$[z]_+ = \max\{z, 0\}. \tag{1125}$$

Lemma 48 (Relative spectral error implies Loewner control)**.** *Assume*

$$\rho_{\Omega,T} \le \varepsilon \tag{1126}$$

for some

$$0 \le \varepsilon < 1. \tag{1127}$$

Then, for every t,

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega}_t \preceq (1+\varepsilon)\Omega. \tag{1128}$$

Moreover,

$$\frac{1}{1+\varepsilon}\Omega^{-1} \preceq \widehat{\Omega}_t^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \tag{1129}$$

Finally, for every candidate and round,

$$V_{a,t}^{\widehat{\Omega}} \;\succeq\; \frac{1}{1+\varepsilon} V_{a,t}^{\Omega}. \tag{1130}$$

Proof. The definition of $\rho_{\Omega,T}$ implies

$$-\varepsilon I \preceq \Omega^{-1/2}\left(\widehat{\Omega}_t - \Omega\right)\Omega^{-1/2} \preceq \varepsilon I. \tag{1131}$$

Adding I gives

$$(1-\varepsilon)I \preceq \Omega^{-1/2}\widehat{\Omega}_t\Omega^{-1/2} \preceq (1+\varepsilon)I. \tag{1132}$$

Congruence by $\Omega^{1/2}$ gives

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega}_t \preceq (1+\varepsilon)\Omega. \tag{1133}$$

Inverting reverses the Loewner order:

$$\frac{1}{1+\varepsilon}\Omega^{-1} \preceq \widehat{\Omega}_t^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \tag{1134}$$

Therefore

$$\begin{aligned}
V_{a,t}^{\widehat{\Omega}} &= X_{a,t}^\top X_{a,t} + \lambda\widehat{\Omega}_t^{-1} && (1135)\\
&\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1+\varepsilon}\Omega^{-1} && (1136)\\
&\succeq \frac{1}{1+\varepsilon}\left(X_{a,t}^\top X_{a,t} + \lambda\Omega^{-1}\right) && (1137)\\
&= \frac{1}{1+\varepsilon}V_{a,t}^{\Omega}. && (1138)
\end{aligned}$$

□

Lemma 49 (Width and effective-dimension stability)**.** *Under the assumptions of Lemma 48, every feature vector x satisfies*

$$x^\top\left(V_{a,t}^{\widehat{\Omega}}\right)^{-1}x \le (1+\varepsilon)x^\top\left(V_{a,t}^{\Omega}\right)^{-1}x. \tag{1139}$$

For any fixed design matrix X, the estimated-geometry effective dimension satisfies

$$d_{\text{eff}}(\widehat{\Omega}_t; X) \le (1+\varepsilon)d_{\text{eff}}(\Omega; X). \tag{1140}$$

Proof. From equation 1130, inversion gives

$$\left(V_{a,t}^{\widehat{\Omega}}\right)^{-1} \preceq (1+\varepsilon)\left(V_{a,t}^{\Omega}\right)^{-1}. \tag{1141}$$

Taking the quadratic form in direction x gives

$$x^\top\left(V_{a,t}^{\widehat{\Omega}}\right)^{-1}x \le (1+\varepsilon)x^\top\left(V_{a,t}^{\Omega}\right)^{-1}x. \tag{1142}$$

For the effective dimension, write

$$A = X^\top X. \tag{1143}$$

Then

$$\begin{aligned}
d_{\text{eff}}(\widehat{\Omega}_t; X) &= \operatorname{tr}\left[A\left(A+\lambda\widehat{\Omega}_t^{-1}\right)^{-1}\right] && (1144)\\
&\le (1+\varepsilon)\operatorname{tr}\left[A\left(A+\lambda\Omega^{-1}\right)^{-1}\right] && (1145)\\
&= (1+\varepsilon)d_{\text{eff}}(\Omega; X). && (1146)
\end{aligned}$$

□

The spectral comparison controls confidence widths. A regret statement also needs valid confidence intervals in the estimated geometry.

Proposition 50 (Conditional perturbation bound for estimated hyper-ridge)**.** *Assume Assumption 17, suppose at most $K_{\max}$ candidates appear through round T, and let $\widehat{\Omega}_t \succ 0$ for every t. Let an estimated-covariance router use a multiplier $\widetilde{\alpha}_T \geq 1$, and suppose the event $\mathcal{E}_{\text{conf}}(\widehat{\Omega})$ gives simultaneous confidence:*

$$\left| x^\top \left(\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a \right) \right| \leq \widetilde{\alpha}_T \sqrt{x^\top \left(V^{\widehat{\Omega}}_{a,t} \right)^{-1} x} \tag{1147}$$

for every available candidate a, every $t \leq T$, and every predictable test feature x. Assume also that

$$\rho_{\Omega,T} \leq \varepsilon < 1. \tag{1148}$$

On the intersection of these two events,

$$\mathrm{WidthErr}_T(\widehat{\Omega}, \Omega) \leq 2\widetilde{\alpha}_T \left(\sqrt{1+\varepsilon} - 1 \right) \sqrt{2K_{\max} T \, c^{\Omega}_T \bar{d}_{\text{eff},T}(\Omega)}, \tag{1149}$$

and

$$\sum_{t=1}^{T} L_t(U) \leq 2\widetilde{\alpha}_T \sqrt{1+\varepsilon} \sqrt{2K_{\max} T \, c^{\Omega}_T \bar{d}_{\text{eff},T}(\Omega)}. \tag{1150}$$

If the intersection has probability at least $1-\eta$, then the right-hand side of equation 1150 plus ηT bounds the expected cumulative exploration cost.

Proof. Lemma 49 implies, for every selected round,

$$\sqrt{x_t(a_t)^\top \left(V^{\widehat{\Omega}}_{a_t,t} \right)^{-1} x_t(a_t)} \leq \sqrt{1+\varepsilon} \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t} \right)^{-1} x_t(a_t)}. \tag{1151}$$

The map $z \mapsto \min\{1, z\}$ is one-Lipschitz and monotone, so

$$\left[\min\left\{ 1, \sqrt{x_t(a_t)^\top \left(V^{\widehat{\Omega}}_{a_t,t} \right)^{-1} x_t(a_t)} \right\} - \min\left\{ 1, \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t} \right)^{-1} x_t(a_t)} \right\} \right]_+ \tag{1152}$$

$$\leq \left(\sqrt{1+\varepsilon} - 1 \right) \min\left\{ 1, \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t} \right)^{-1} x_t(a_t)} \right\}. \tag{1153}$$

Summing and using the oracle width-sum bound from the proof of Proposition 47 gives

$$\mathrm{WidthErr}_T(\widehat{\Omega}, \Omega) \leq 2\widetilde{\alpha}_T \left(\sqrt{1+\varepsilon} - 1 \right) \sum_{t=1}^{T} \min\left\{ 1, \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t} \right)^{-1} x_t(a_t)} \right\} \tag{1154}$$

$$\leq 2\widetilde{\alpha}_T \left(\sqrt{1+\varepsilon} - 1 \right) \sqrt{2K_{\max} T \, c^{\Omega}_T \bar{d}_{\text{eff},T}(\Omega)}. \tag{1155}$$

On $\mathcal{E}_{\text{conf}}(\widehat{\Omega})$, optimism gives

$$L_t(U) \leq 2\widetilde{\alpha}_T \min\left\{ 1, \sqrt{x_t(a_t)^\top \left(V^{\widehat{\Omega}}_{a_t,t} \right)^{-1} x_t(a_t)} \right\}.$$

Apply equation 1151, sum over t, and use the same candidate-wise potential bound to obtain equation 1150. On the complement of an event with probability $1-\eta$, bounded rewards give $\sum_{t=1}^{T} L_t(U) \leq T$, proving the expectation statement. □

A numerical perturbation example clarifies the scale. If

$$\varepsilon = 0.21, \tag{1156}$$

then

$$\sqrt{1+\varepsilon} = 1.10. \tag{1157}$$

Thus the estimated-geometry width is at most 1.10 times the oracle-geometry width on the spectral event. Turning this comparison into an exploration bound also requires the confidence event. Neither event has been proved for the periodic estimator.

Conjecture 1 (Sharper estimated-covariance exploration cost)**.** *There is an estimated-covariance routing rule for which*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le C\sqrt{K_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T\log(1+T)} + o(\sqrt{T}), \tag{1158}$$

under assumptions comparable to those used here. The explicit frequentist radius in Theorem 57 does not prove this rate.

The conjecture requires a sharper confidence analysis or a different routing rule. The periodic estimator and its diagnostics do not establish it.

K.2 Map of hyper-ridge claims to proofs and open steps

Table 33 summarizes which parts of the Hyper-Ridge PersonalEvo theory are proved in this section and which are conditional. The inherited four-term decomposition remains the base accounting theorem; the new analysis only sharpens the exploration-cost term $\bar{L}_T$.

Claim	Mathematical object	Status
Exact PersonalEvo accounting	$V_{\mathrm{static}} + \Delta_{\mathrm{route}} + \bar{G}_T - \bar{L}_T$	inherited from Theorem 2
Oracle hyper-ridge confidence	$V_{a,t}^{\Omega} = X_{a,t}^{\top}X_{a,t} + \lambda\Omega^{-1}$	Lemma 45
Effective-dimension potential	$\sum_s \min\{1, q_s\}$	Lemma 46
Known-Ω exploration cost	explicit radius times effective-dimension potential	Proposition 47
Estimated-Ω width stability	$\mathrm{WidthErr}_T$ under spectral error and confidence	Proposition 50
Frozen calibration estimate	finite-sample covariance and exploration bounds	Theorem 57
Periodic covariance estimate	confidence and covariance rates	open
Empirical validation of hyper-ridge gains	oracle, estimated, diagonal, and base routers	diagnostic only; full test remains open

Table 33: Theory map for Hyper-Ridge PersonalEvo. The oracle and calibration-split variants have separate guarantees; the periodic estimator does not.

L Estimated-Covariance Control: the Calibration-Split Theorem

This section analyzes a separate calibration-split router. It estimates Ω from fresh candidates, freezes the estimate, and then routes over a fixed archive. The result does not cover the periodic estimator in Algorithm 1.

Assumption 19 (Exogenous uniform-routing calibration diversity)**.** *Let $\Omega \succ 0$. In the calibration phase, the algorithm uses N calibration candidates indexed by $i \in \{1, \dots, N\}$, and it collects m calibration rewards for each calibration candidate. The calibration feature at calibration pull j of calibration candidate i is $x_{i,j} \in \mathbb{R}^d$, and we write*

$$z_{i,j} = \Omega^{1/2}x_{i,j}.$$

The calibration features are generated by the environment under an exogenous uniform-routing calibration policy before the reward noises of the corresponding calibration block are revealed. The following conditions hold.

First, the feature arrays

$$X_i = \begin{bmatrix} x_{i,1}^{\top} \\ \vdots \\ x_{i,m}^{\top} \end{bmatrix}, \qquad i = 1, \dots, N,$$

are independent across i. Within each calibration block, the vectors $z_{i,1}, \ldots, z_{i,m}$ are independent. For every i, j,

$$\mathbb{E}\big[z_{i,j} z_{i,j}^\top\big] \succeq \kappa_{\mathrm{cal}} I_d, \qquad \|z_{i,j}\|_2^2 \le L_{\mathrm{cal}}^2 \quad \textit{almost surely},$$

for deterministic constants $\kappa_{\mathrm{cal}} > 0$ and $L_{\mathrm{cal}} < \infty$. The full calibration feature array is independent of the calibration candidate effects.

Second, the calibration candidate effects $\theta_1, \ldots, \theta_N$ are independent and identically distributed and satisfy

$$\mathbb{E}\theta_i = 0, \qquad \mathbb{E}\theta_i\theta_i^\top = \Omega, \qquad \|\theta_i\|_{\Omega^{-1}} \le S_\Omega \quad \textit{almost surely}.$$

The observed calibration reward and centered response are

$$r_{i,j} = b(u_{i,j}) + x_{i,j}^\top \theta_i + \xi_{i,j}, \qquad y_{i,j} := r_{i,j} - b(u_{i,j}).$$

Third, conditional on the full calibration feature array, the pairs (θ_i, ξ_i), where $\xi_i = (\xi_{i,1}, \ldots, \xi_{i,m})^\top$, are independent across i. Within each block there is a filtration

$$\mathcal{H}_{i,0} \subseteq \mathcal{H}_{i,1} \subseteq \cdots \subseteq \mathcal{H}_{i,m}$$

such that the full calibration feature array and θ_i are $\mathcal{H}_{i,0}$-measurable, $\xi_{i,j}$ is $\mathcal{H}_{i,j}$-measurable, and, for every $\eta \in \mathbb{R}$,

$$\mathbb{E}[\exp(\eta\xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{\eta^2\sigma^2}{2}\right) \quad \textit{almost surely}.$$

The filtration $\mathcal{H}_{i,j}$ is used only for conditioning. The calibration estimator observes X_i and y_i, not θ_i.

Lemma 51 (Net reduction for symmetric quadratic forms). *Let $A \in \mathbb{R}^{d\times d}$ be symmetric. Let $\mathcal{N} \subset \mathbb{S}^{d-1}$ be a $1/4$-net of the Euclidean unit sphere. Then*

$$\|A\|_{\mathrm{op}} \le 2 \sup_{v\in\mathcal{N}} |v^\top A v|.$$

Moreover, there exists such a $1/4$-net with cardinality at most 9^d.

Proof. Define

$$M = \sup_{v\in\mathcal{N}} |v^\top A v|.$$

Since A is symmetric,

$$\|A\|_{\mathrm{op}} = \sup_{u\in\mathbb{S}^{d-1}} |u^\top A u|.$$

Fix $u \in \mathbb{S}^{d-1}$. Choose $v \in \mathcal{N}$ satisfying

$$\|u - v\|_2 \le \frac{1}{4}.$$

Then

$$u^\top A u - v^\top A v = (u-v)^\top A u + v^\top A(u-v).$$

Using $\|u\|_2 = \|v\|_2 = 1$, we obtain

$$|u^\top A u - v^\top A v| \le \|u-v\|_2\|A\|_{\mathrm{op}}\|u\|_2 + \|v\|_2\|A\|_{\mathrm{op}}\|u-v\|_2 \le \frac{1}{2}\|A\|_{\mathrm{op}}.$$

Therefore

$$|u^\top A u| \le M + \frac{1}{2}\|A\|_{\mathrm{op}}.$$

Taking the supremum over $u \in \mathbb{S}^{d-1}$ yields

$$\|A\|_{\mathrm{op}} \le M + \frac{1}{2}\|A\|_{\mathrm{op}}.$$

Rearranging proves

$$\|A\|_{\mathrm{op}} \le 2M.$$

For the covering-number claim, let $\mathcal{N}$ be a maximal subset of $\mathbb{S}^{d-1}$ such that distinct points of $\mathcal{N}$ have Euclidean distance strictly larger than $1/4$. Maximality implies that $\mathcal{N}$ is a $1/4$-net, because any point of the sphere whose distance from every point of $\mathcal{N}$ were larger than $1/4$ could be added to $\mathcal{N}$. The closed Euclidean balls

$$B(v, 1/8) = \{w \in \mathbb{R}^d : \|w - v\|_2 \leq 1/8\}, \qquad v \in \mathcal{N},$$

are pairwise disjoint. Each is contained in $B(0, 9/8)$, since $\|v\|_2 = 1$. Comparing d-dimensional Euclidean volumes gives

$$|\mathcal{N}| \left(\frac{1}{8}\right)^d \leq \left(\frac{9}{8}\right)^d.$$

Thus $|\mathcal{N}| \leq 9^d$. □

Lemma 52 (Independent sub-Gaussian sample covariance)**.** *There is a universal numerical constant $C_{\mathrm{cov}} > 0$ with the following property. Let $Z_1, \ldots, Z_N \in \mathbb{R}^d$ be independent mean-zero random vectors such that, for every i, every $u \in \mathbb{S}^{d-1}$, and every $s \in \mathbb{R}$,*

$$\mathbb{E} \exp(s u^\top Z_i) \leq \exp\left(\frac{s^2 K^2}{2}\right).$$

Let

$$\Sigma_i = \mathbb{E}[Z_i Z_i^\top].$$

Then, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left\| \frac{1}{N} \sum_{i=1}^{N} \left(Z_i Z_i^\top - \Sigma_i\right) \right\|_{\mathrm{op}} \leq C_{\mathrm{cov}} K^2 \left[\sqrt{\frac{d + \log(2/\delta)}{N}} + \frac{d + \log(2/\delta)}{N} \right].$$

The same conclusion holds conditionally on any sigma-field with respect to which the displayed independence and moment-generating-function hypotheses hold with the same deterministic value of K.

Proof. Fix $u \in \mathbb{S}^{d-1}$. The displayed moment-generating-function bound implies

$$\Pr(u^\top Z_i \geq t) \leq \inf_{s>0} \exp\left(-st + \frac{s^2 K^2}{2}\right) = \exp\left(-\frac{t^2}{2K^2}\right) \qquad t \geq 0.$$

Applying the same argument to $-u$ gives

$$\Pr(|u^\top Z_i| \geq t) \leq 2 \exp\left(-\frac{t^2}{2K^2}\right) \qquad t \geq 0.$$

Equivalently, $u^\top Z_i$ is sub-Gaussian with sub-Gaussian norm bounded by a universal multiple of K. Lemma 2.7.7 of Vershynin, *High-Dimensional Probability*, Cambridge University Press, 2018, shows that its square is sub-exponential. Centering changes the sub-exponential norm by at most a universal factor, so

$$Y_i = (u^\top Z_i)^2 - \mathbb{E}(u^\top Z_i)^2$$

is sub-exponential with sub-exponential norm bounded by a universal multiple of K^2. Theorem 2.8.1 of the same reference, Bernstein's inequality for sums of independent centered sub-exponential random variables, gives a universal constant $c_1 > 0$ such that, for every $t > 0$,

$$\Pr\left(\left| \frac{1}{N} \sum_{i=1}^{N} \left[(u^\top Z_i)^2 - \mathbb{E}(u^\top Z_i)^2 \right] \right| > c_1 K^2 \left[\sqrt{\frac{t}{N}} + \frac{t}{N} \right] \right) \leq 2e^{-t}.$$

Let $\mathcal{N}$ be a $1/4$-net of $\mathbb{S}^{d-1}$ with $|\mathcal{N}| \leq 9^d$, whose existence follows from Lemma 51. Applying the preceding scalar inequality to every $u \in \mathcal{N}$, with

$$t = d \log 9 + \log(2/\delta),$$

and taking a union bound gives an event of probability at least $1 - \delta$ on which

$$\sup_{u \in \mathcal{N}} \left| u^\top \left[\frac{1}{N} \sum_{i=1}^{N} \left(Z_i Z_i^\top - \Sigma_i\right) \right] u \right| \leq c_1 K^2 \left[\sqrt{\frac{d \log 9 + \log(2/\delta)}{N}} + \frac{d \log 9 + \log(2/\delta)}{N} \right].$$

Applying Lemma 51 to the symmetric matrix inside the brackets yields

$$\left\| \frac{1}{N} \sum_{i=1}^{N} \left(Z_i Z_i^\top - \Sigma_i \right) \right\|_{\text{op}} \leq 2c_1 K^2 \left[\sqrt{\frac{d \log 9 + \log(2/\delta)}{N}} + \frac{d \log 9 + \log(2/\delta)}{N} \right].$$

Since $\log 9 < 3$, increasing the universal constant proves the stated bound.

For the conditional version, fix a regular conditional distribution given the conditioning sigma-field. Under that conditional distribution, the same proof applies with the same deterministic K and the same constants. Integrating the conditional failure probability over the conditioning sigma-field gives the unconditional conditional-form statement. ☐

Lemma 53 (Calibration Gram lower tail under exogenous diversity)**.** *Assume Assumption 19. For each calibration candidate define*

$$G_i = X_i^\top X_i, \qquad H_i = \Omega^{1/2} G_i \Omega^{1/2} = \sum_{j=1}^{m} z_{i,j} z_{i,j}^\top.$$

For every $\delta_{\text{gram}} \in (0, 1)$*, if*

$$m \geq \frac{8L_{\text{cal}}^2}{\kappa_{\text{cal}}} \log\left(\frac{dN}{\delta_{\text{gram}}} \right),$$

then, with probability at least $1 - \delta_{\text{gram}}$*,*

$$H_i \succeq \frac{\kappa_{\text{cal}} m}{2} I_d \qquad \textit{for every } i = 1, \ldots, N.$$

On the same event every G_i *is invertible and*

$$\left\| \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} \right\|_{\text{op}} \leq \frac{2}{\kappa_{\text{cal}} m} \qquad \textit{for every } i = 1, \ldots, N.$$

Proof. Fix i. Define

$$A_{i,j} = z_{i,j} z_{i,j}^\top.$$

By Assumption 19, the matrices $A_{i,1}, \ldots, A_{i,m}$ are independent positive-semidefinite random matrices and satisfy

$$0 \preceq A_{i,j} \preceq L_{\text{cal}}^2 I_d$$

almost surely. Moreover,

$$\lambda_{\min}\left(\sum_{j=1}^{m} \mathbb{E} A_{i,j} \right) \geq \kappa_{\text{cal}} m.$$

Tropp's matrix Chernoff lower-tail inequality for independent positive-semidefinite summands bounded above by RI_d states that, with

$$\mu_{\min} = \lambda_{\min}\left(\sum_{j=1}^{m} \mathbb{E} A_{i,j} \right),$$

for every $\varepsilon \in [0, 1]$,

$$\Pr\left(\lambda_{\min}\left(\sum_{j=1}^{m} A_{i,j} \right) \leq (1 - \varepsilon)\mu_{\min} \right) \leq d \exp\left(-\frac{\varepsilon^2 \mu_{\min}}{2R} \right).$$

This is Theorem 5.1.1 of Tropp, *An Introduction to Matrix Concentration Inequalities*, Foundations and Trends in Machine Learning, 2015, applied with $R = L_{\text{cal}}^2$. Taking $\varepsilon = 1/2$ gives

$$\Pr\left(H_i \not\succeq \frac{\kappa_{\text{cal}} m}{2} I_d \right) \leq d \exp\left(-\frac{\kappa_{\text{cal}} m}{8L_{\text{cal}}^2} \right).$$

The assumed lower bound on m implies

$$d \exp\left(-\frac{\kappa_{\text{cal}} m}{8L_{\text{cal}}^2} \right) \leq \frac{\delta_{\text{gram}}}{N}.$$

A union bound over $i = 1, \ldots, N$ proves the simultaneous Gram event.

On this event,

$$\Omega^{1/2} G_i \Omega^{1/2} \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d.$$

Thus G_i is invertible. Inverting the Loewner inequality gives

$$(\Omega^{1/2} G_i \Omega^{1/2})^{-1} \preceq \frac{2}{\kappa_{\mathrm{cal}} m} I_d.$$

Since

$$(\Omega^{1/2} G_i \Omega^{1/2})^{-1} = \Omega^{-1/2} G_i^{-1} \Omega^{-1/2},$$

the asserted operator-norm bound follows. □

Lemma 54 (Conditional calibration-block covariance control)**.** *Assume Assumption 19. Write $X_{1:N} = (X_1, \ldots, X_N)$. Condition on a realization of the full calibration feature array for which*

$$H_i = \Omega^{1/2} X_i^\top X_i \Omega^{1/2} \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \qquad \textit{for every } i = 1, \ldots, N.$$

For each i, define

$$\widetilde{\theta}_i = G_i^{-1} X_i^\top y_i, \qquad y_i = \begin{bmatrix} r_{i,1} - b(u_{i,1}) \\ \vdots \\ r_{i,m} - b(u_{i,m}) \end{bmatrix} = X_i \theta_i + \xi_i,$$

and define

$$Z_i = \Omega^{-1/2} \widetilde{\theta}_i, \qquad \tau_m = \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m.$$

Then, conditional on the feature array, $Z_1, \ldots, Z_N$ are independent mean-zero random vectors. Moreover,

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \qquad R_i \succeq 0, \qquad \|R_i\|_{\mathrm{op}} \le \tau_m,$$

and, for every $u \in \mathbb{S}^{d-1}$ and every $s \in \mathbb{R}$,

$$\mathbb{E}\big[\exp(s u^\top Z_i) \mid X_{1:N}\big] \le \exp\left(\frac{s^2 K_m^2}{2}\right).$$

Proof. The least-squares identity gives

$$\widetilde{\theta}_i = G_i^{-1} X_i^\top (X_i \theta_i + \xi_i) = \theta_i + G_i^{-1} X_i^\top \xi_i.$$

Set

$$U_i = \Omega^{-1/2} \theta_i, \qquad E_i = \Omega^{-1/2} G_i^{-1} X_i^\top \xi_i.$$

Then

$$Z_i = U_i + E_i.$$

The random-effects assumptions imply, conditional on the full calibration feature array,

$$\mathbb{E}[U_i \mid X_{1:N}] = 0, \qquad \mathbb{E}[U_i U_i^\top \mid X_{1:N}] = I_d, \qquad \|U_i\|_2 \le S_\Omega \quad \text{almost surely}.$$

The conditional sub-Gaussian noise assumption implies

$$\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] = 0.$$

Indeed, for $h > 0$, Jensen's inequality gives

$$\exp(h\, \mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}]) \le \mathbb{E}[\exp(h \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{h^2 \sigma^2}{2}\right),$$

so $\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] \le h\sigma^2/2$. Letting $h \downarrow 0$ gives the upper bound 0. Applying the same argument to $-\xi_{i,j}$ gives the lower bound 0. Therefore the conditional mean is zero. By iterated conditioning,

$$\mathbb{E}[\xi_i \mid X_{1:N}, \theta_i] = 0.$$

Consequently,

$$\mathbb{E}[E_i \mid X_{1:N}, \theta_i] = 0,$$

and the conditional cross terms satisfy

$$\mathbb{E}[U_i E_i^\top \mid X_{1:N}] = 0, \qquad \mathbb{E}[E_i U_i^\top \mid X_{1:N}] = 0.$$

Thus

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \qquad R_i = \mathbb{E}[E_i E_i^\top \mid X_{1:N}] \succeq 0.$$

We bound R_i. Fix $v \in \mathbb{S}^{d-1}$, and set

$$b_i = X_i G_i^{-1} \Omega^{-1/2} v \in \mathbb{R}^m.$$

Conditional on the feature array, b_i is deterministic. Then

$$v^\top R_i v = \mathbb{E}\left[(b_i^\top \xi_i)^2 \mid X_{1:N}\right].$$

For $j < k$, the random variable $\xi_{i,j}$ is $\mathcal{H}_{i,k-1}$-measurable. Hence

$$\mathbb{E}[\xi_{i,j}\xi_{i,k} \mid X_{1:N}] = \mathbb{E}[\xi_{i,j}\, \mathbb{E}[\xi_{i,k} \mid \mathcal{H}_{i,k-1}] \mid X_{1:N}] = 0.$$

The conditional sub-Gaussian inequality also implies

$$\mathbb{E}[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}] \le \sigma^2.$$

Indeed, for $h > 0$, applying the moment-generating-function bound at h and $-h$ gives

$$\mathbb{E}[\cosh(h\xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{h^2\sigma^2}{2}\right).$$

Conditional Fatou's lemma and $2(\cosh(hx) - 1)/h^2 \to x^2$ yield

$$\mathbb{E}\left[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}\right] \le \liminf_{h \downarrow 0} \frac{2}{h^2}\left[\exp\left(\frac{h^2\sigma^2}{2}\right) - 1\right] = \sigma^2.$$

Therefore

$$v^\top R_i v \le \sigma^2 \|b_i\|_2^2.$$

Moreover,

$$\|b_i\|_2^2 = v^\top \Omega^{-1/2} G_i^{-1} X_i^\top X_i G_i^{-1} \Omega^{-1/2} v = v^\top \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} v.$$

On the conditioned Gram event,

$$\Omega^{-1/2} G_i^{-1} \Omega^{-1/2} = (\Omega^{1/2} G_i \Omega^{1/2})^{-1} \preceq \frac{2}{\kappa_{\mathrm{cal}} m} I_d.$$

Thus

$$v^\top R_i v \le \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m} = \tau_m.$$

Taking the supremum over $v \in \mathbb{S}^{d-1}$ gives

$$\|R_i\|_{\mathrm{op}} \le \tau_m.$$

It remains to prove the one-dimensional moment-generating-function bound. Fix $u \in \mathbb{S}^{d-1}$. Since

$$\mathbb{E}[u^\top U_i \mid X_{1:N}] = 0$$

and

$$|u^\top U_i| \le \|U_i\|_2 \le S_\Omega \quad \text{almost surely},$$

Hoeffding's lemma gives

$$\mathbb{E}\left[\exp(s u^\top U_i) \mid X_{1:N}\right] \le \exp\left(\frac{s^2 S_\Omega^2}{2}\right).$$

For the noise part, set

$$b_i = X_i G_i^{-1} \Omega^{-1/2} u.$$

Conditional on the full feature array, b_i is deterministic. Iterating the conditional sub-Gaussian inequality over $j = m, m-1, \dots, 1$ gives

$$\mathbb{E}\big[\exp(s b_i^\top \xi_i) \mid X_{1:N}, \theta_i\big] \le \exp\left(\frac{s^2\sigma^2\|b_i\|_2^2}{2}\right).$$

The Gram-event calculation above gives

$$\|b_i\|_2^2 \le \frac{2}{\kappa_{\mathrm{cal}} m}.$$

Therefore

$$\mathbb{E}\big[\exp(s u^\top E_i) \mid X_{1:N}, \theta_i\big] \le \exp\left(\frac{s^2\tau_m}{2}\right).$$

Since U_i is a function of θ_i,

$$\begin{aligned}
\mathbb{E}\big[\exp(s u^\top Z_i) \mid X_{1:N}\big] &= \mathbb{E}\big[\exp(s u^\top U_i)\mathbb{E}\big[\exp(s u^\top E_i) \mid X_{1:N}, \theta_i\big] \mid X_{1:N}\big] \\
&\le \exp\left(\frac{s^2\tau_m}{2}\right)\mathbb{E}\big[\exp(s u^\top U_i) \mid X_{1:N}\big] \\
&\le \exp\left(\frac{s^2(S_\Omega^2 + \tau_m)}{2}\right).
\end{aligned}$$

Finally, Assumption 19 makes the pairs (θ_i, ξ_i) independent across i, conditional on the feature array. Since Z_i is a function of (X_i, θ_i, ξ_i), the Z_i are conditionally independent. □

Lemma 55 (Relative covariance error for the calibration estimator)**.** *Assume Assumption 19. Define*

$$\widehat{\Omega} = \frac{1}{N}\sum_{i=1}^{N} \widetilde{\theta}_i\widetilde{\theta}_i^\top + \gamma I_d, \qquad \gamma > 0,$$

where $\widetilde{\theta}_i = G_i^{-1} X_i^\top y_i$ on the event that G_i is invertible, and $\widetilde{\theta}_i = 0$ otherwise. Let

$$\tau_m = \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m.$$

For $\delta_{\mathrm{gram}}, \delta_{\mathrm{cov}} \in (0,1)$, suppose

$$m \ge \frac{8L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}} \log\left(\frac{dN}{\delta_{\mathrm{gram}}}\right).$$

Define

$$\varepsilon_{\mathrm{cov}}(N, m, \gamma, \delta_{\mathrm{cov}}) = C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d + \log(2/\delta_{\mathrm{cov}})}{N}\right] + \tau_m + \gamma\|\Omega^{-1}\|_{\mathrm{op}}.$$

Then, with probability at least $1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}}$,

$$\left\|\Omega^{-1/2}(\widehat{\Omega} - \Omega)\Omega^{-1/2}\right\|_{\mathrm{op}} \le \varepsilon_{\mathrm{cov}}(N, m, \gamma, \delta_{\mathrm{cov}}).$$

Proof. Let

$$\mathcal{E}_{\mathrm{gram}} = \left\{H_i \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \text{ for every } i = 1, \dots, N\right\}.$$

By Lemma 53,

$$\Pr(\mathcal{E}_{\mathrm{gram}}) \ge 1 - \delta_{\mathrm{gram}}.$$

Condition on a realization of the full calibration feature array in $\mathcal{E}_{\mathrm{gram}}$. By Lemma 54, the vectors

$$Z_i = \Omega^{-1/2}\widetilde{\theta}_i$$

are conditionally independent and satisfy

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \qquad R_i \succeq 0, \qquad \|R_i\|_{\mathrm{op}} \le \tau_m,$$

and, for every $u \in \mathbb{S}^{d-1}$ and every $s \in \mathbb{R}$,

$$\mathbb{E}\big[\exp(su^\top Z_i) \mid X_{1:N}\big] \le \exp\left(\frac{s^2 K_m^2}{2}\right).$$

Applying the conditional form of Lemma 52 gives, conditional on this feature realization, an event of conditional probability at least $1 - \delta_{\mathrm{cov}}$ on which

$$\left\| \frac{1}{N} \sum_{i=1}^{N} \big[Z_i Z_i^\top - \mathbb{E}(Z_i Z_i^\top \mid X_{1:N})\big] \right\|_{\mathrm{op}} \le C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d + \log(2/\delta_{\mathrm{cov}})}{N} \right].$$

On the intersection of $\mathcal{E}_{\mathrm{gram}}$ and this conditional covariance event,

$$\begin{aligned} \left\| \frac{1}{N} \sum_{i=1}^{N} Z_i Z_i^\top - I_d \right\|_{\mathrm{op}} &\le \left\| \frac{1}{N} \sum_{i=1}^{N} \big[Z_i Z_i^\top - \mathbb{E}(Z_i Z_i^\top \mid X_{1:N})\big] \right\|_{\mathrm{op}} + \left\| \frac{1}{N} \sum_{i=1}^{N} R_i \right\|_{\mathrm{op}} \\ &\le C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d + \log(2/\delta_{\mathrm{cov}})}{N} \right] + \tau_m. \end{aligned}$$

The last inequality uses

$$\left\| \frac{1}{N} \sum_{i=1}^{N} R_i \right\|_{\mathrm{op}} \le \frac{1}{N} \sum_{i=1}^{N} \|R_i\|_{\mathrm{op}} \le \tau_m.$$

Since

$$\Omega^{-1/2} \widehat{\Omega} \Omega^{-1/2} = \frac{1}{N} \sum_{i=1}^{N} Z_i Z_i^\top + \gamma \Omega^{-1},$$

we have

$$\Omega^{-1/2} (\widehat{\Omega} - \Omega) \Omega^{-1/2} = \left(\frac{1}{N} \sum_{i=1}^{N} Z_i Z_i^\top - I_d \right) + \gamma \Omega^{-1}.$$

The triangle inequality gives

$$\left\| \Omega^{-1/2} (\widehat{\Omega} - \Omega) \Omega^{-1/2} \right\|_{\mathrm{op}} \le \varepsilon_{\mathrm{cov}}(N, m, \gamma, \delta_{\mathrm{cov}}).$$

The unconditional probability of the intersection of the Gram event and the conditional covariance event is at least $1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}}$ by the law of total probability and a union bound. □

Lemma 56 (Confidence under a frozen estimated covariance)**.** *Assume the post-calibration routing model of Assumption 17. Assume the post-calibration routing archive is fixed before routing begins and contains at most $K_{\max}$ candidates. Let $\mathcal{F}_0^{\mathrm{route}}$ contain all calibration data, the fixed routing archive, and the latent parameter of every routing candidate. Let $\widehat{\Omega} \succ 0$ be $\mathcal{F}_0^{\mathrm{route}}$-measurable and frozen during routing. Assume the post-routing conditional sub-Gaussian bound equation 1002 holds for a routing filtration that contains $\mathcal{F}_0^{\mathrm{route}}$. Suppose*

$$\left\| \Omega^{-1/2} (\widehat{\Omega} - \Omega) \Omega^{-1/2} \right\|_{\mathrm{op}} \le \varepsilon \qquad \textit{for some } 0 \le \varepsilon < 1.$$

For candidate a, define

$$V_{a,t}^{\widehat{\Omega}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}^{-1}, \qquad \widehat{\theta}_{a,t}^{\widehat{\Omega}} = (V_{a,t}^{\widehat{\Omega}})^{-1} X_{a,t}^\top y_{a,t}.$$

Define

$$\alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\mathrm{route}}) = \max\left\{ 1,\ \sigma \sqrt{d \log\left(1 + \frac{(1+\varepsilon) T L_{\Omega}^2}{\lambda}\right) + 2 \log\left(\frac{K_{\max}}{\delta_{\mathrm{route}}}\right)} + \sqrt{\frac{\lambda}{1-\varepsilon}}\, S_{\Omega} \right\}.$$

Then, conditional on $\mathcal{F}_0^{\mathrm{route}}$, with probability at least $1 - \delta_{\mathrm{route}}$, uniformly over all routing candidates a, all $t \le T$, and all predictable test features x,

$$\left| x^\top (\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a) \right| \le \alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\mathrm{route}}) \sqrt{x^\top (V_{a,t}^{\widehat{\Omega}})^{-1} x}.$$

Proof. Fix a routing candidate a. Conditional on $\mathcal{F}_0^{\mathrm{route}}$, set

$$A = \lambda\widehat{\Omega}^{-1}.$$

Then A is deterministic and positive definite. Define

$$S_{a,t} = \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\xi_s, \qquad V_{a,t}^{\widehat{\Omega}} = A + \sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top.$$

Using the reward model,

$$\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a = (V_{a,t}^{\widehat{\Omega}})^{-1}S_{a,t} - (V_{a,t}^{\widehat{\Omega}})^{-1}A\theta_a.$$

Hence

$$\|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} \le \|S_{a,t}\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} + \|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}}.$$

Because $V_{a,t}^{\widehat{\Omega}} \succeq A$,

$$\|A\theta_a\|^2_{(V_{a,t}^{\widehat{\Omega}})^{-1}} \le \theta_a^\top A\theta_a = \lambda\theta_a^\top\widehat{\Omega}^{-1}\theta_a.$$

The relative error assumption implies

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega} \preceq (1+\varepsilon)\Omega,$$

and therefore

$$\widehat{\Omega}^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}.$$

Using $\|\theta_a\|_{\Omega^{-1}} \le S_\Omega$, we get

$$\|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} \le \sqrt{\frac{\lambda}{1-\varepsilon}}\,S_\Omega.$$

It remains to control the martingale term. If $\sigma = 0$, conditional sub-Gaussianity implies $S_{a,t} = 0$ almost surely and the claim follows from the bias bound. Suppose $\sigma > 0$. For fixed $q \in \mathbb{R}^d$, define

$$M_t(q) = \exp\left(\frac{q^\top S_{a,t}}{\sigma} - \frac{1}{2}q^\top\left(V_{a,t}^{\widehat{\Omega}} - A\right)q\right).$$

If round t selects candidate a, conditional sub-Gaussianity gives

$$\mathbb{E}\left[\exp\left(\frac{q^\top x_t(a)\xi_t}{\sigma} - \frac{(q^\top x_t(a))^2}{2}\right)\middle|\mathcal{F}_{t-1}\right] \le 1.$$

If round t does not select a, then $S_{a,t}$ and $V_{a,t}^{\widehat{\Omega}}$ do not change. Therefore $(M_t(q))_{t\le T}$ is a nonnegative supermartingale with initial value one.

Let Q be Gaussian with precision A, so that Q has density

$$\varphi_A(q) = \frac{\det(A)^{1/2}}{(2\pi)^{d/2}}\exp\left(-\frac{1}{2}q^\top Aq\right).$$

Define

$$\mathcal{M}_t = \int_{\mathbb{R}^d} M_t(q)\,\varphi_A(q)\,dq.$$

Tonelli's theorem and the supermartingale property of $M_t(q)$ imply that $(\mathcal{M}_t)_{t\le T}$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Completing the square in the Gaussian integral gives

$$\mathcal{M}_t = \frac{\det(A)^{1/2}}{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}}\exp\left(\frac{\|S_{a,t}\|^2_{(V_{a,t}^{\widehat{\Omega}})^{-1}}}{2\sigma^2}\right).$$

Ville's inequality gives

$$\Pr\left(\sup_{t\le T}\mathcal{M}_t \ge \frac{1}{\delta_a}\right) \le \delta_a.$$

Thus, with probability at least $1-\delta_a$, uniformly over $t \le T$,

$$\|S_{a,t}\|_{(V^{\widehat{\Omega}}_{a,t})^{-1}} \le \sigma \sqrt{2\log\left(\frac{\det(V^{\widehat{\Omega}}_{a,t})^{1/2}}{\det(A)^{1/2}\delta_a}\right)}.$$

We next bound the determinant ratio. Since $A = \lambda\widehat{\Omega}^{-1}$,

$$\frac{\det(V^{\widehat{\Omega}}_{a,t})}{\det(A)} = \det\Big(I + \lambda^{-1}\widehat{\Omega}^{1/2}X_{a,t}^\top X_{a,t}\widehat{\Omega}^{1/2}\Big).$$

The relation $\widehat{\Omega} \preceq (1+\varepsilon)\Omega$ implies

$$\operatorname{tr}\Big(\lambda^{-1}\widehat{\Omega}^{1/2}X_{a,t}^\top X_{a,t}\widehat{\Omega}^{1/2}\Big) = \lambda^{-1}\operatorname{tr}(X_{a,t}\widehat{\Omega}X_{a,t}^\top) \le \frac{1+\varepsilon}{\lambda}\sum_{s\in\mathcal{T}_{a,t}} x_s(a)^\top \Omega x_s(a).$$

By the feature-radius assumption,

$$\sum_{s\in\mathcal{T}_{a,t}} x_s(a)^\top \Omega x_s(a) \le T L_\Omega^2.$$

For any positive-semidefinite $d \times d$ matrix M with eigenvalues $\nu_1, \dots, \nu_d$,

$$\det(I+M) = \prod_{k=1}^d (1+\nu_k) \le \left(1 + \sum_{k=1}^d \nu_k\right)^d = (1+\operatorname{tr} M)^d.$$

Therefore

$$2\log\left(\frac{\det(V^{\widehat{\Omega}}_{a,t})^{1/2}}{\det(A)^{1/2}\delta_a}\right) \le d\log\left(1 + \frac{(1+\varepsilon)TL_\Omega^2}{\lambda}\right) + 2\log(1/\delta_a).$$

Taking $\delta_a = \delta_{\mathrm{route}}/K_{\max}$ and applying a union bound over the fixed routing archive gives, with probability at least $1-\delta_{\mathrm{route}}$, the simultaneous bound

$$\|\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a\|_{V^{\widehat{\Omega}}_{a,t}} \le \alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\mathrm{route}})$$

for every candidate a and every $t \le T$. Finally, for every predictable test feature x, Cauchy–Schwarz in the $V^{\widehat{\Omega}}_{a,t}$-inner product gives

$$\left|x^\top(\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a)\right| \le \|\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a\|_{V^{\widehat{\Omega}}_{a,t}} \sqrt{x^\top (V^{\widehat{\Omega}}_{a,t})^{-1} x}.$$

Substituting the preceding simultaneous parameter bound proves the claim. □

Theorem 57 (Calibration-split frozen-covariance Hyper-Ridge)**.** *Fix a post-calibration routing horizon T. Consider the following modified Hyper-Ridge PersonalEvo procedure.*

For the version combined with the four-term decomposition, draw one routing user U before the post-calibration phase and hold that user fixed for all T routing rounds.

Before routing begins, the algorithm runs a calibration phase on $N = N_T$ calibration candidates that are not used in the subsequent routing archive. For each calibration candidate i, the algorithm collects $m = m_T$ calibration rewards under the exogenous uniform-routing calibration policy:

$$y_{i,j} := r_{i,j} - b(u_{i,j}) = x_{i,j}^\top \theta_i + \xi_{i,j}, \qquad j = 1, \dots, m.$$

Let

$$X_i = \begin{bmatrix} x_{i,1}^\top \\ \vdots \\ x_{i,m}^\top \end{bmatrix}, \qquad G_i = X_i^\top X_i.$$

On the event that G_i is invertible, set

$$\widetilde{\theta}_i = G_i^{-1} X_i^\top y_i;$$

otherwise set $\widetilde{\theta}_i = 0$. *For a deterministic ridge floor* $\gamma = \gamma_T > 0$, *define*

$$\widehat{\Omega} = \frac{1}{N}\sum_{i=1}^{N} \widetilde{\theta}_i \widetilde{\theta}_i^\top + \gamma I_d.$$

During all T *post-calibration routing rounds, the router uses the frozen estimate*

$$\widehat{\Omega}_t = \widehat{\Omega}.$$

The post-calibration routing archive is fixed before routing begins and has cardinality at most $K_{\max}$. *The routing UCB rule uses*

$$\alpha_T = \alpha_{\widehat{\Omega},T}(\varepsilon_*, \delta_{\text{route}}),$$

where $\alpha_{\widehat{\Omega},T}$ *is defined in Lemma 56, and where* ε_* *is the deterministic quantity defined below.*

Assume Assumption 19 and the post-calibration routing assumptions of Assumption 17. Let

$$\tau_m = \frac{2\sigma^2}{\kappa_{\text{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m.$$

For $\delta_{\text{gram}}, \delta_{\text{cov}}, \delta_{\text{route}} \in (0, 1)$, *suppose*

$$m \geq \frac{8L_{\text{cal}}^2}{\kappa_{\text{cal}}} \log\left(\frac{dN}{\delta_{\text{gram}}}\right).$$

Define

$$\varepsilon_* = C_{\text{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\text{cov}})}{N}} + \frac{d + \log(2/\delta_{\text{cov}})}{N}\right] + \tau_m + \gamma \|\Omega^{-1}\|_{\text{op}}.$$

Computing this radius requires supplied deterministic bounds for $S_\Omega, \sigma, \kappa_{\text{cal}}, L_{\text{cal}}, L_\Omega$, *and* $\|\Omega^{-1}\|_{\text{op}}$. *Without them, the certificate is oracle-tuned. Let* $\mathcal{F}_0^{\text{route}}$ *contain all calibration data, the fixed post-calibration archive, and the latent parameter of every routing candidate. Assume the post-routing conditional sub-Gaussian bound holds after this sigma-field is added to the routing filtration, and assume* $\varepsilon_* < 1$. *Define*

$$c_T^\Omega = 1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right)$$

and

$$B_{T,0}^{\text{cal}} = 2\alpha_T \sqrt{2K_{\max} T\, c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)}.$$

Then, with probability at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}}$,

$$\rho_{\Omega,T} := \max_{1 \leq t \leq T} \left\| \Omega^{-1/2}(\widehat{\Omega}_t - \Omega)\Omega^{-1/2} \right\|_{\text{op}} \leq \varepsilon_*.$$

In particular, taking $\delta_{\text{gram}} = \delta_{\text{cov}} = \delta/2$ *gives the covariance-error control with probability at least* $1 - \delta$.

With probability at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}} - \delta_{\text{route}}$,

$$\sum_{t=1}^{T} L_t(U) \leq \sqrt{1 + \varepsilon_*}\, B_{T,0}^{\text{cal}}.$$

Define

$$\text{Rem}_T^{\text{cal}} = \left(\sqrt{1 + \varepsilon_*} - 1\right) B_{T,0}^{\text{cal}} + (\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}})T.$$

Then

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \leq B_{T,0}^{\text{cal}} + \text{Rem}_T^{\text{cal}}.$$

If each of the Nm *calibration pulls is assigned a worst-case unit opportunity-cost charge, define*

$$\text{Rem}_T^{\text{charged}} = Nm + \text{Rem}_T^{\text{cal}}.$$

The expected post-calibration exploration cost plus this calibration charge is at most

$$B_{T,0}^{\mathrm{cal}} + \mathrm{Rem}_T^{\mathrm{charged}}.$$

For fixed

$$d,\ K_{\max},\ S_\Omega,\ \sigma,\ \kappa_{\mathrm{cal}},\ L_{\mathrm{cal}},\ L_\Omega,\ \lambda,\ \|\Omega^{-1}\|_{\mathrm{op}},$$

and a uniformly bounded sequence $\bar{d}_{\mathrm{eff},T}(\Omega)$*, take*

$$N = \lceil T^{1/5} \rceil, \qquad m = \lceil T^{1/5} \rceil, \qquad \gamma = T^{-1/5}, \qquad \delta_{\mathrm{gram}} = \delta_{\mathrm{cov}} = \delta_{\mathrm{route}} = T^{-2}.$$

Then, for all sufficiently large T*,*

$$\varepsilon_* < 1, \qquad \varepsilon_* = O\Big(T^{-1/10}\sqrt{\log T}\Big).$$

Moreover,

$$\alpha_T = O(\sqrt{d \log T}), \qquad B_{T,0}^{\mathrm{cal}} = O\Bigg(\sqrt{dK_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T}\,\log T\Bigg),$$

and

$$\mathrm{Rem}_T^{\mathrm{cal}} = O\Big(T^{2/5}(\log T)^{3/2}\Big) = o(\sqrt{T}).$$

The same rate holds for $\mathrm{Rem}_T^{\mathrm{charged}}$.

Proof. Because $\widehat{\Omega}_t = \widehat{\Omega}$ for every post-calibration routing round,

$$\rho_{\Omega,T} = \Big\|\Omega^{-1/2}(\widehat{\Omega} - \Omega)\Omega^{-1/2}\Big\|_{\mathrm{op}}.$$

The high-probability covariance bound is exactly Lemma 55. Therefore, with probability at least $1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}}$,

$$\rho_{\Omega,T} \le \varepsilon_*.$$

Assume now that $\varepsilon_* < 1$ and work on the covariance event

$$\mathcal{E}_{\mathrm{cov}} = \{\rho_{\Omega,T} \le \varepsilon_*\}.$$

On this event,

$$(1 - \varepsilon_*)\Omega \preceq \widehat{\Omega} \preceq (1 + \varepsilon_*)\Omega.$$

By Lemma 56, conditional on $\mathcal{F}_0^{\mathrm{route}}$, with probability at least $1 - \delta_{\mathrm{route}}$, the following simultaneous routing confidence event holds:

$$\Big|x^\top(\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a)\Big| \le \alpha_T\sqrt{x^\top(V_{a,t}^{\widehat{\Omega}})^{-1}x}$$

for every routing candidate a, every $t \le T$, and every predictable feature x. The conditional probability statement integrates to an unconditional statement. A union bound gives an event $\mathcal{E}$ of probability at least

$$1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}} - \delta_{\mathrm{route}}$$

on which both the covariance event and the routing confidence event hold.

On $\mathcal{E}$, fix a routing round t. Let

$$a_t^\star \in \arg\max_{a \in A_t(U)} x_t(a)^\top \theta_a$$

and let a_t be the action selected by the frozen estimated-covariance UCB rule. The confidence event applied to $a_t^\star$ gives

$$x_t(a_t^\star)^\top \theta_{a_t^\star} \le x_t(a_t^\star)^\top \widehat{\theta}_{a_t^\star,t}^{\widehat{\Omega}} + \alpha_T\sqrt{x_t(a_t^\star)^\top(V_{a_t^\star,t}^{\widehat{\Omega}})^{-1}x_t(a_t^\star)}.$$

Since a_t maximizes the same UCB score,

$$\begin{aligned} &x_t(a_t^\star)^\top \widehat{\theta}_{a_t^\star,t}^{\widehat{\Omega}} + \alpha_T\sqrt{x_t(a_t^\star)^\top(V_{a_t^\star,t}^{\widehat{\Omega}})^{-1}x_t(a_t^\star)} \\ &\le x_t(a_t)^\top \widehat{\theta}_{a_t,t}^{\widehat{\Omega}} + \alpha_T\sqrt{x_t(a_t)^\top(V_{a_t,t}^{\widehat{\Omega}})^{-1}x_t(a_t)}. \end{aligned}$$

The confidence event applied to a_t gives

$$x_t(a_t)^\top \widehat{\theta}^{\widehat{\Omega}}_{a_t,t} \le x_t(a_t)^\top \theta_{a_t} + \alpha_T \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}.$$

Combining the last three inequalities yields

$$L_t(U) = x_t(a_t^\star)^\top \theta_{a_t^\star} - x_t(a_t)^\top \theta_{a_t} \le 2\alpha_T \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}.$$

Since $0 \le L_t(U) \le 1$ and $\alpha_T \ge 1$,

$$L_t(U) \le 2\alpha_T \min\left\{1, \sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)}\right\}.$$

The covariance event implies

$$\widehat{\Omega}^{-1} \succeq \frac{1}{1+\varepsilon_*}\Omega^{-1}.$$

Therefore

$$\begin{aligned} V^{\widehat{\Omega}}_{a,t} &= X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}^{-1} \\ &\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1+\varepsilon_*}\Omega^{-1} \\ &\succeq \frac{1}{1+\varepsilon_*}\left(X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}\right) = \frac{1}{1+\varepsilon_*} V^{\Omega}_{a,t}. \end{aligned}$$

Inverting gives

$$(V^{\widehat{\Omega}}_{a,t})^{-1} \preceq (1+\varepsilon_*)(V^{\Omega}_{a,t})^{-1}.$$

Consequently,

$$\sqrt{x_t(a_t)^\top (V^{\widehat{\Omega}}_{a_t,t})^{-1} x_t(a_t)} \le \sqrt{1+\varepsilon_*}\sqrt{x_t(a_t)^\top (V^{\Omega}_{a_t,t})^{-1} x_t(a_t)}.$$

For $c \ge 1$ and $z \ge 0$,

$$\min\{1, cz\} \le c \min\{1, z\}.$$

Applying this with $c = \sqrt{1+\varepsilon_*}$ gives

$$L_t(U) \le 2\alpha_T \sqrt{1+\varepsilon_*} \min\left\{1, \sqrt{x_t(a_t)^\top (V^{\Omega}_{a_t,t})^{-1} x_t(a_t)}\right\}.$$

Summing over t, partitioning selected rounds by candidate, applying Cauchy–Schwarz, and then applying the already-proved Ω-elliptical-potential lemma candidate by candidate yields

$$\sum_{t=1}^{T} \min\left\{1, \sqrt{x_t(a_t)^\top (V^{\Omega}_{a_t,t})^{-1} x_t(a_t)}\right\} \le \sqrt{2K_{\max} T\, c_T^{\Omega} \bar{d}_{\mathrm{eff},T}(\Omega)}.$$

Substituting this bound into the previous display proves the pathwise routing bound

$$\sum_{t=1}^{T} L_t(U) \le \sqrt{1+\varepsilon_*}\, B^{\mathrm{cal}}_{T,0}.$$

On $\mathcal{E}$,

$$\sum_{t=1}^{T} L_t(U) \le \sqrt{1+\varepsilon_*}\, B^{\mathrm{cal}}_{T,0} = B^{\mathrm{cal}}_{T,0} + (\sqrt{1+\varepsilon_*} - 1) B^{\mathrm{cal}}_{T,0}.$$

On $\mathcal{E}^c$, the deterministic bound $0 \le L_t(U) \le 1$ gives

$$\sum_{t=1}^{T} L_t(U) \le T.$$

Taking expectations yields

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B^{\mathrm{cal}}_{T,0} + (\sqrt{1+\varepsilon_*} - 1) B^{\mathrm{cal}}_{T,0} + (\delta_{\mathrm{gram}} + \delta_{\mathrm{cov}} + \delta_{\mathrm{route}})T.$$

This is the stated bound with $\mathrm{Rem}_T^{\mathrm{cal}}$. Under the declared unit opportunity-cost convention, the Nm calibration pulls add at most Nm, giving $\mathrm{Rem}_T^{\mathrm{charged}}$.

It remains to verify the asymptotic schedule. With fixed problem constants and

$$N = \lceil T^{1/5} \rceil, \qquad m = \lceil T^{1/5} \rceil, \qquad \gamma = T^{-1/5}, \qquad \delta_{\mathrm{gram}} = \delta_{\mathrm{cov}} = \delta_{\mathrm{route}} = T^{-2},$$

we have

$$\sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} = O\left(T^{-1/10}\sqrt{\log T}\right),$$
$$\frac{d + \log(2/\delta_{\mathrm{cov}})}{N} = O(T^{-1/5}\log T),$$
$$\tau_m = O(T^{-1/5}), \qquad \gamma\|\Omega^{-1}\|_{\mathrm{op}} = O(T^{-1/5}).$$

The dominant term is $O(T^{-1/10}\sqrt{\log T})$, so

$$\varepsilon_* = O\left(T^{-1/10}\sqrt{\log T}\right).$$

Since $T^{-1/10}\sqrt{\log T} \to 0$, there exists $T_0 < \infty$ such that $\varepsilon_* < 1$ for all $T \geq T_0$. The Gram lower-tail condition also holds for all sufficiently large T, because its right-hand side is $O(\log T)$ whereas $m = \lceil T^{1/5} \rceil$.

The explicit radius in Lemma 56 then satisfies

$$\alpha_T = O(\sqrt{d \log T}).$$

Since $c_T^{\Omega} = O(\log T)$ and the effective-dimension cap is uniformly bounded,

$$B_{T,0}^{\mathrm{cal}} = O\left(\sqrt{dK_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T}\,\log T\right).$$

For $0 \leq \varepsilon_* \leq 1$,

$$\sqrt{1+\varepsilon_*} - 1 = \frac{\varepsilon_*}{\sqrt{1+\varepsilon_*}+1} \leq \varepsilon_*.$$

Therefore

$$(\sqrt{1+\varepsilon_*} - 1)B_{T,0}^{\mathrm{cal}} = O\left(T^{-1/10}\sqrt{\log T}\right) O(\sqrt{T}\log T) = O\left(T^{2/5}(\log T)^{3/2}\right).$$

Also,

$$Nm = O(T^{2/5}),$$

and

$$(\delta_{\mathrm{gram}} + \delta_{\mathrm{cov}} + \delta_{\mathrm{route}})T = 3T^{-1}.$$

Hence

$$\mathrm{Rem}_T^{\mathrm{charged}} = O(T^{2/5}) + O\left(T^{2/5}(\log T)^{3/2}\right) + O(T^{-1}) = o(\sqrt{T}),$$

because $T^{-1/10}(\log T)^{3/2} \to 0$. The uncharged remainder is no larger. □

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